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(54) **HEAT EXCHANGE AND TEMPERATURE SENSING DEVICE AND METHOD OF USE**

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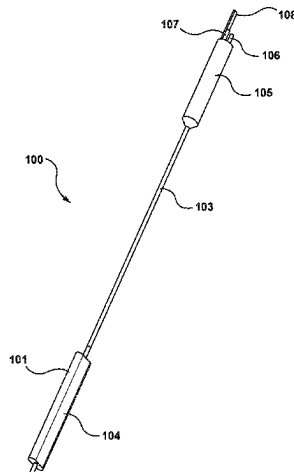
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(57) **ABSTRACT**

A method and apparatus are disclosed for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium, the method including altering a heat

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exchange device from an insertable configuration to a heat exchanging configuration which conforms and corresponds with a cross-section of an inside of the esophagus such that the esophagus is maintained in its natural shape and location. In some embodiments the heat exchange device has a heating/cooling balloon which is inflated to be in the heat exchanging configuration. Some alternative embodiments includes altering the configuration of the balloon to conform to or correspond with the cross section of an esophagus by means other than inflation.

20 Claims, 54 Drawing Sheets

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See application file for complete search history.

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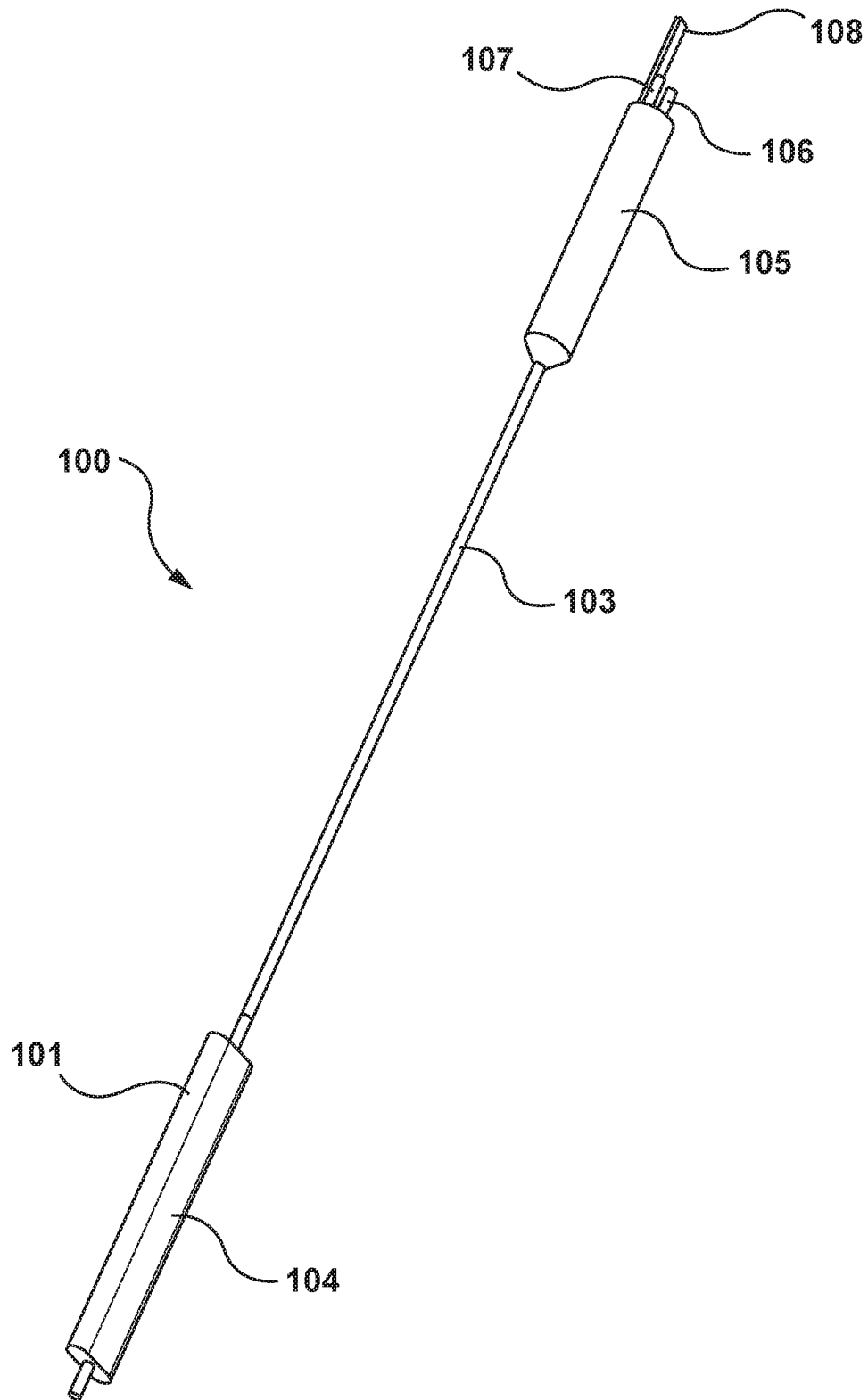
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**FIG. 1**

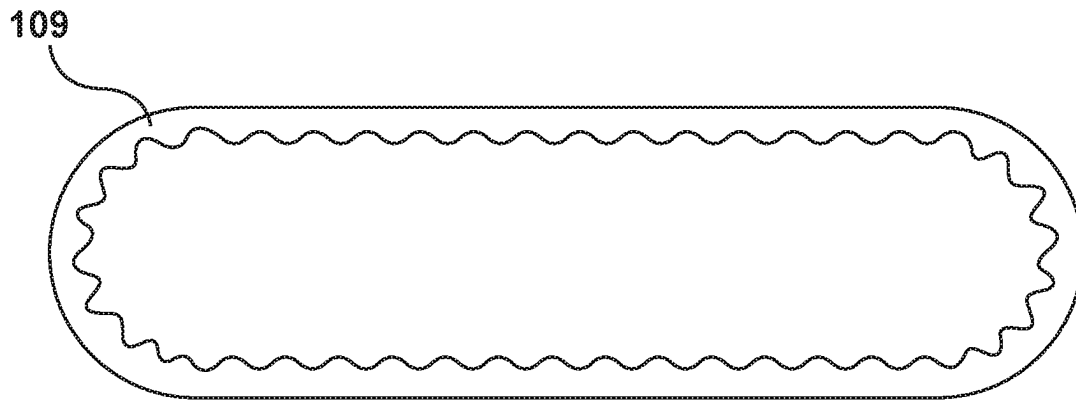


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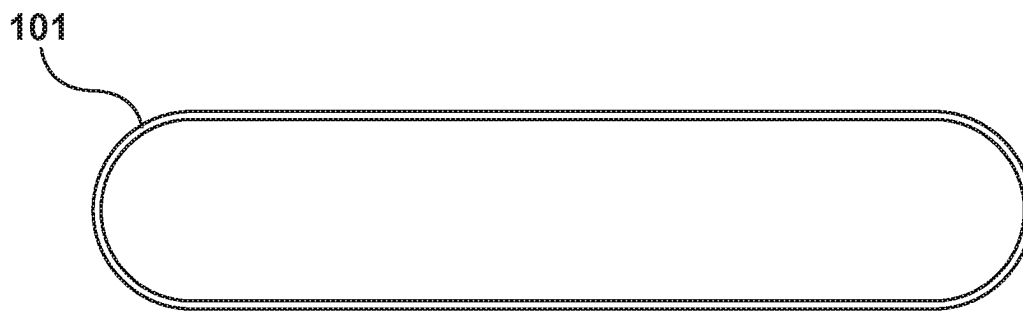


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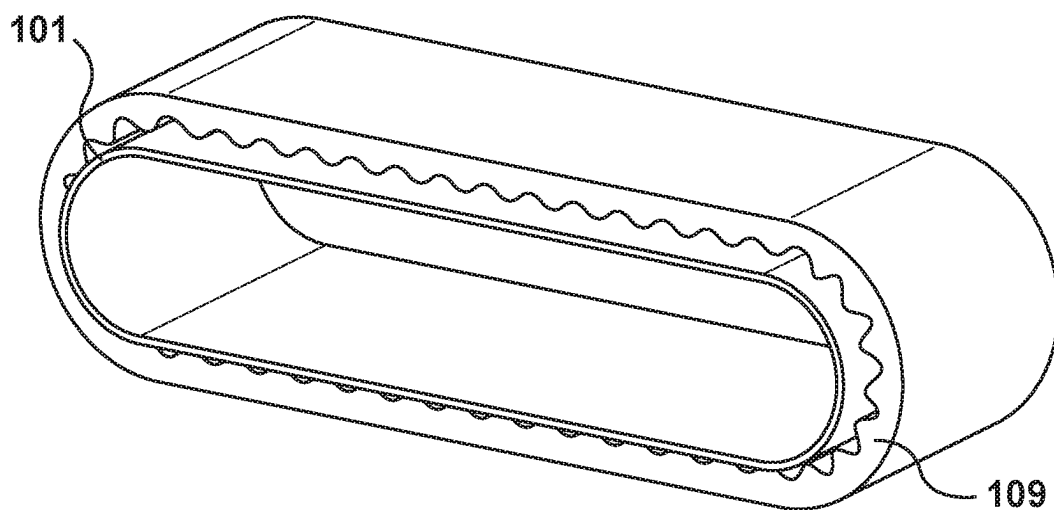


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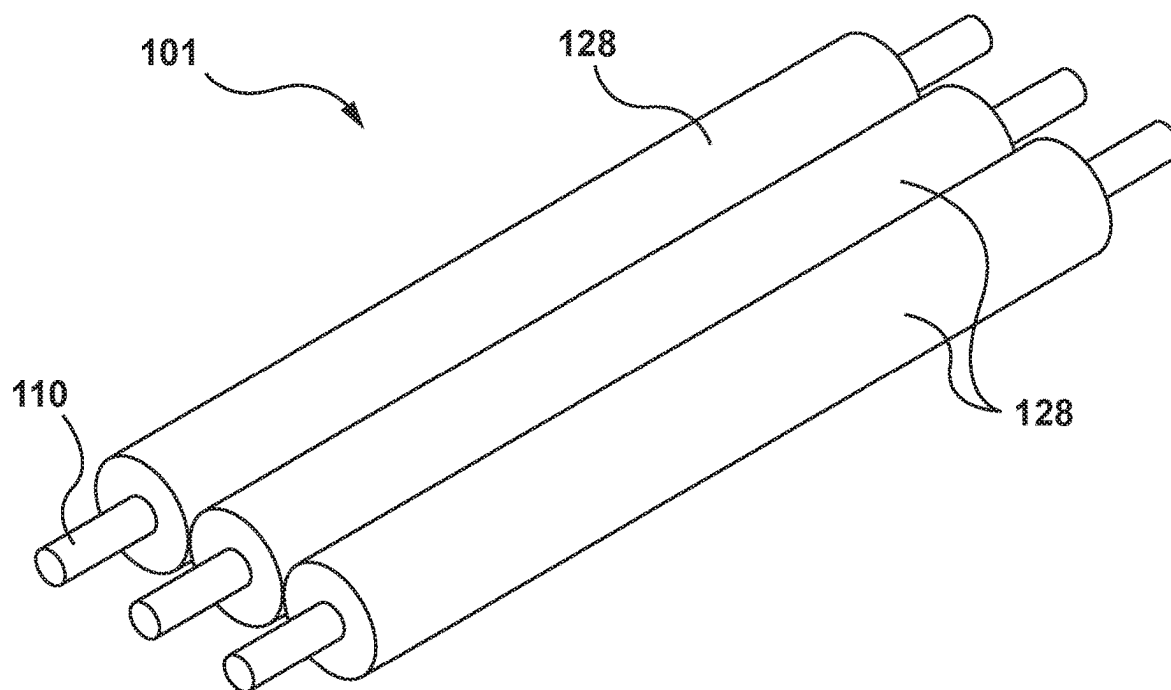


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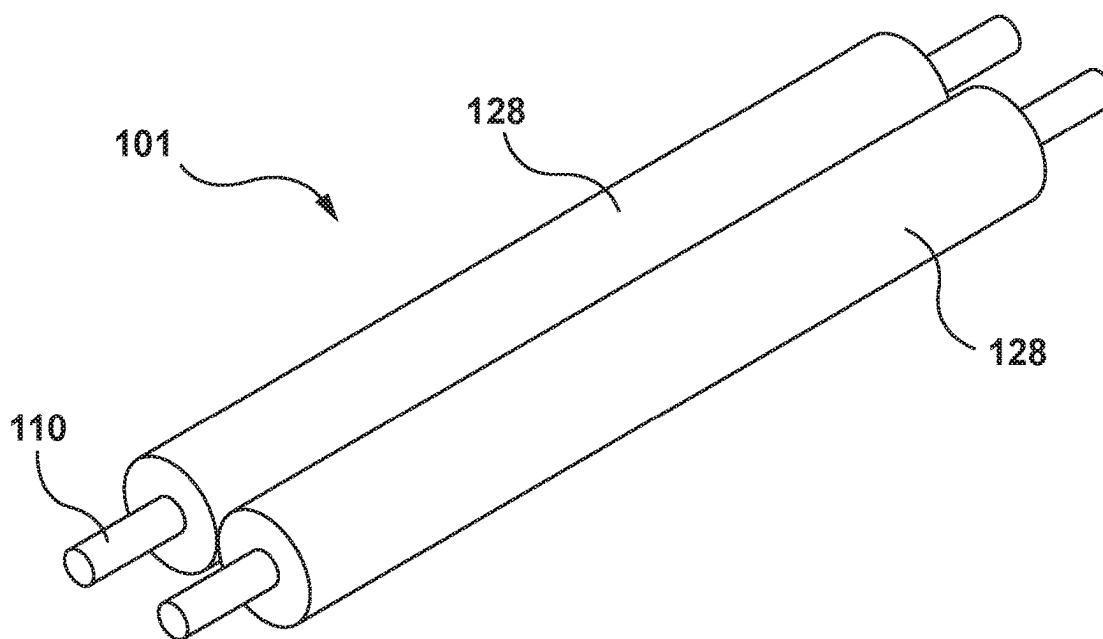


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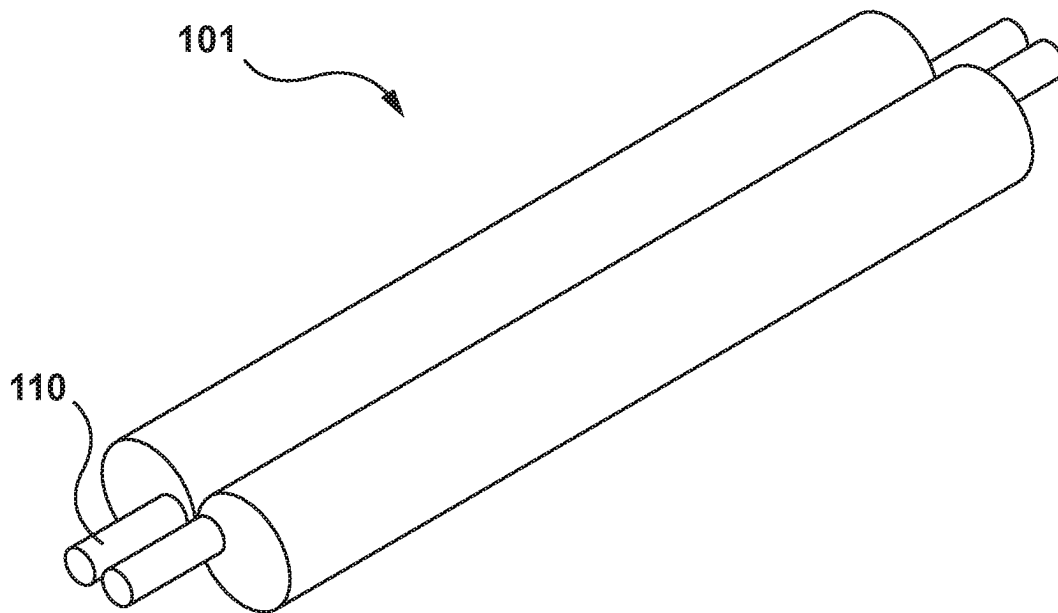


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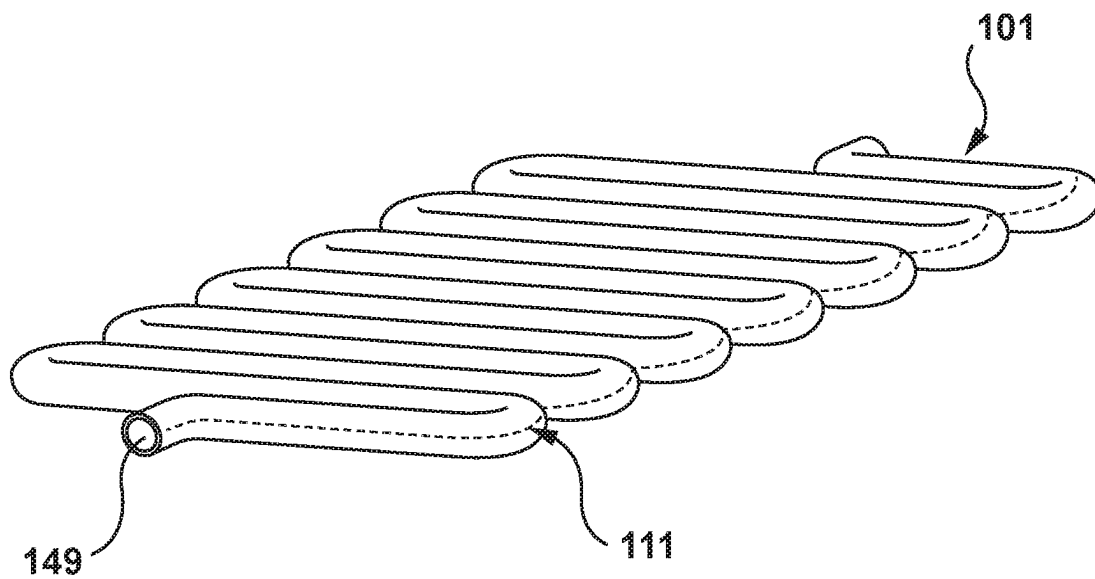


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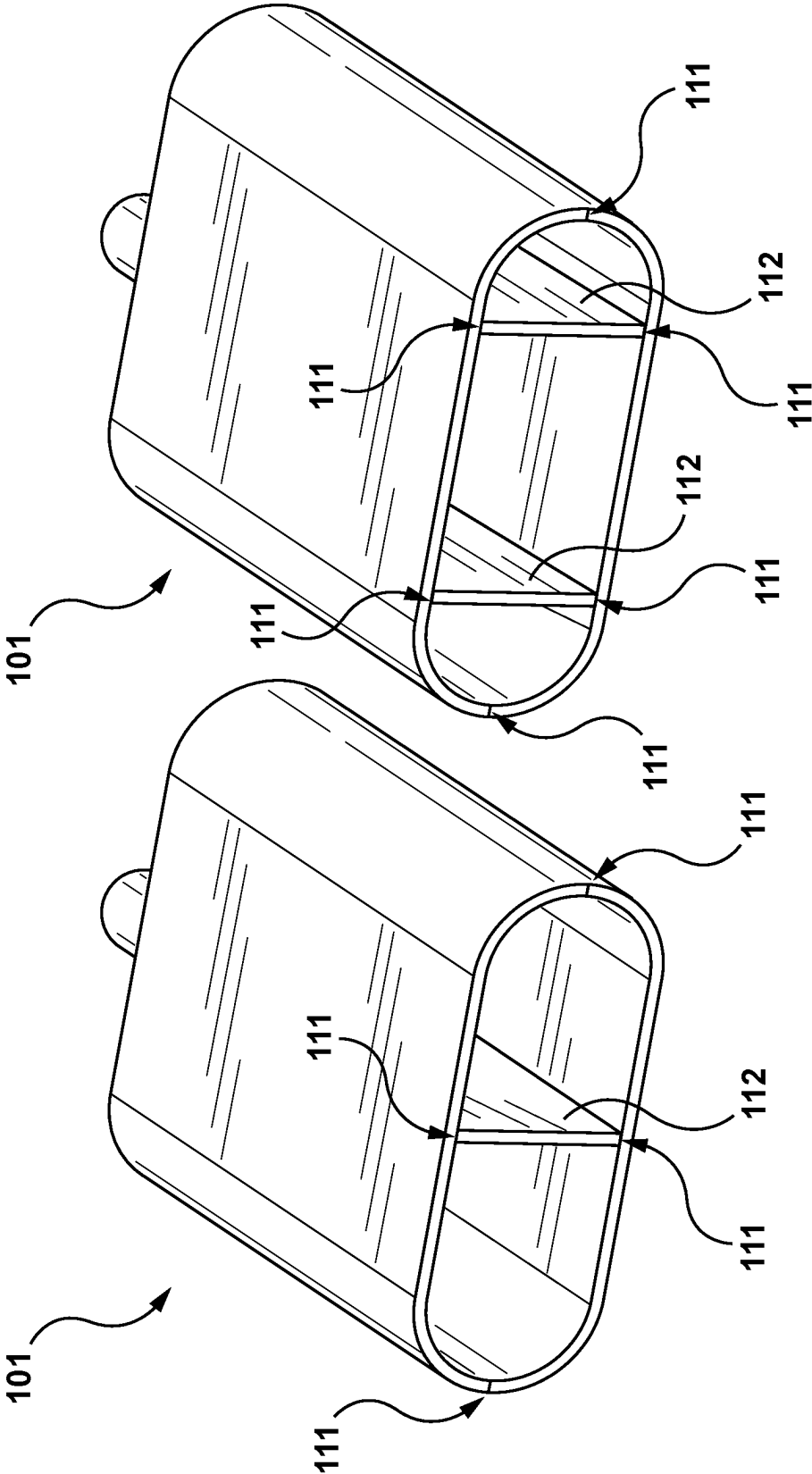


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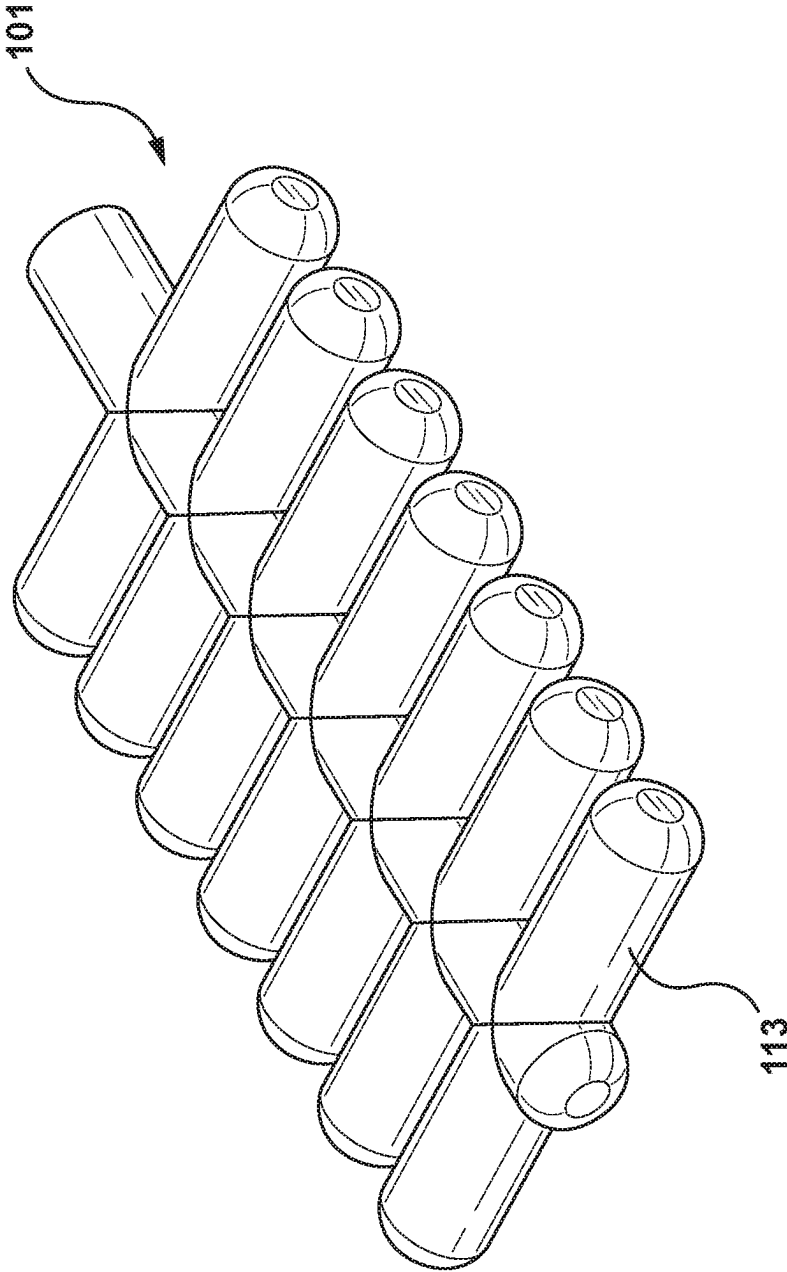


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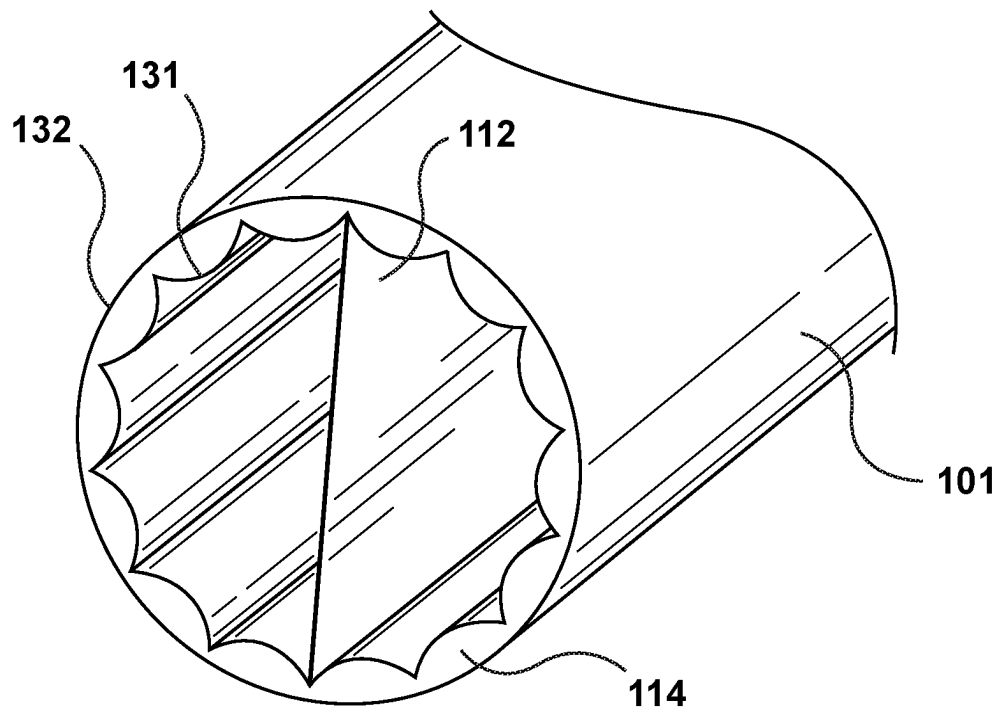


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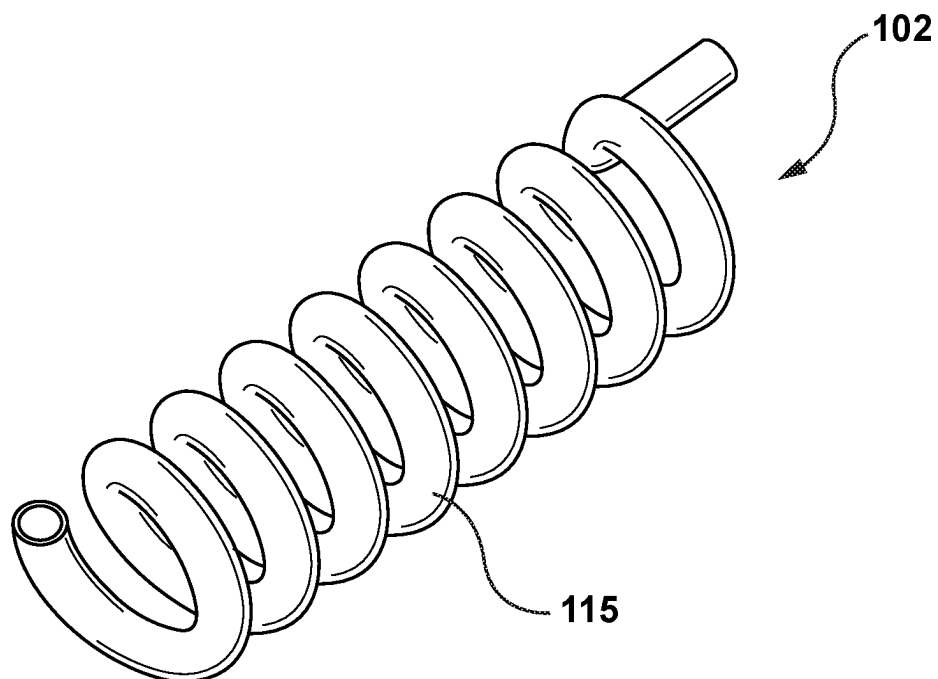


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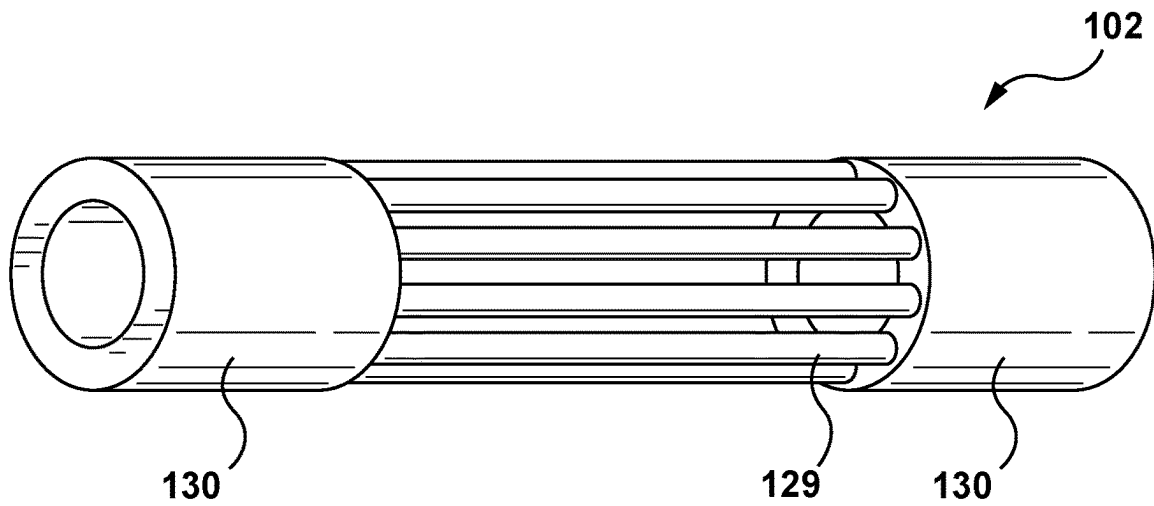


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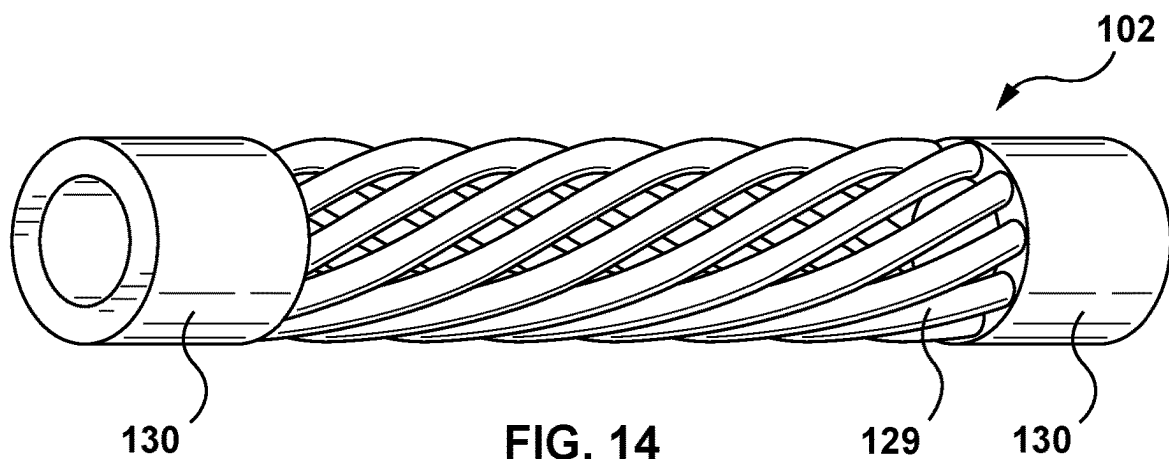


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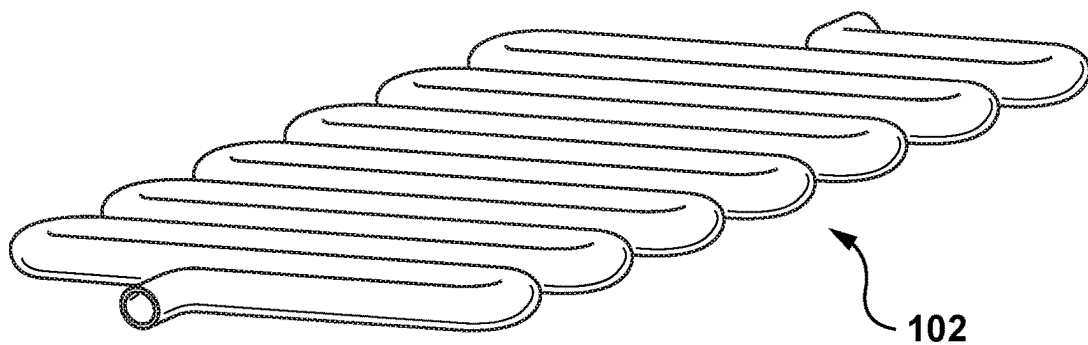


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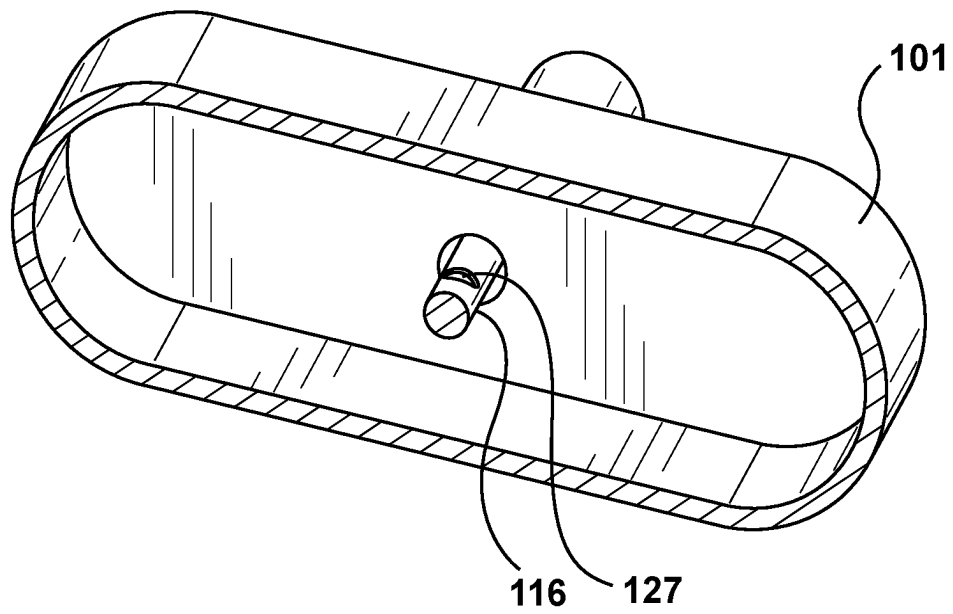


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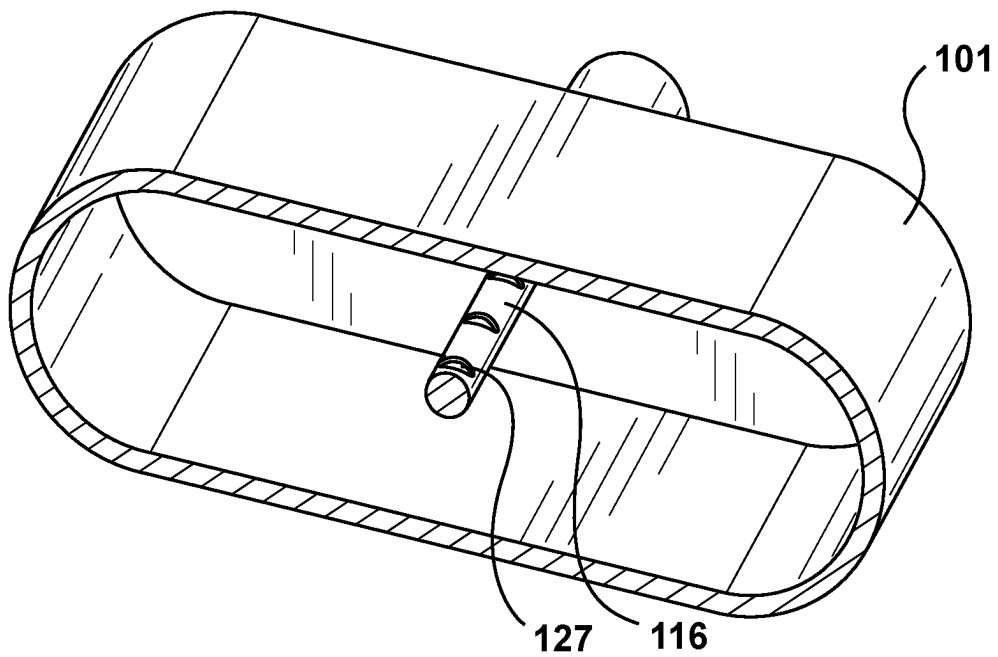


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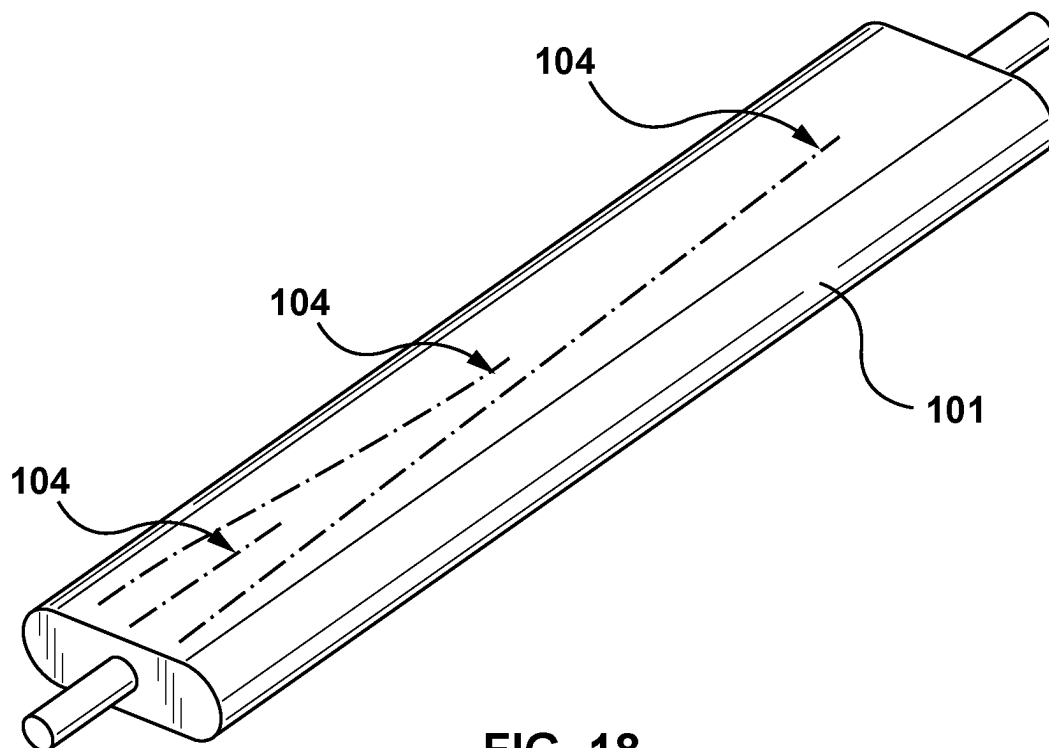


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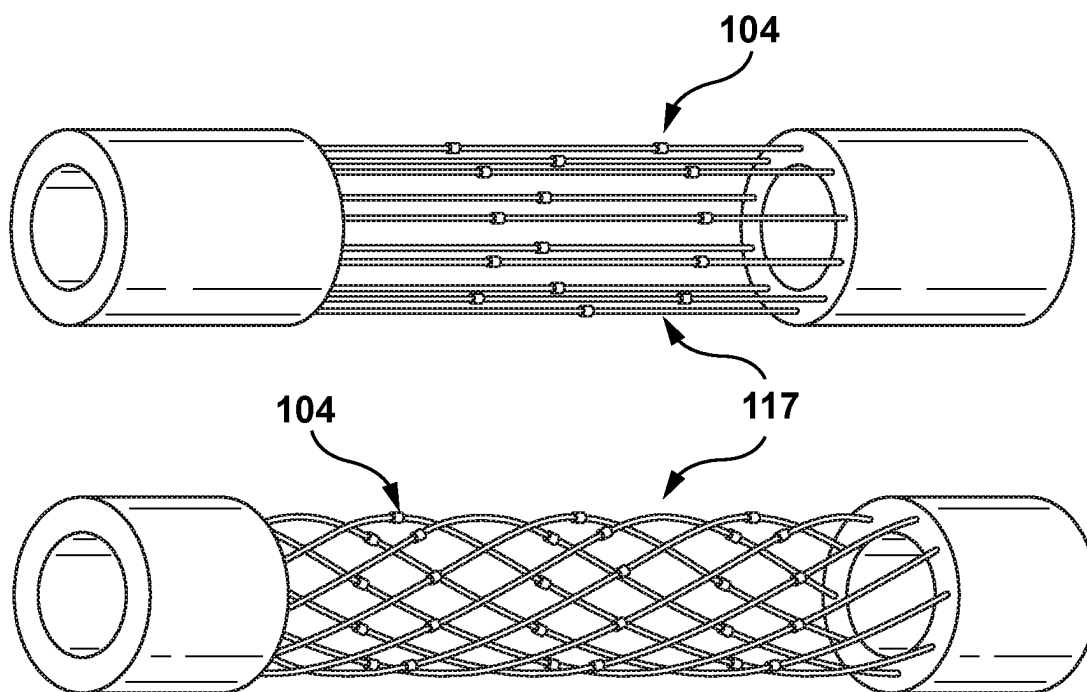


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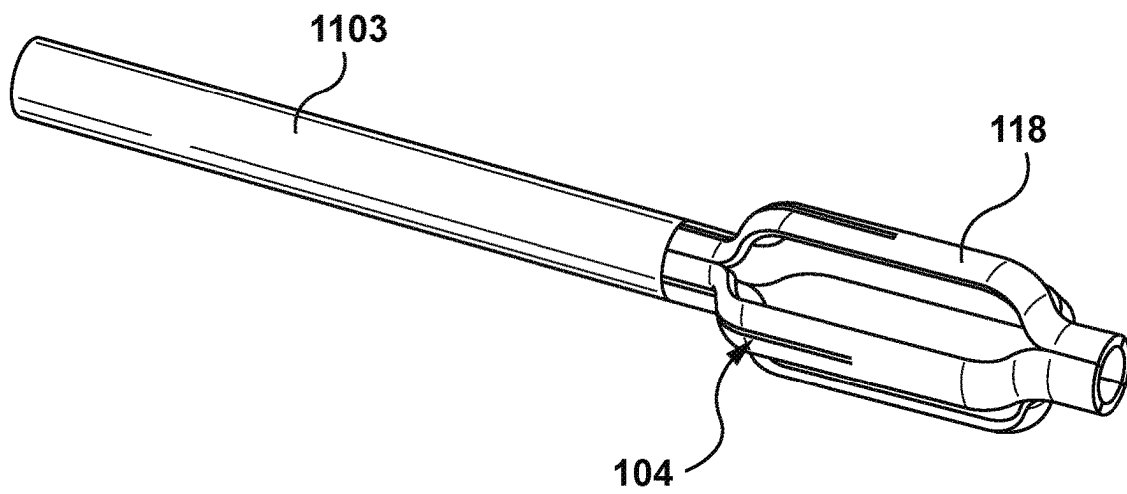


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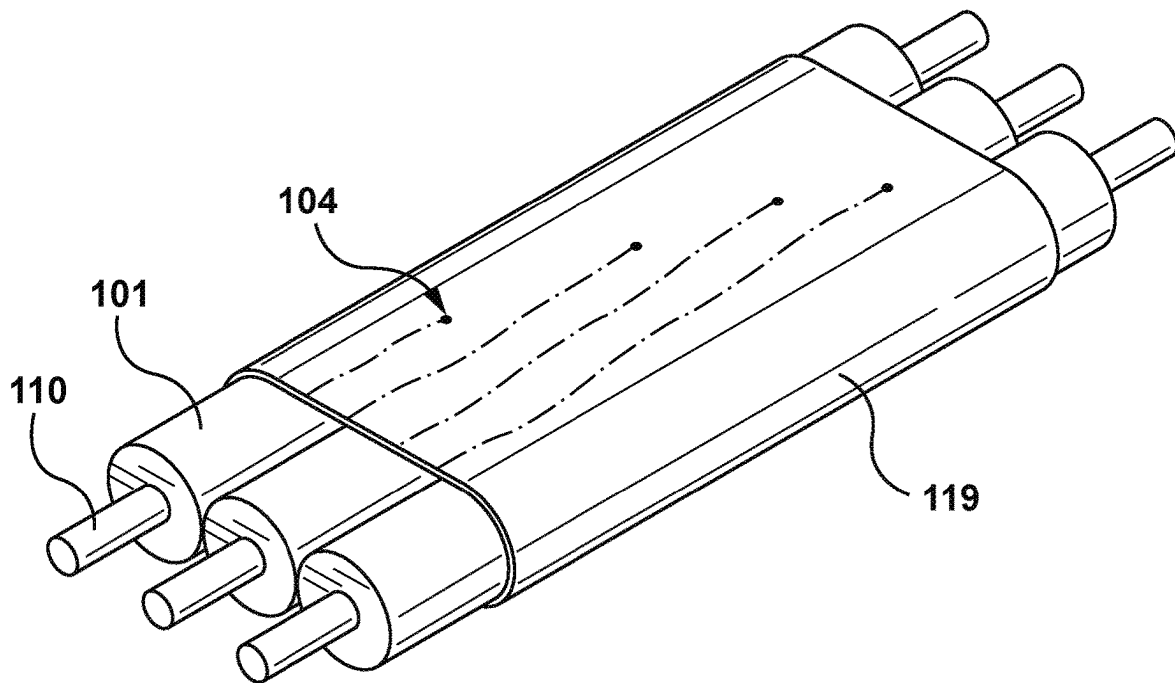


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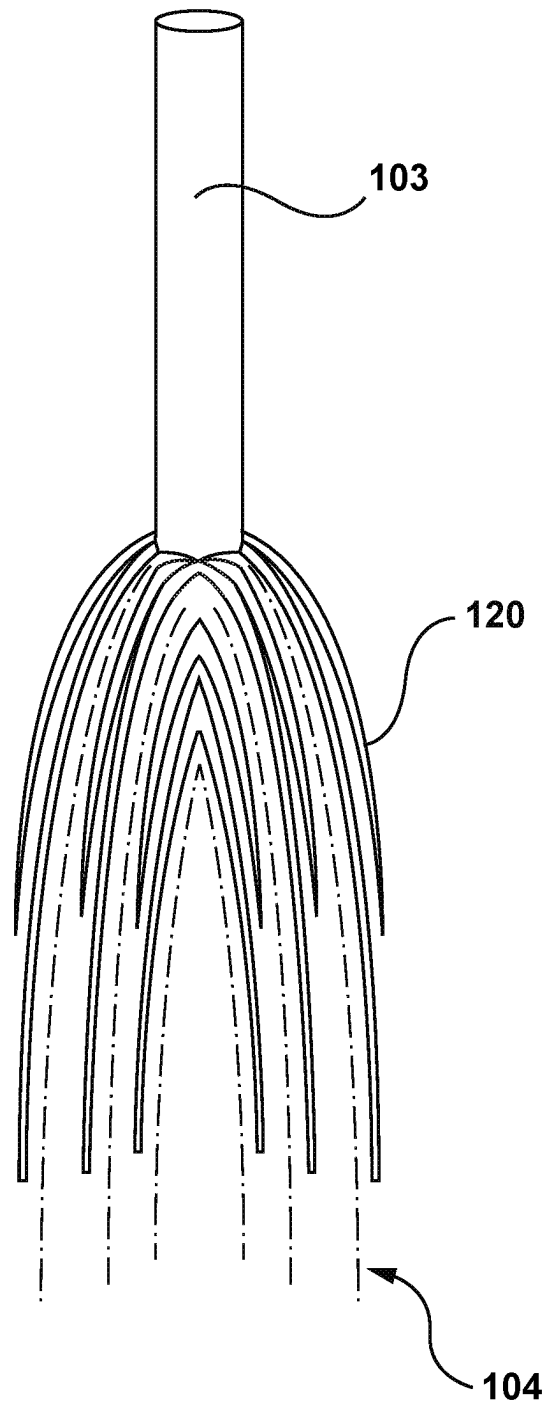


FIG. 22

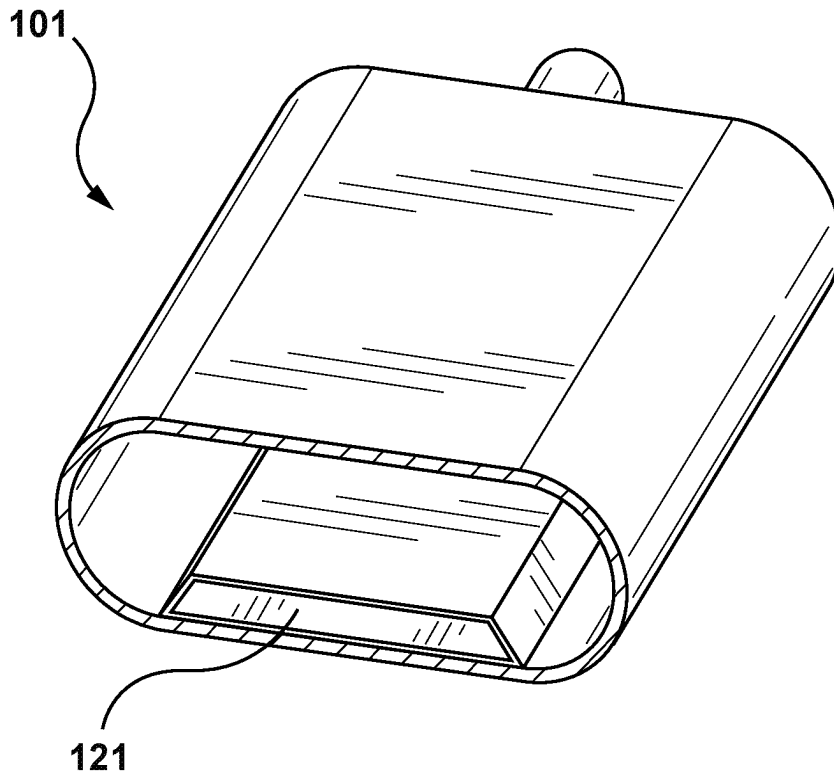


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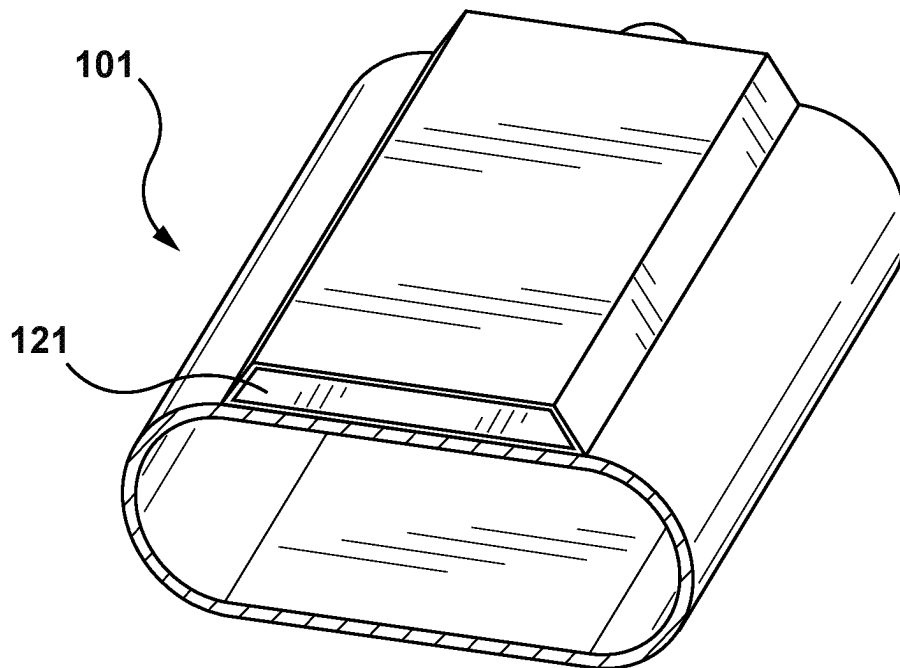


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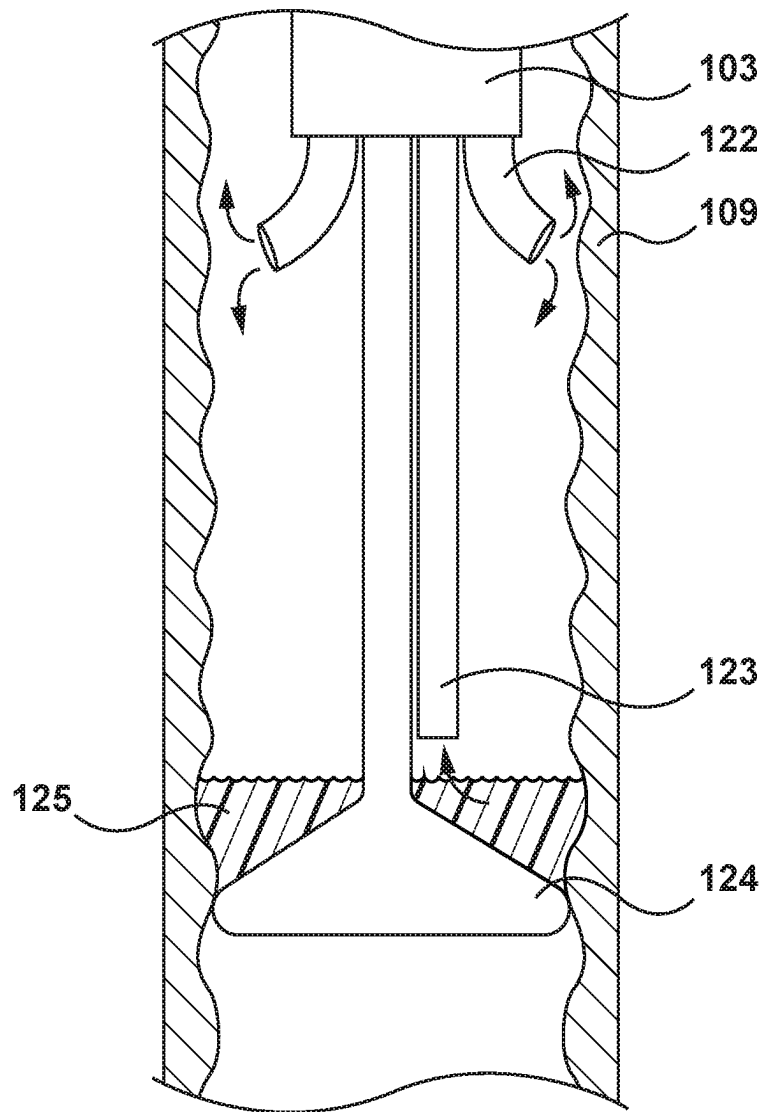
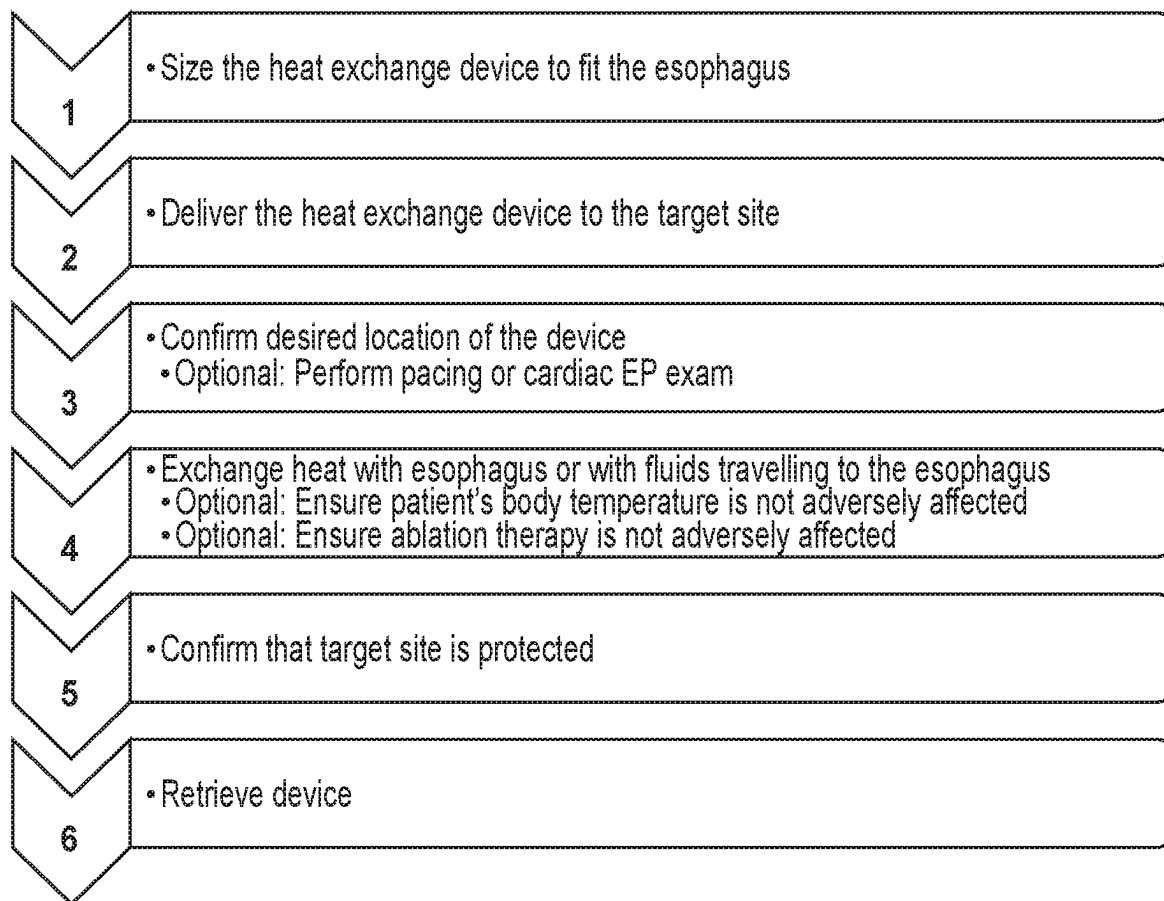


FIG. 25

**FIG. 26**

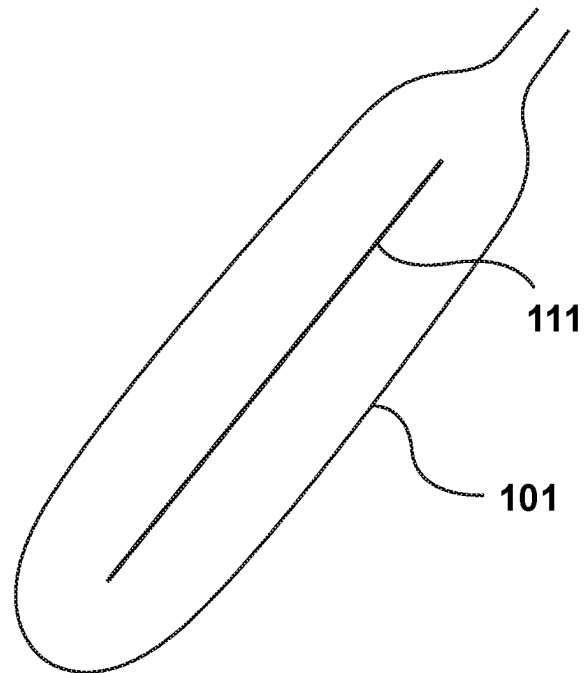


FIG. 27

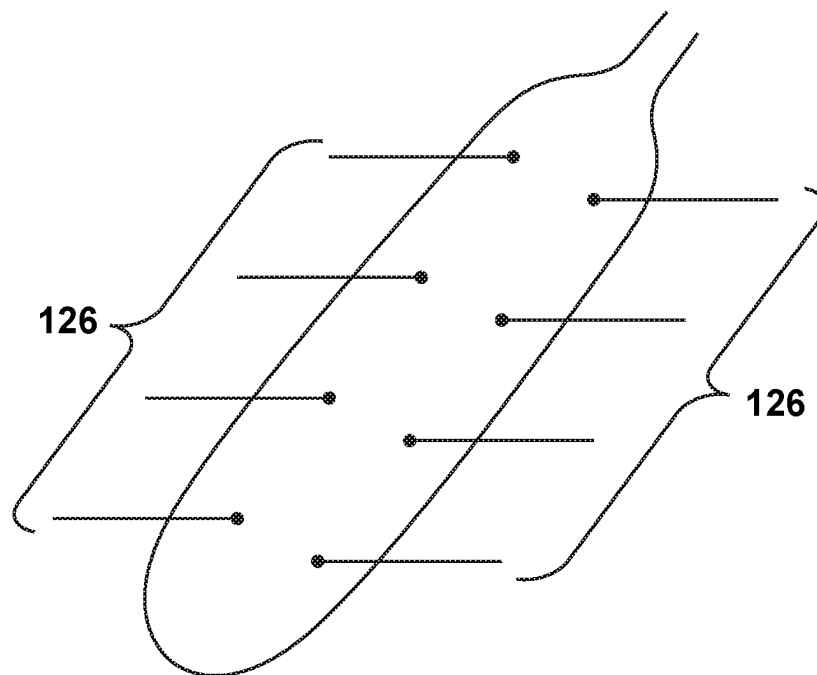


FIG. 28

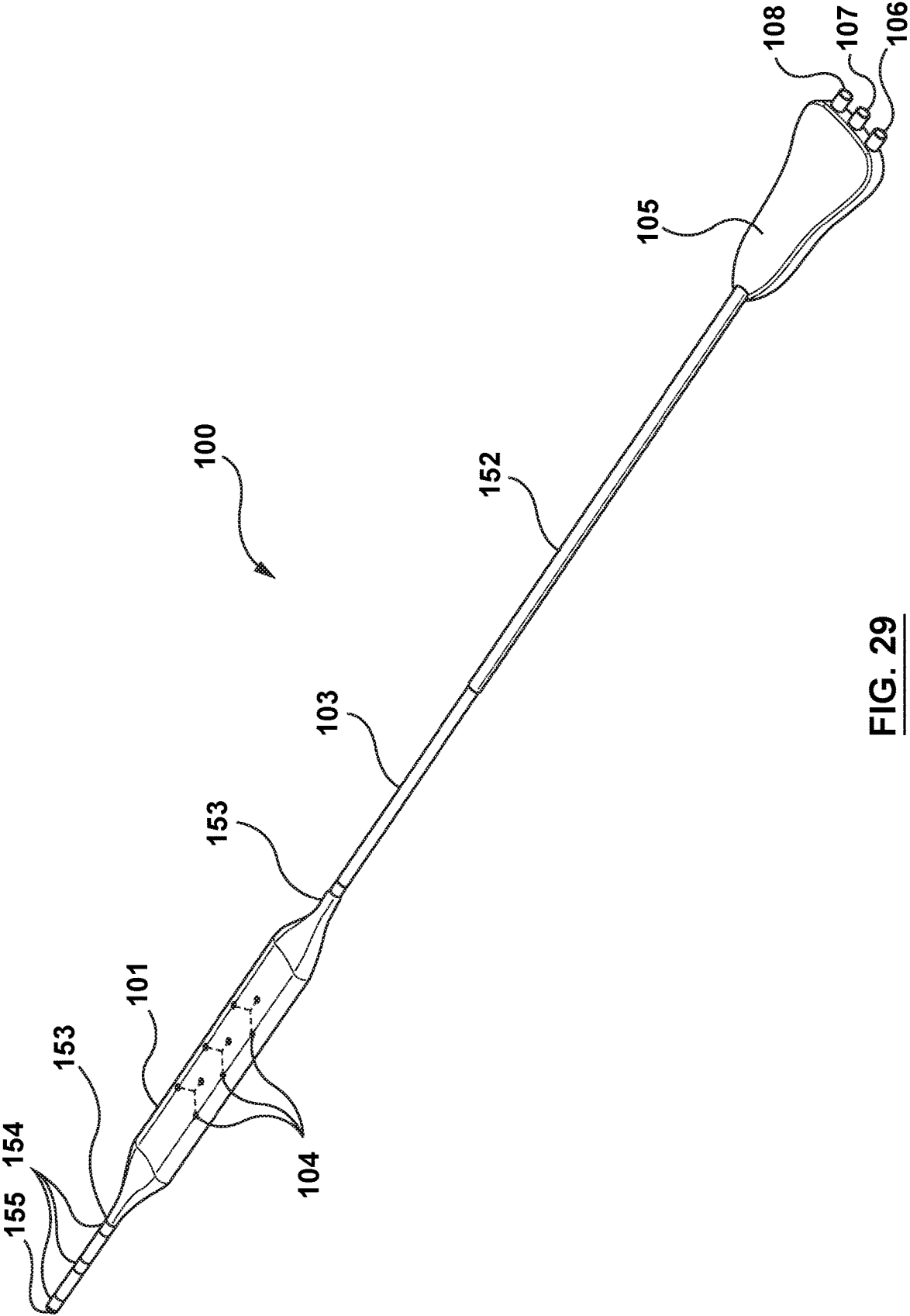


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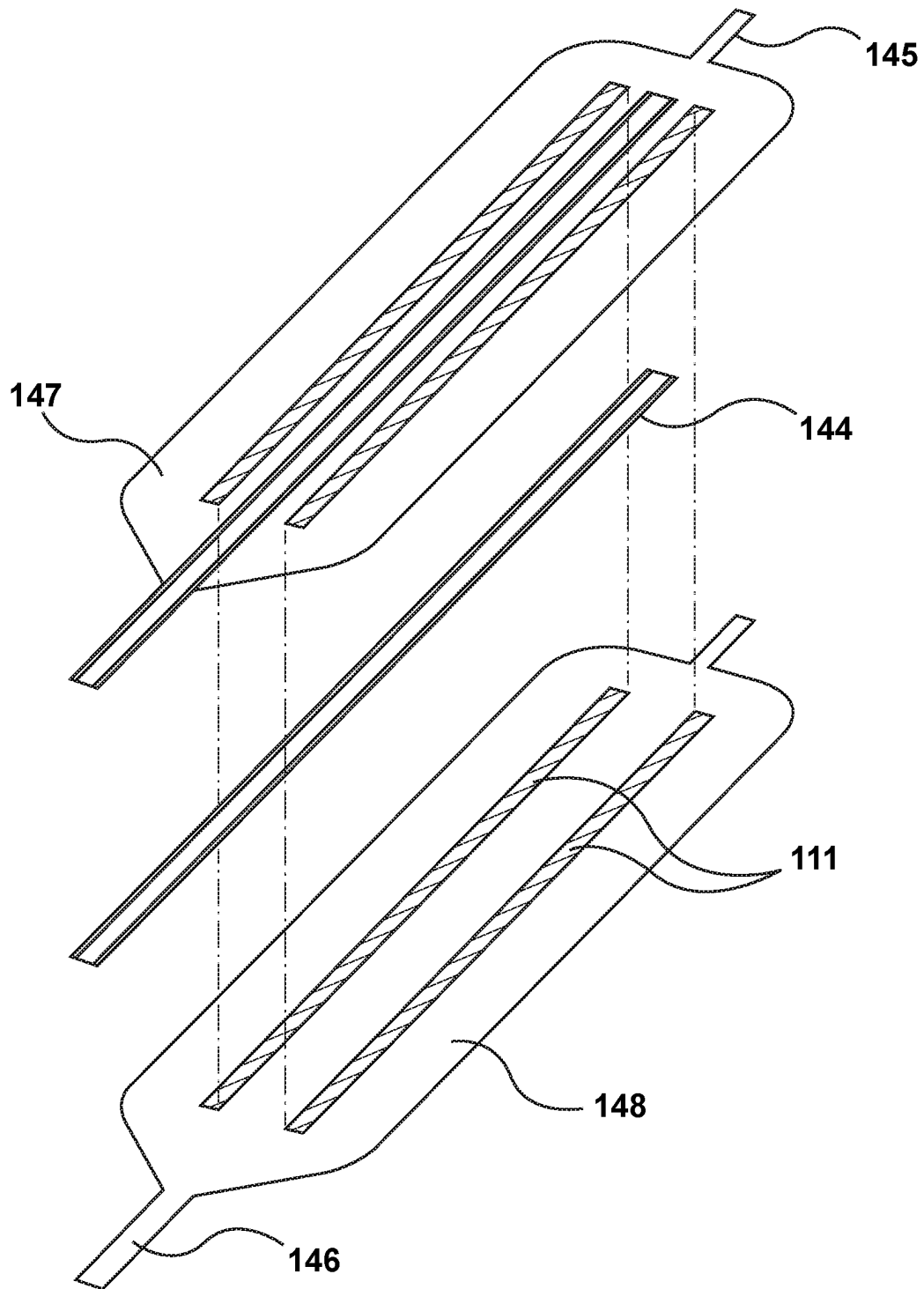


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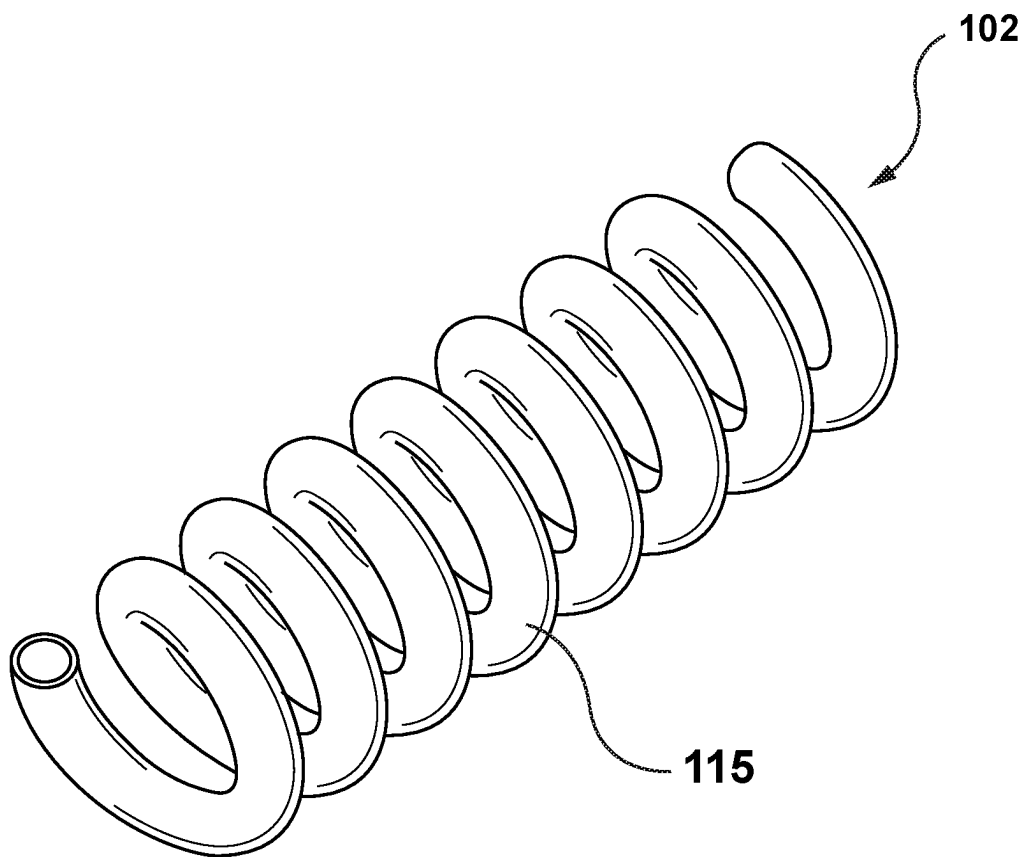


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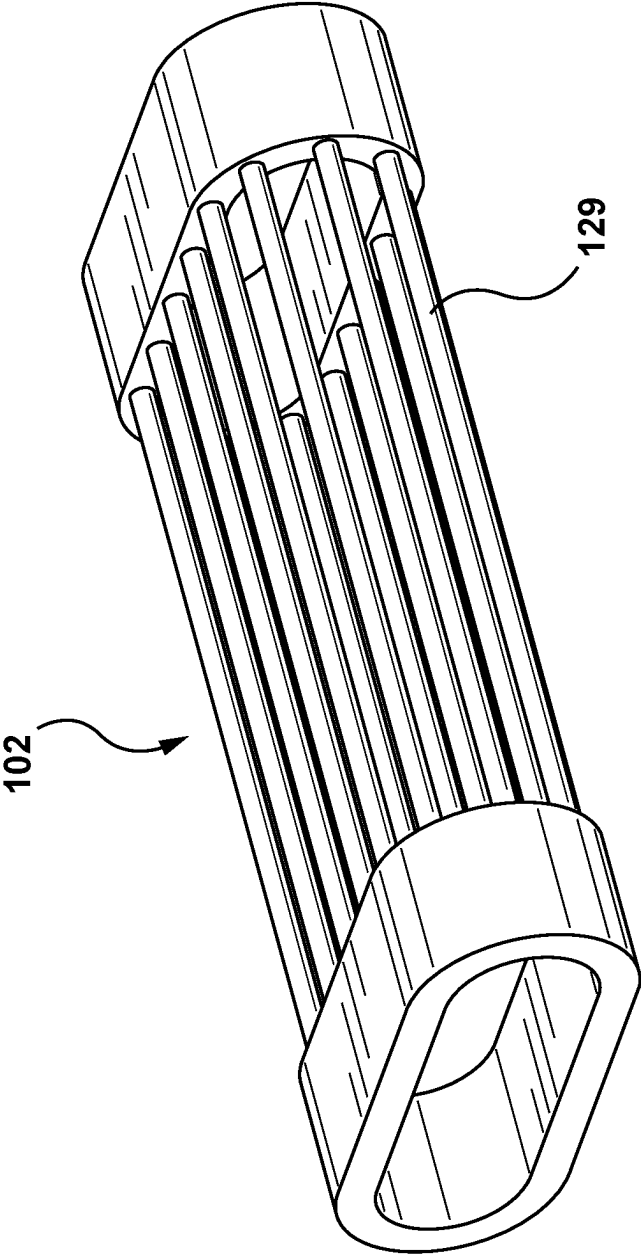


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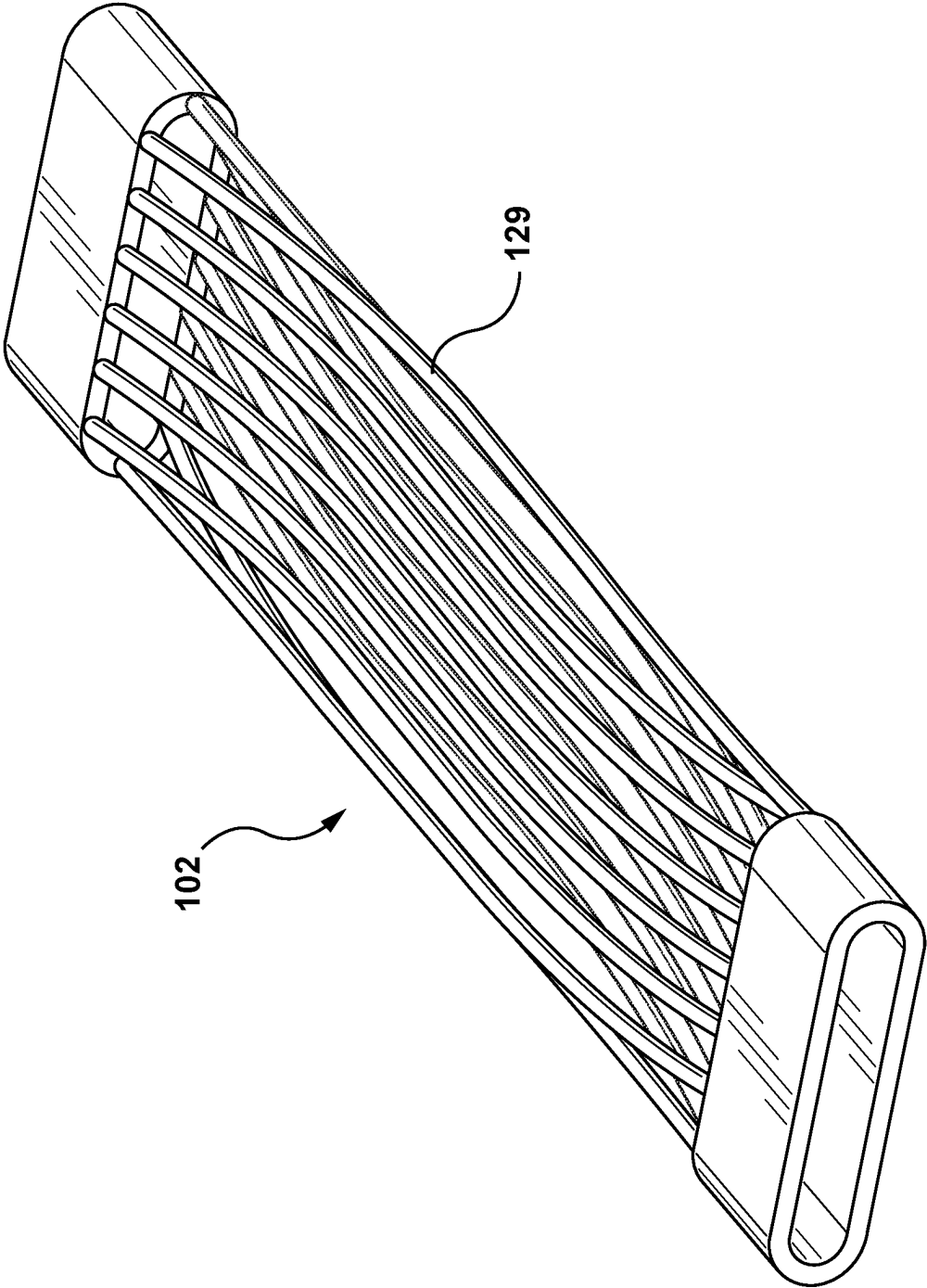


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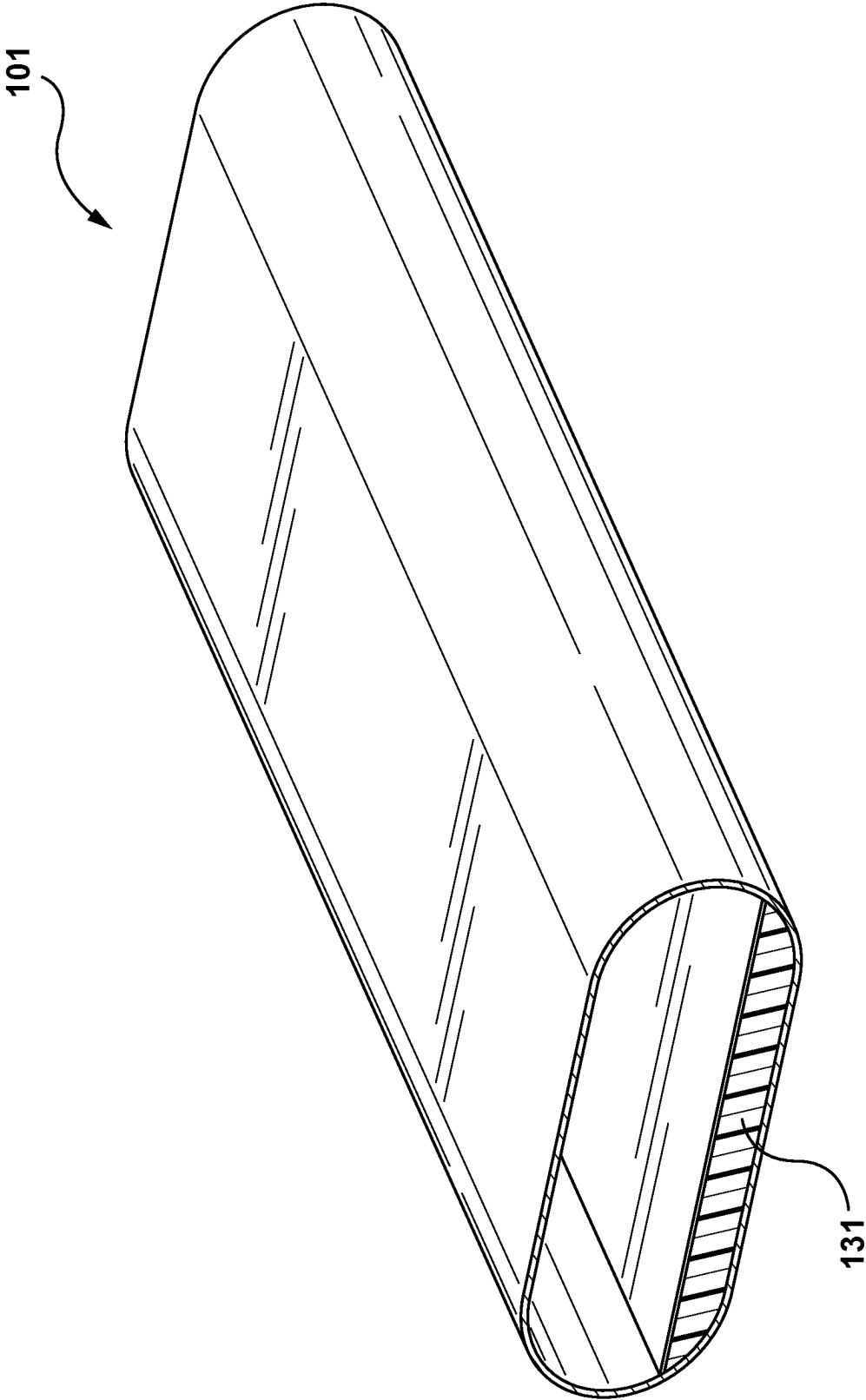


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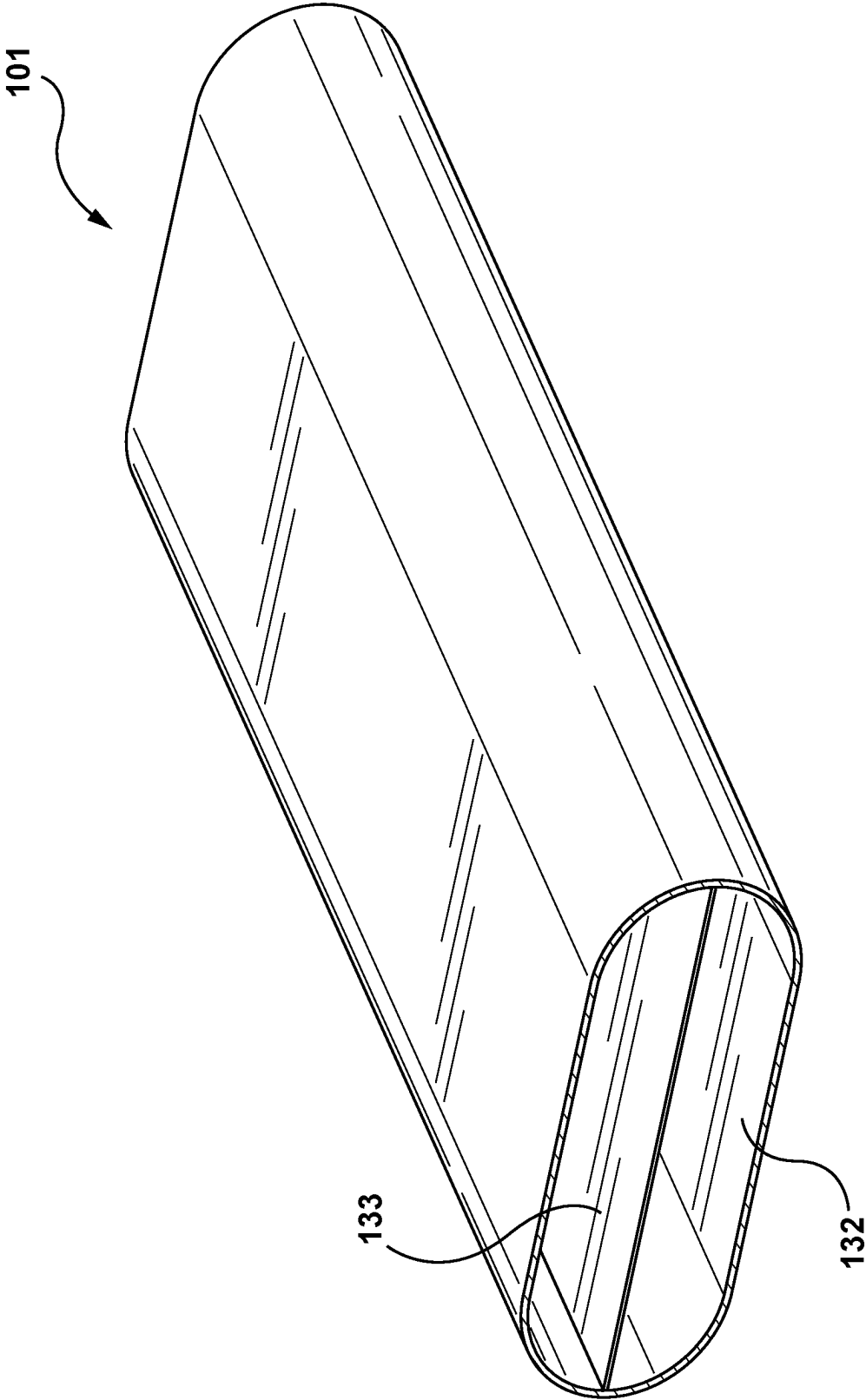


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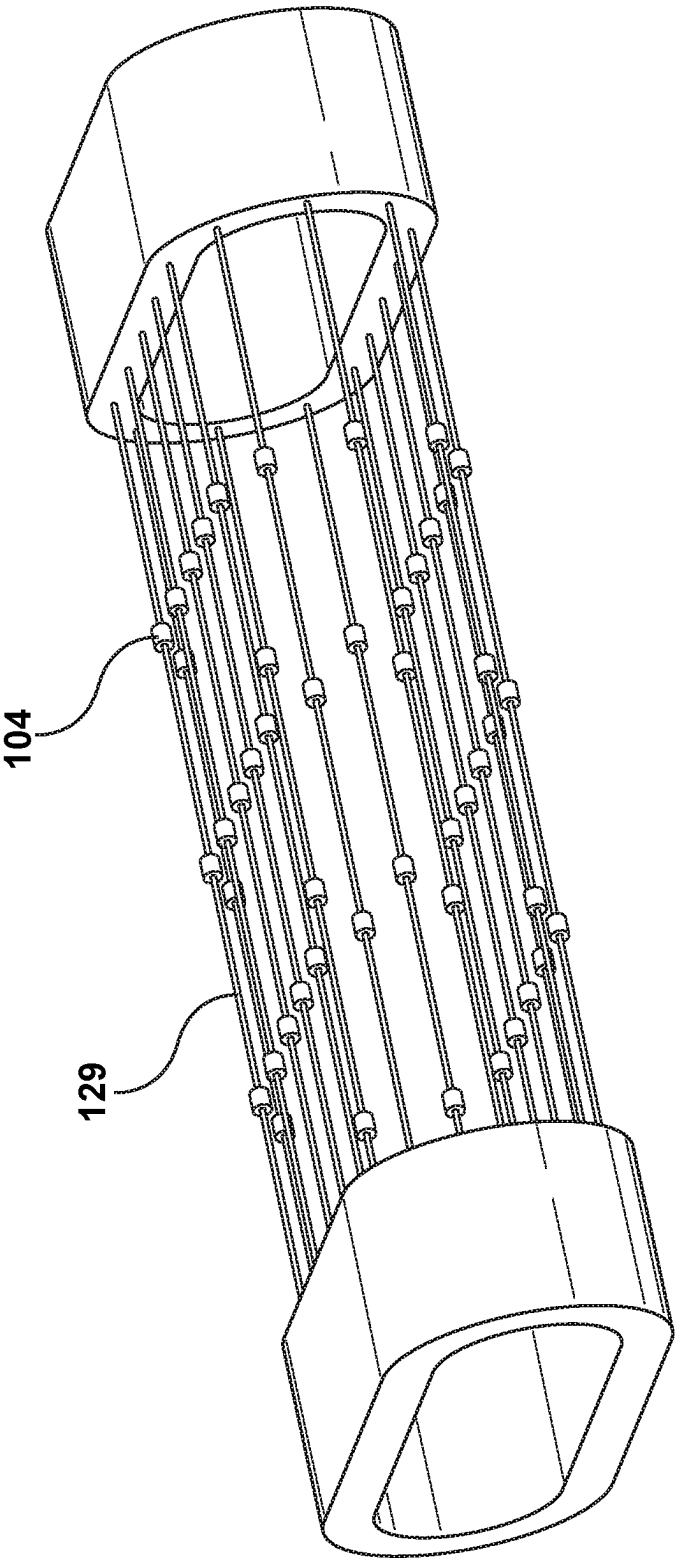


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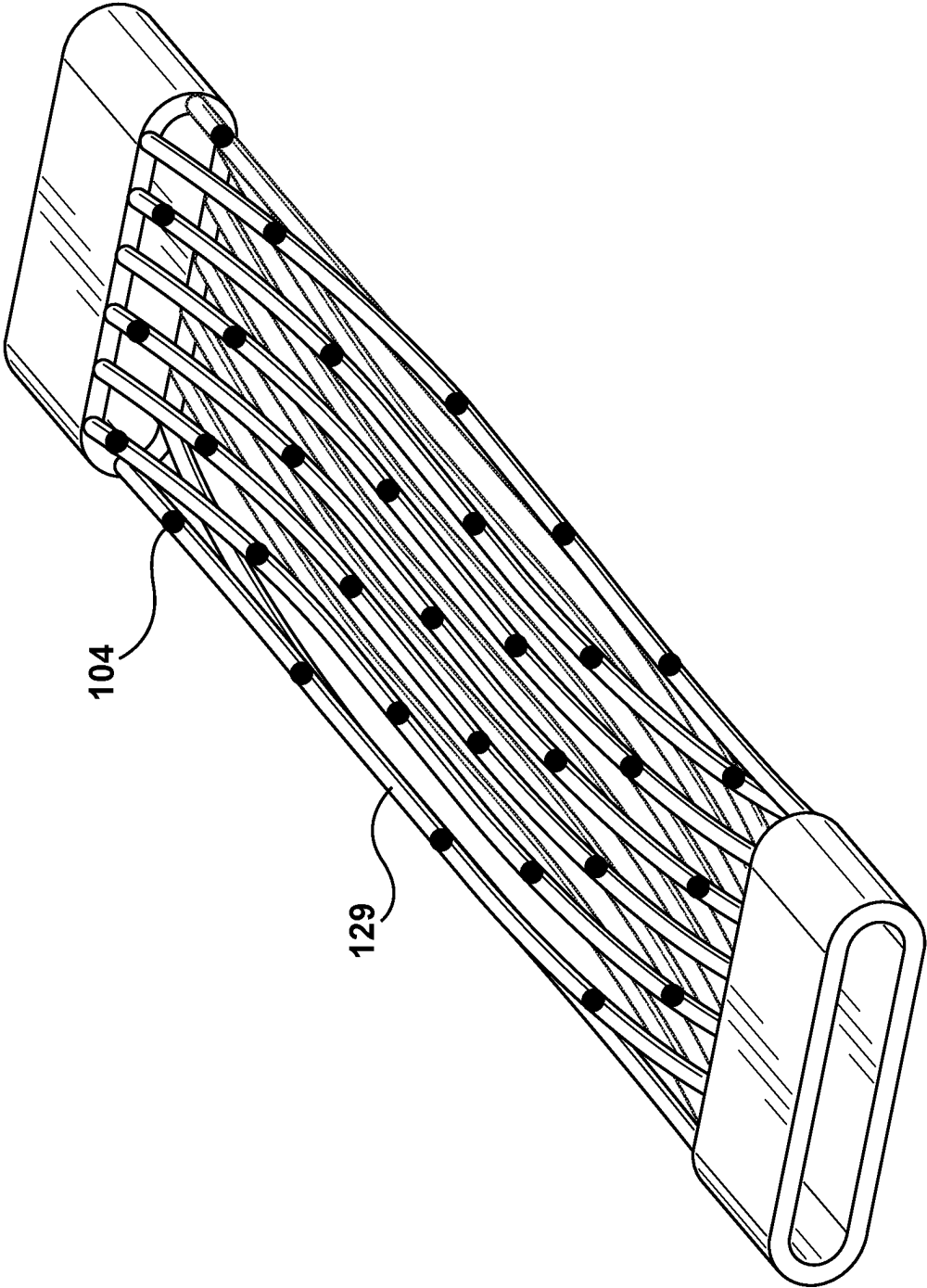


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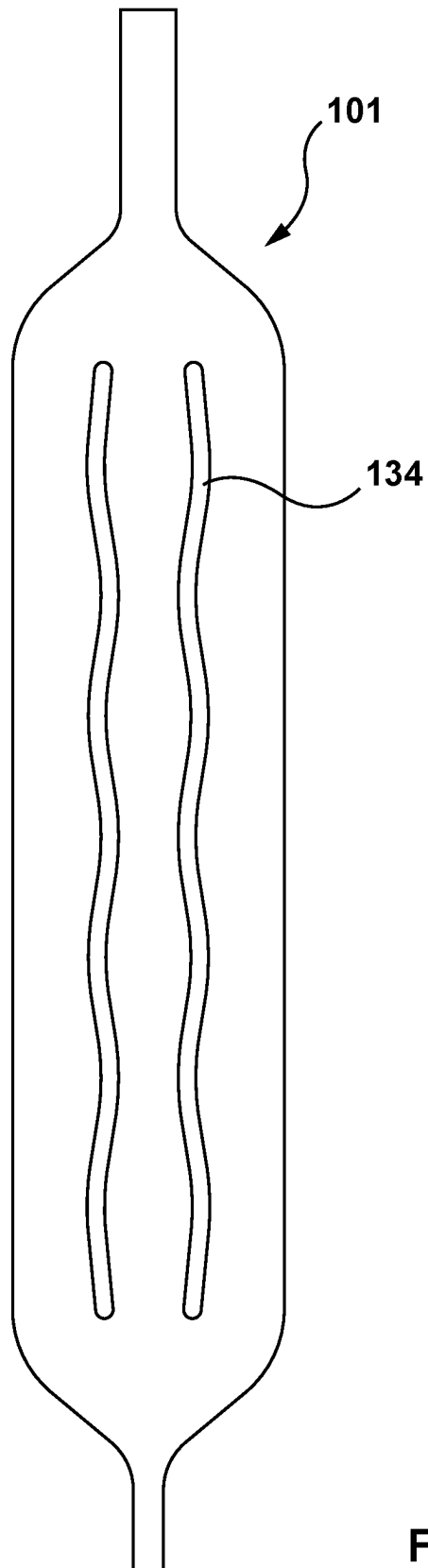


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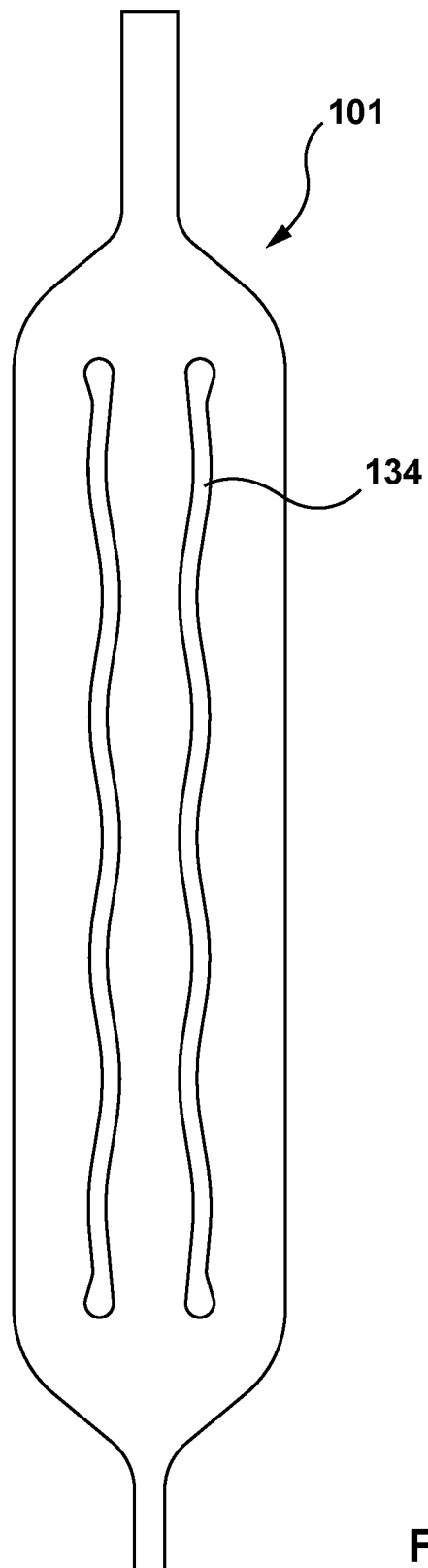


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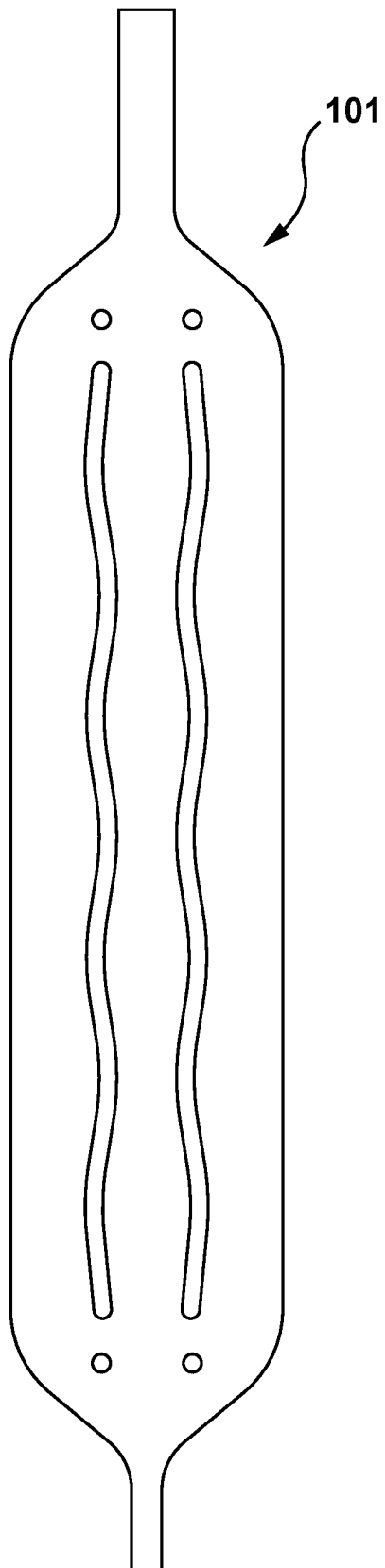


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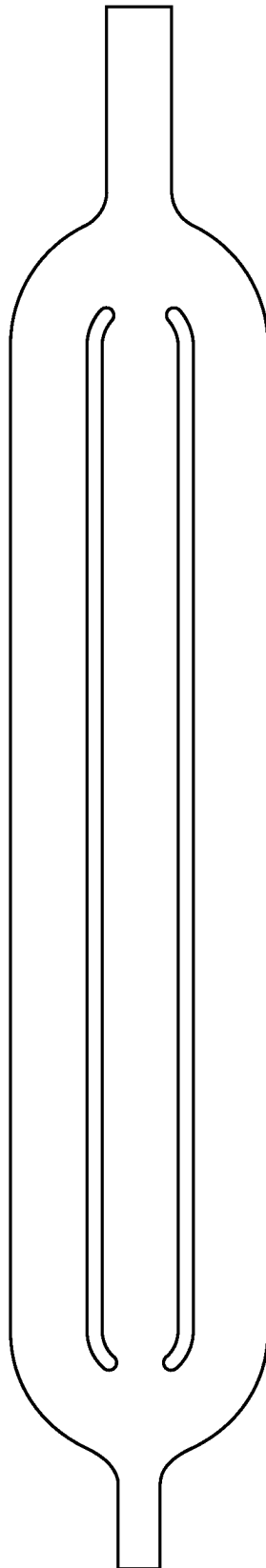


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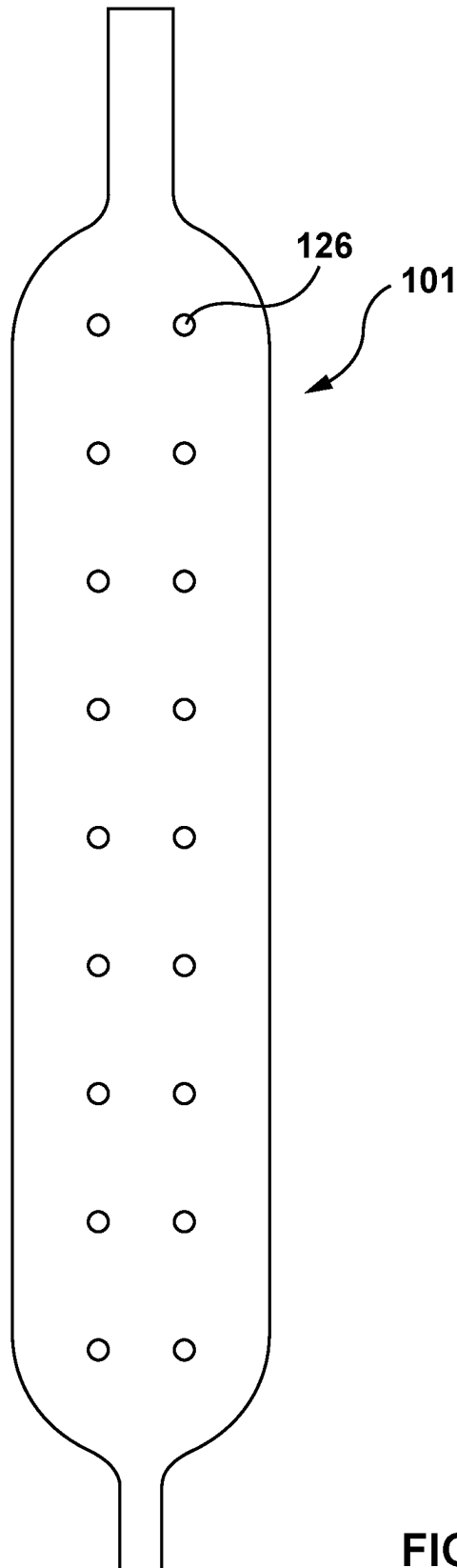


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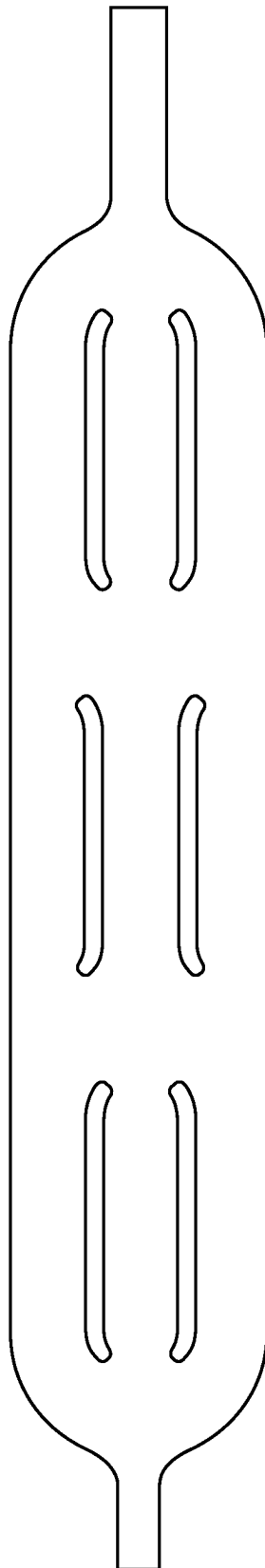


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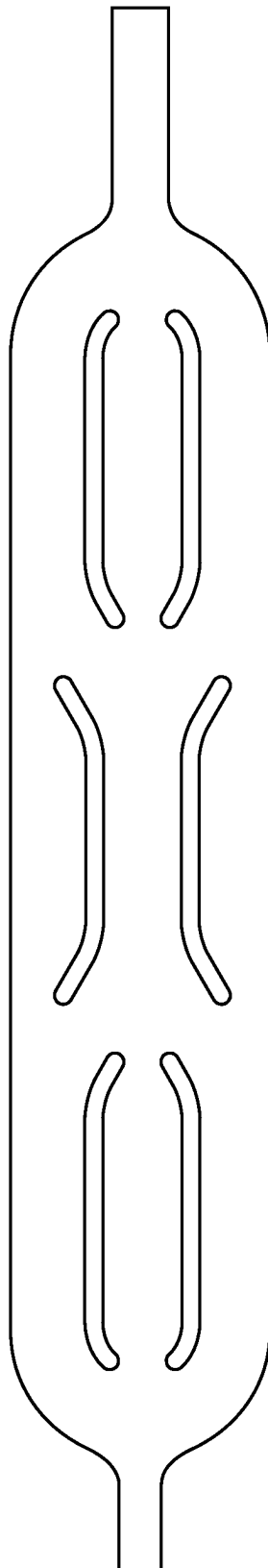


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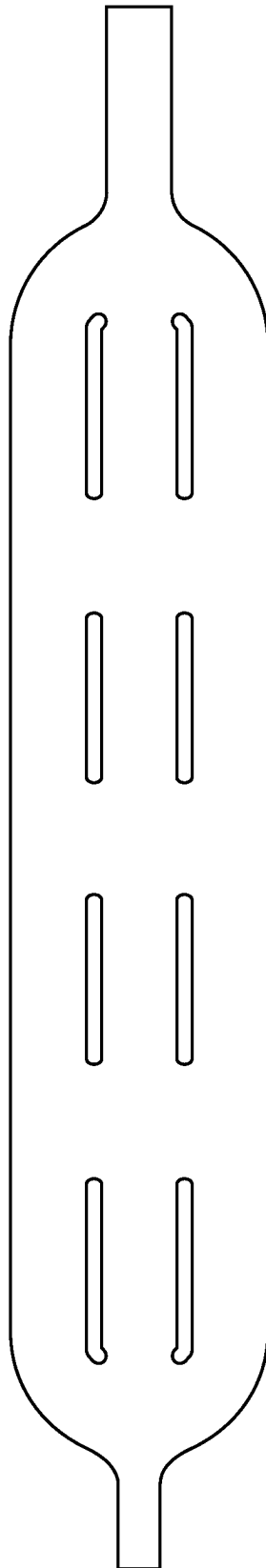


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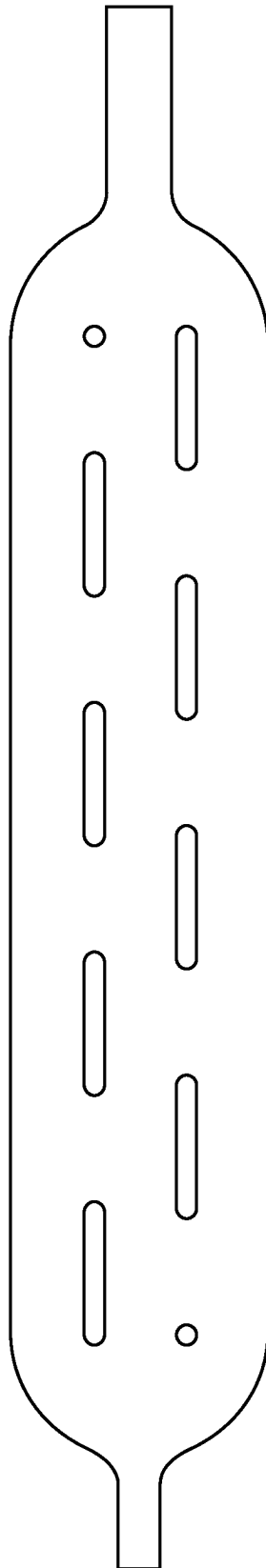


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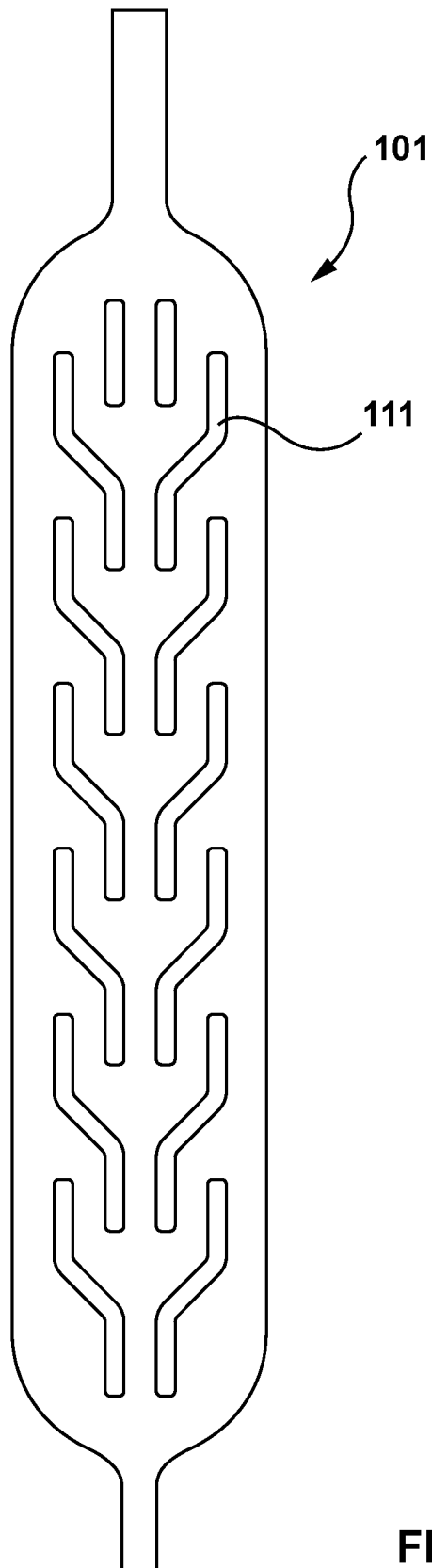


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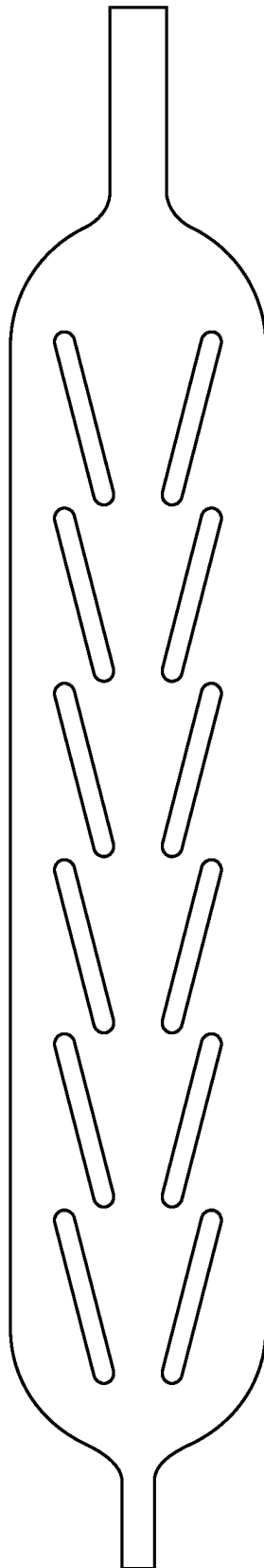


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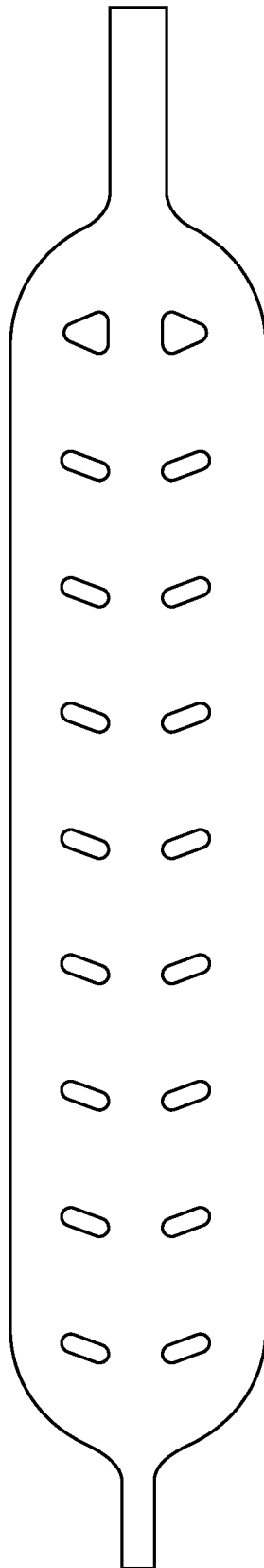


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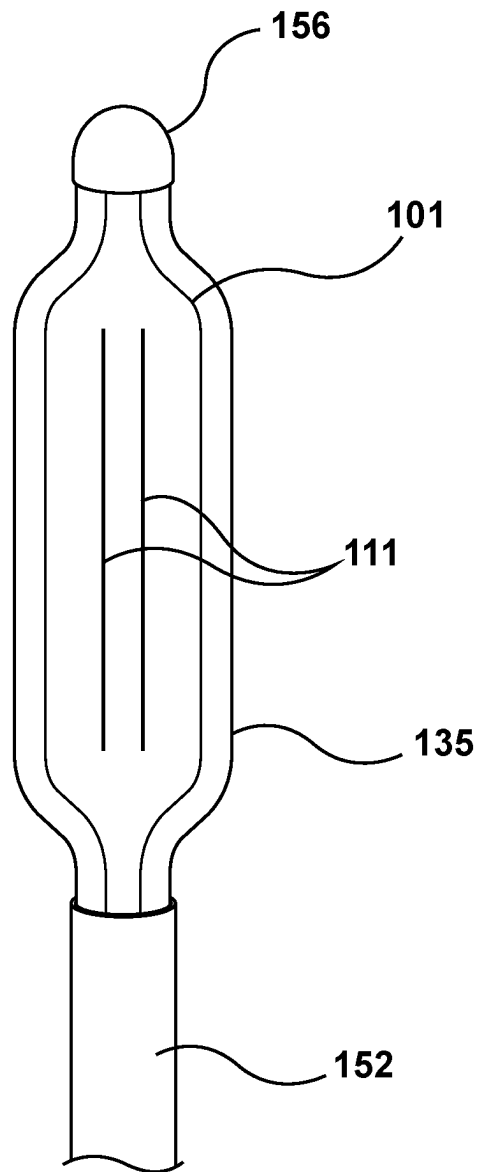


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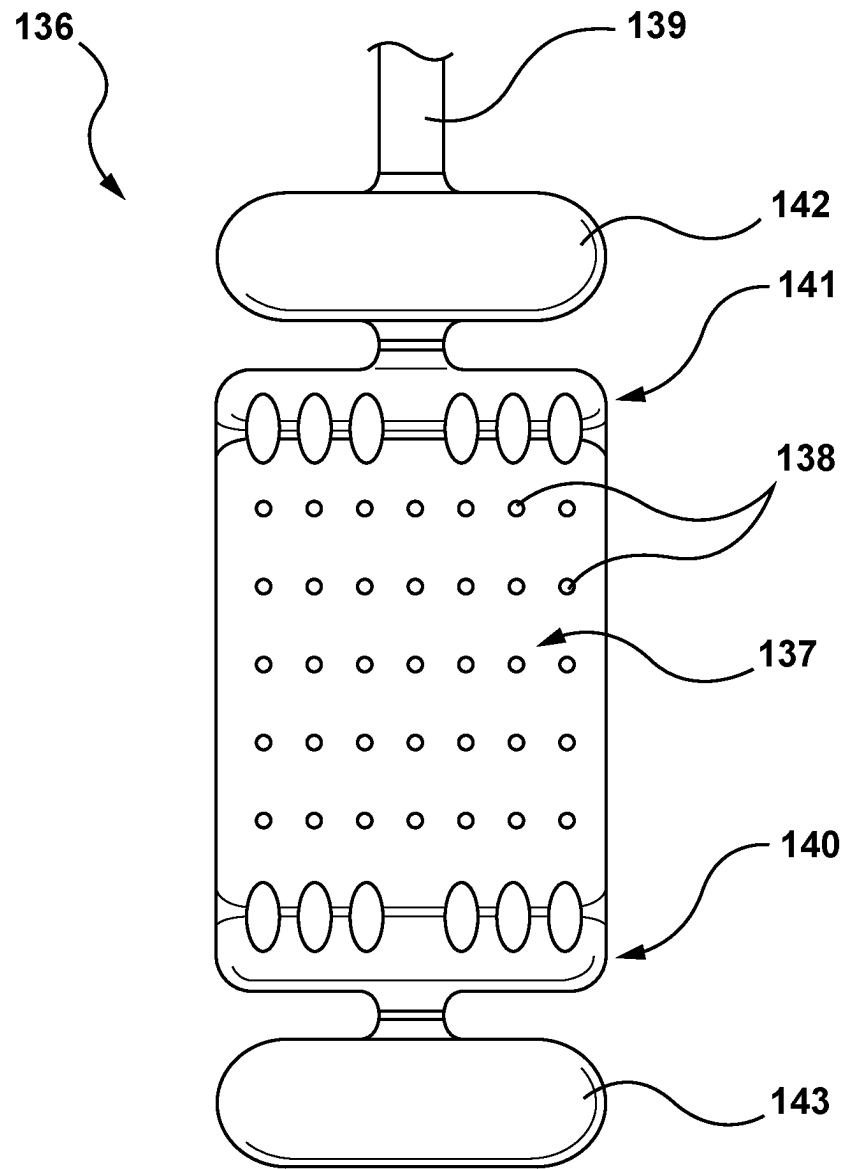


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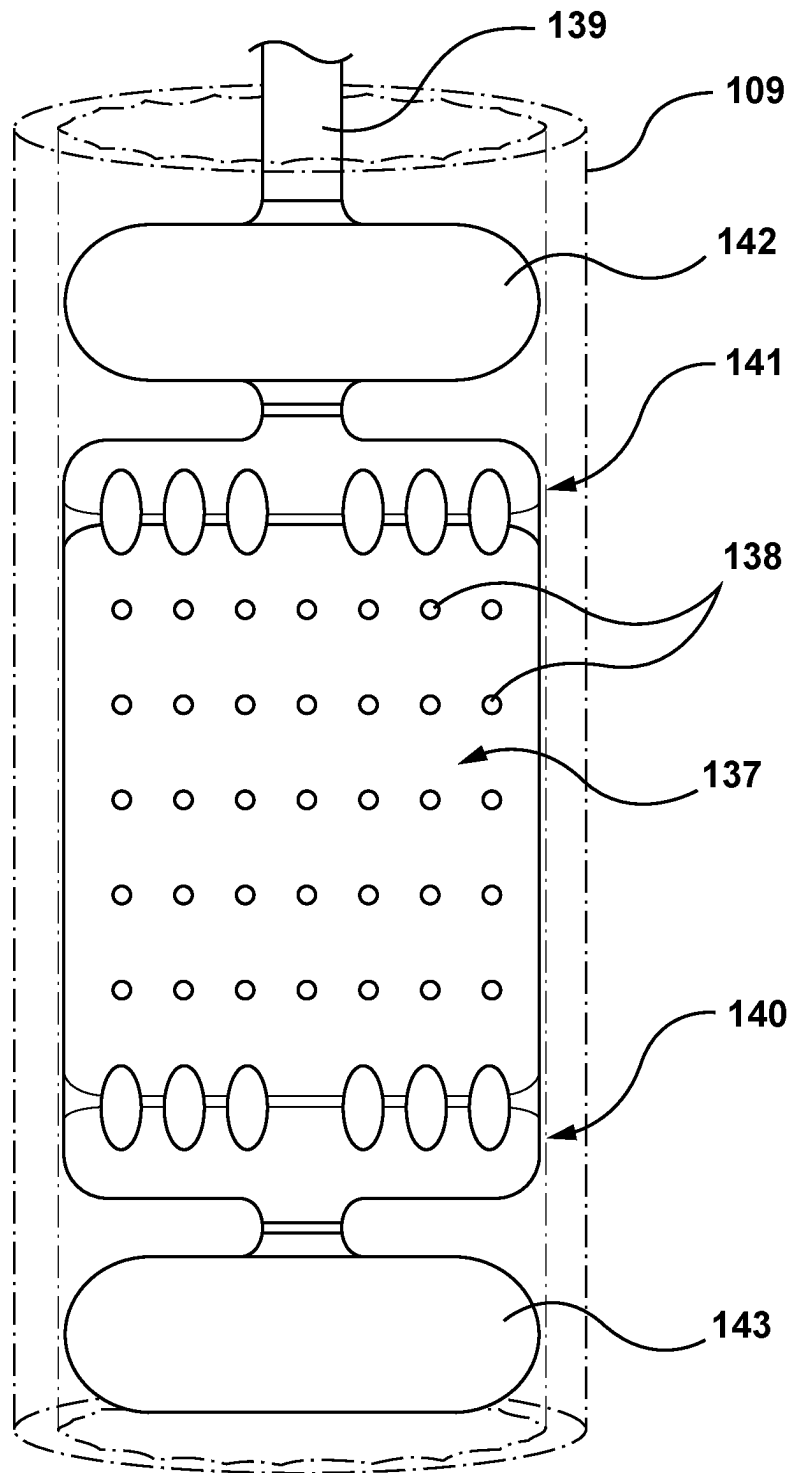


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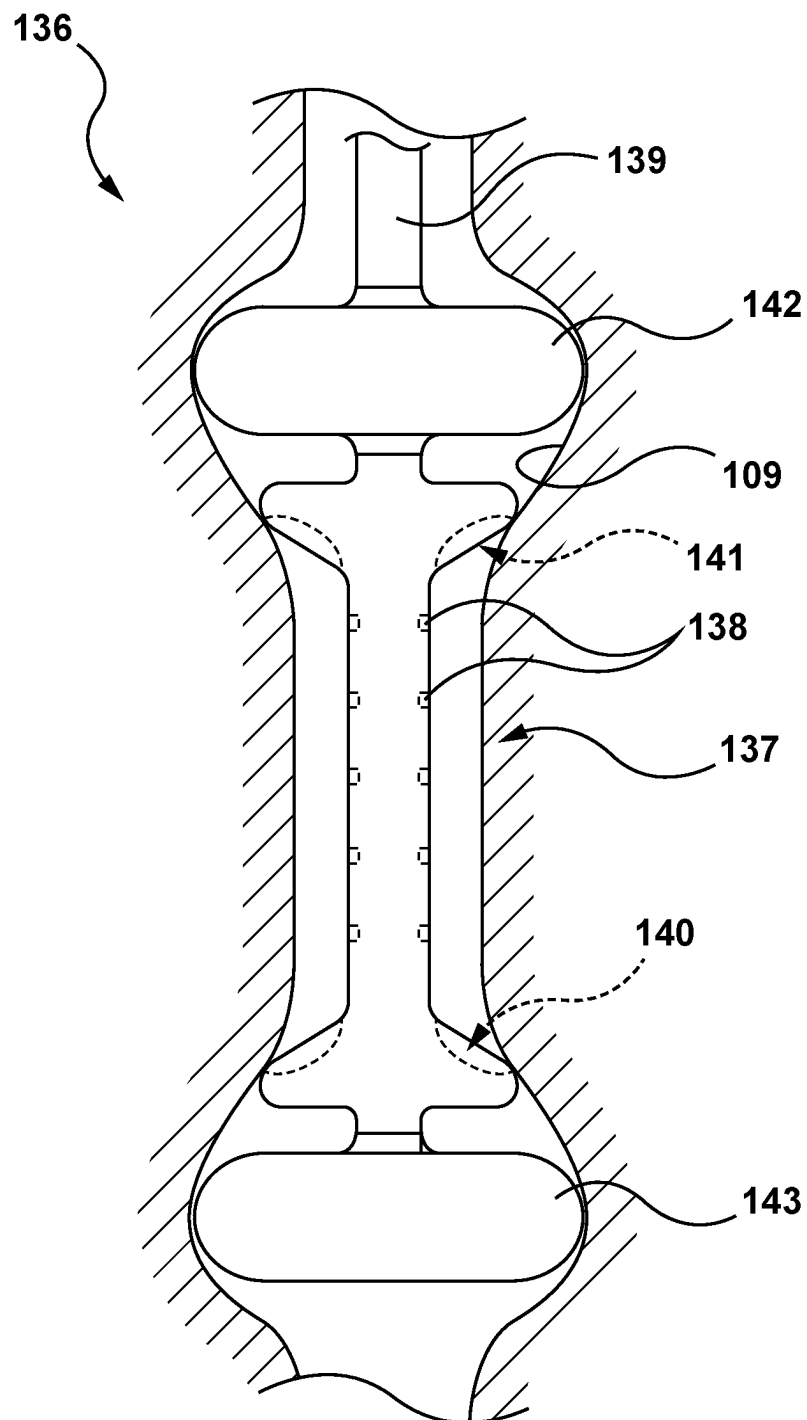


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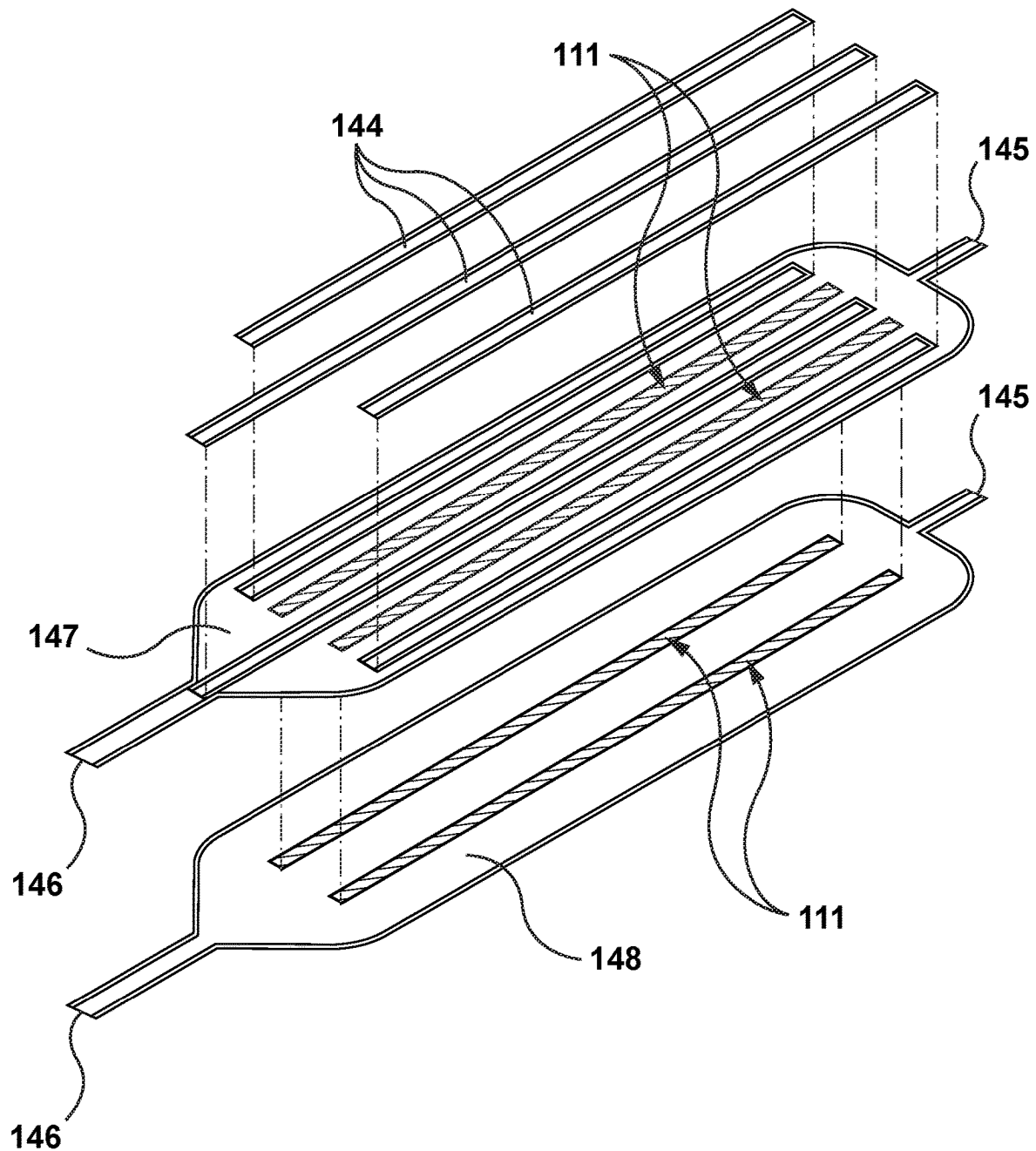
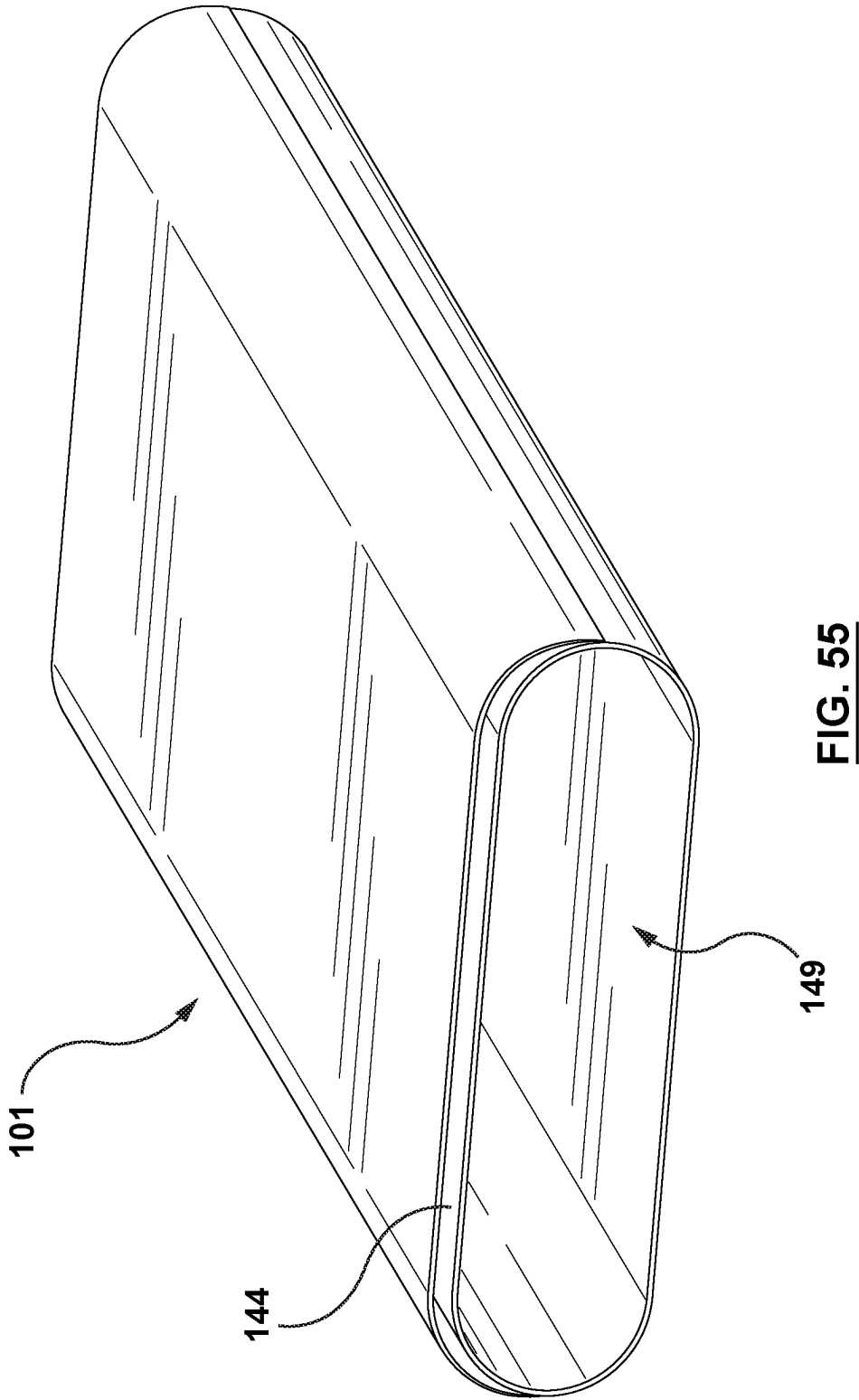


FIG. 54



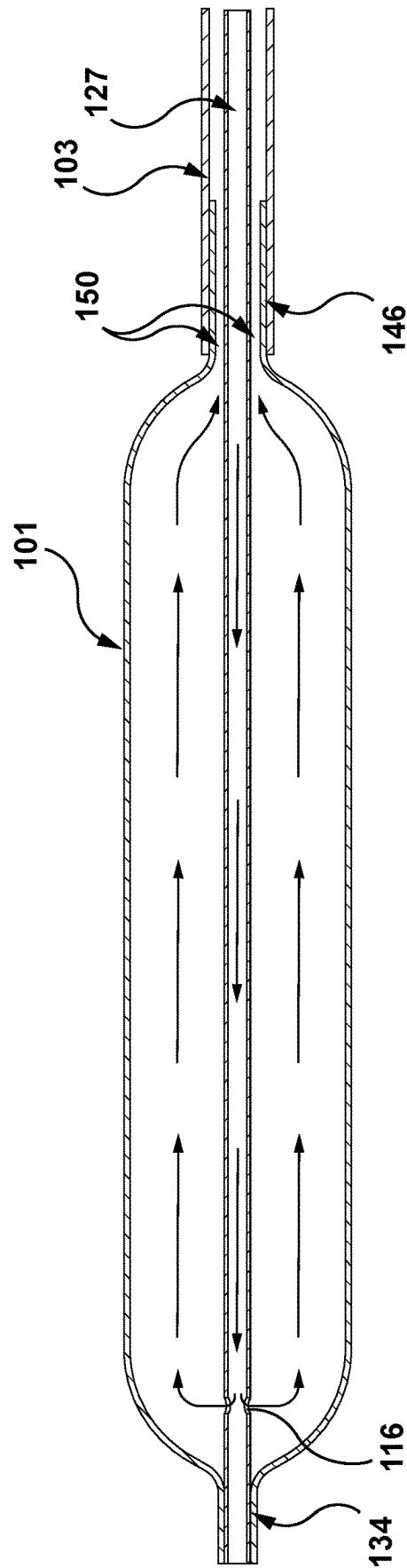


FIG. 56

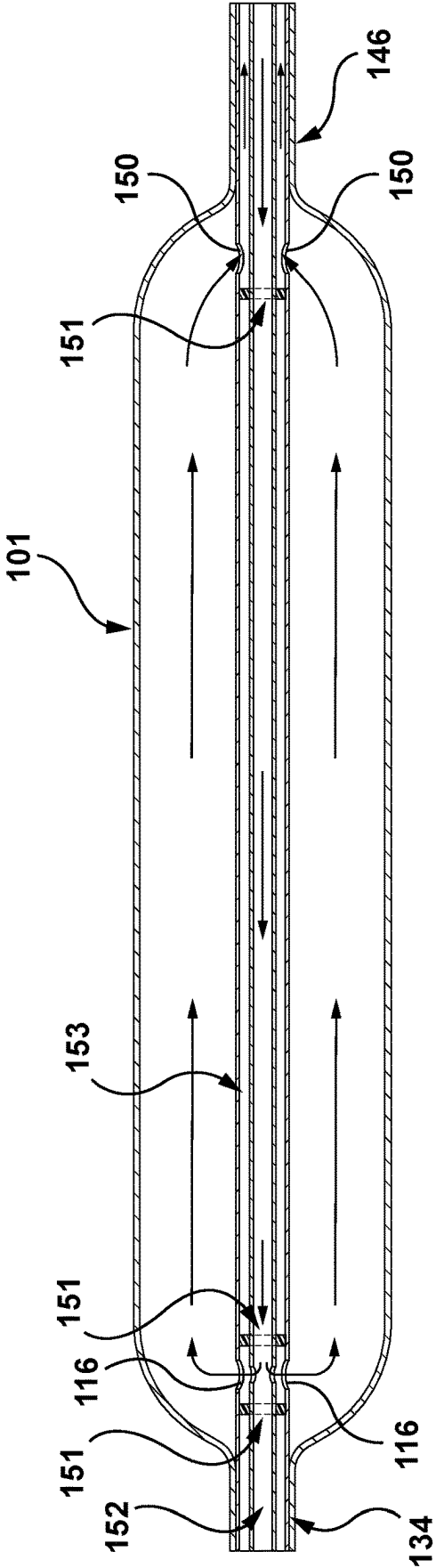


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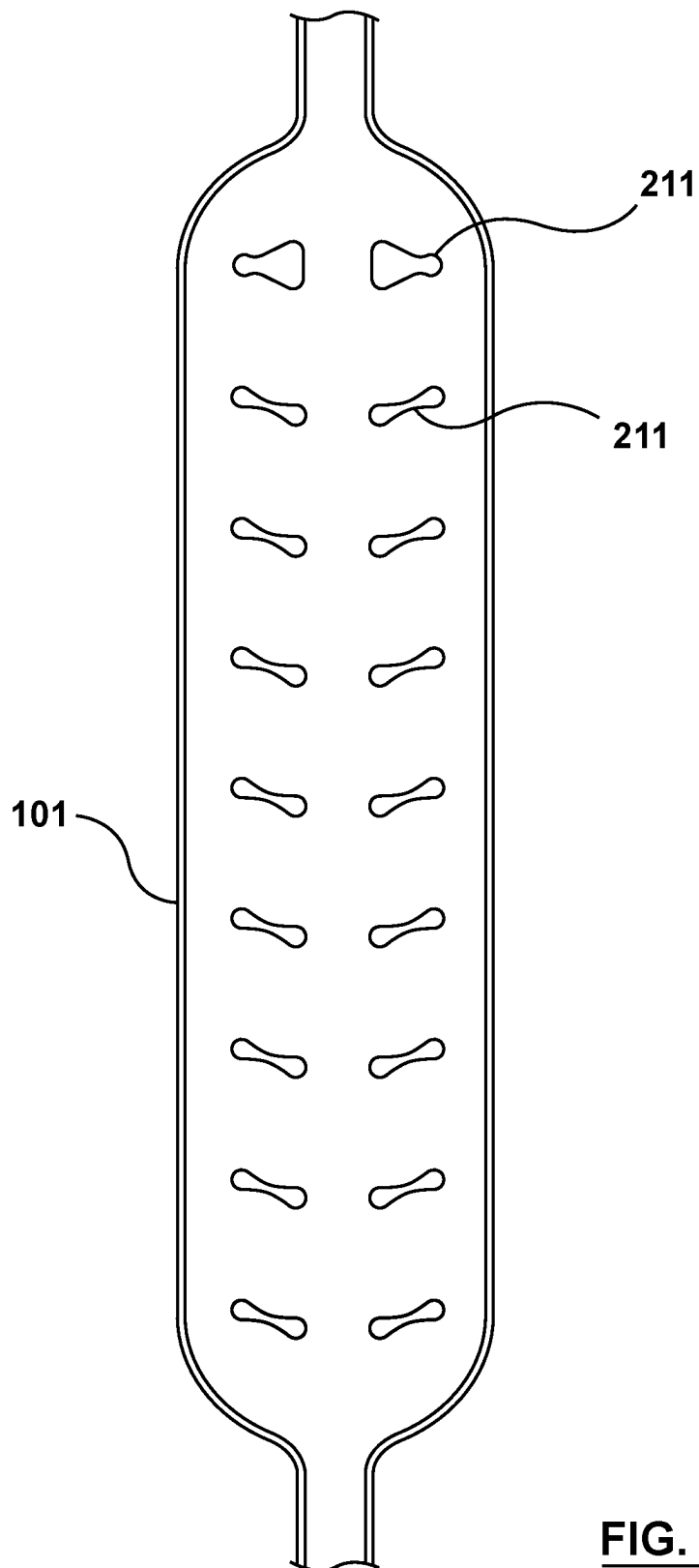


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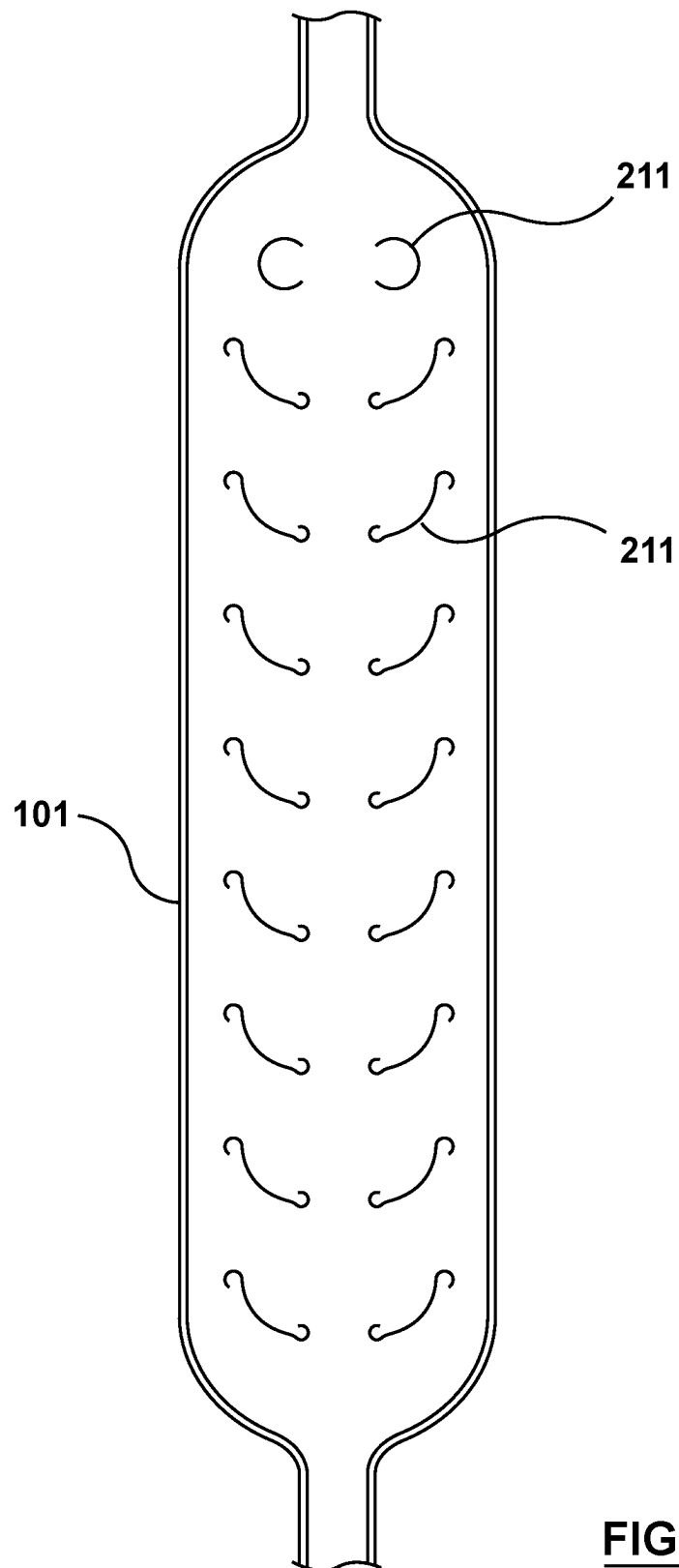


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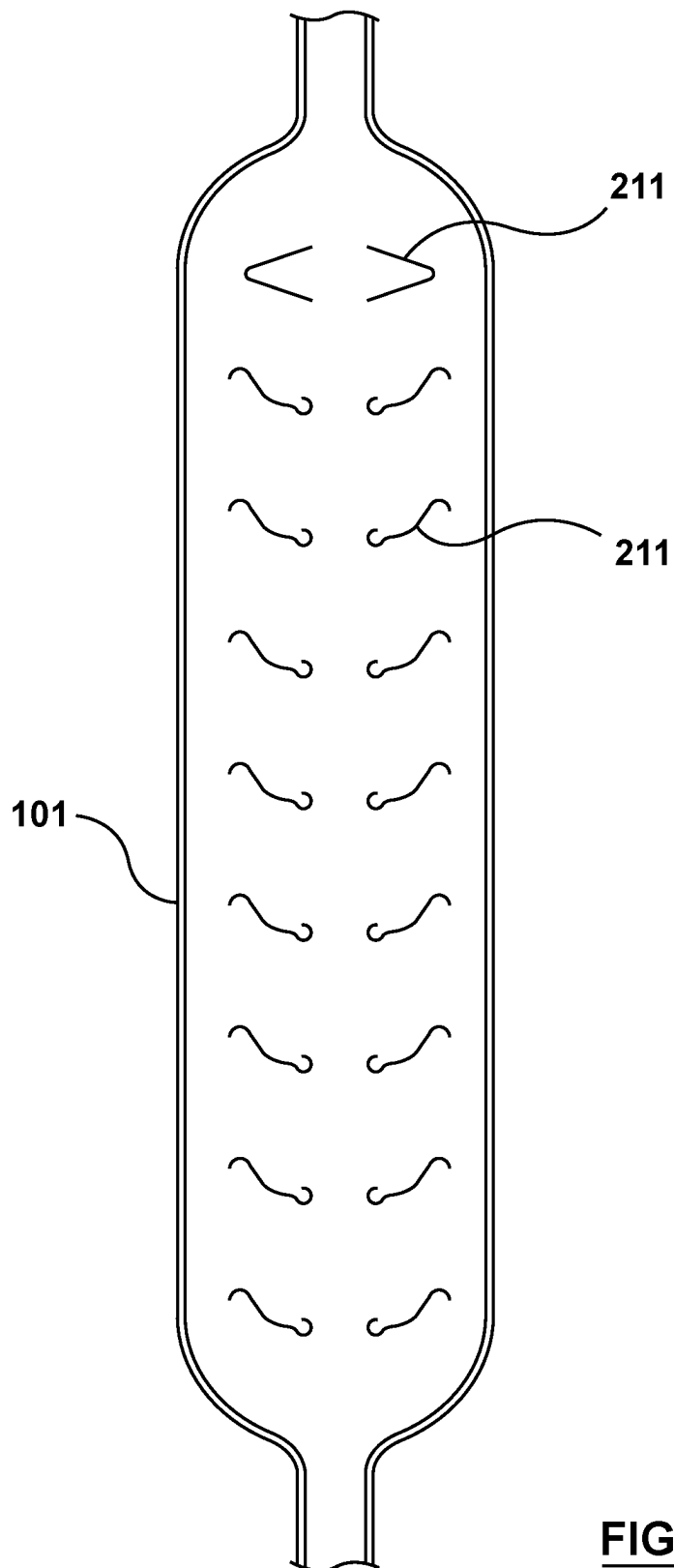
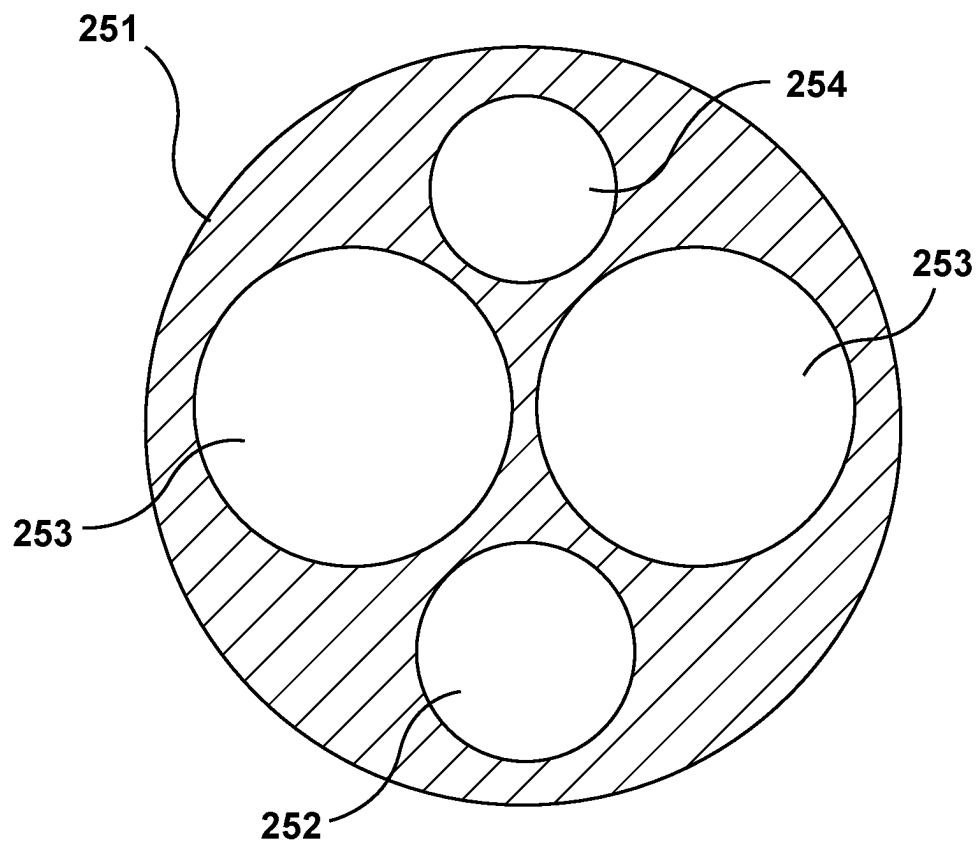


FIG. 60

**FIG. 61**

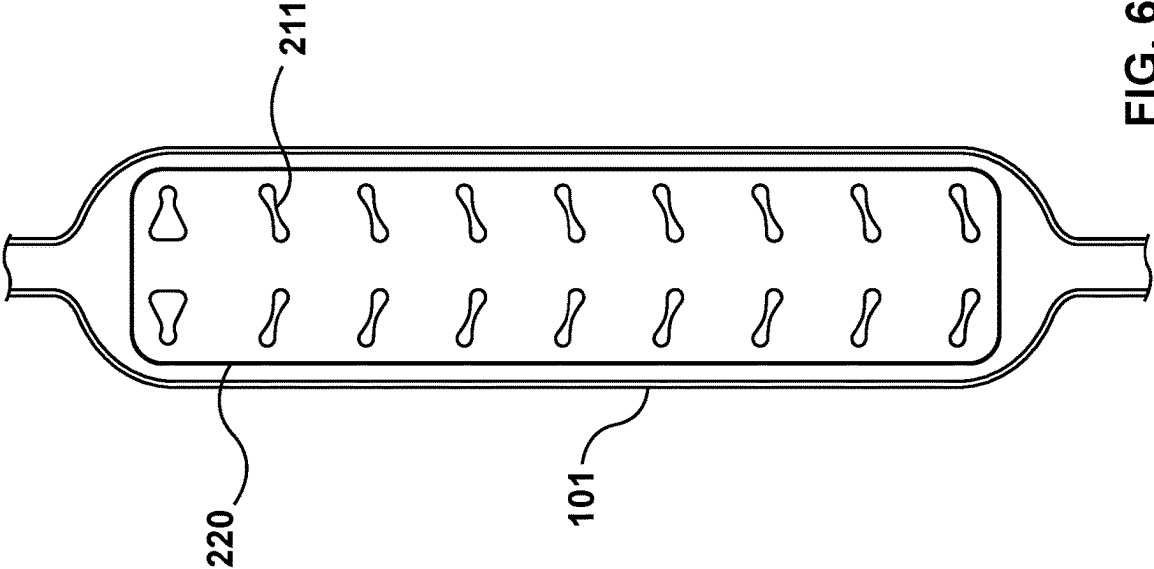


FIG. 63

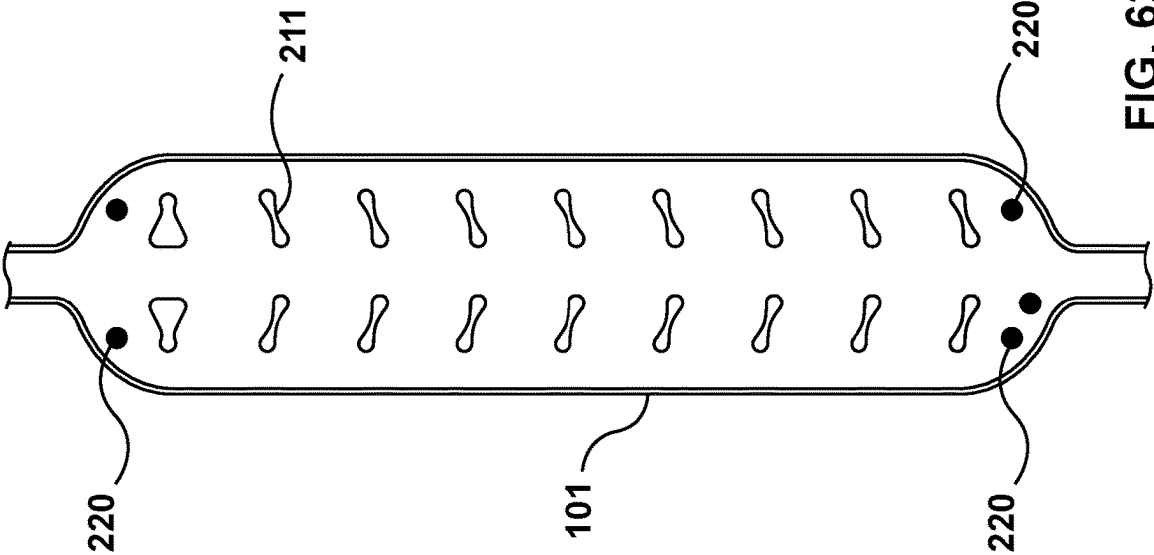


FIG. 62

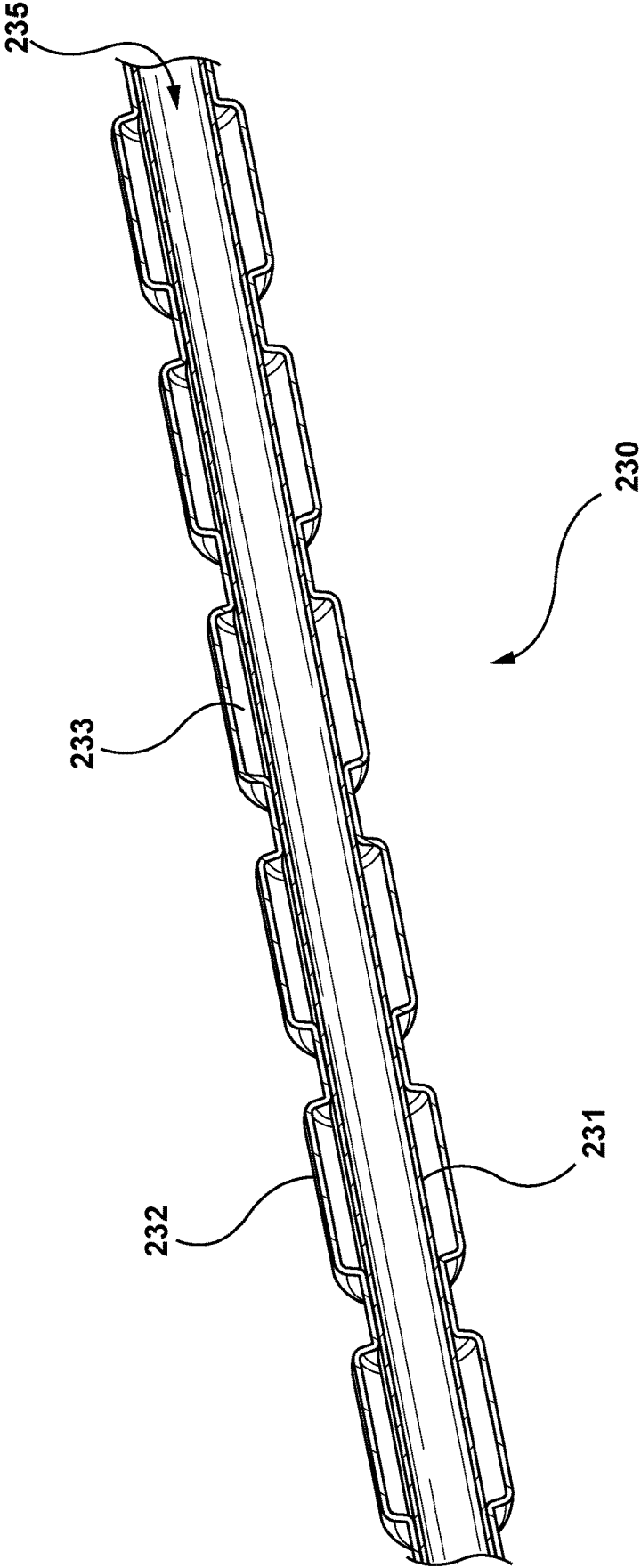


FIG. 64

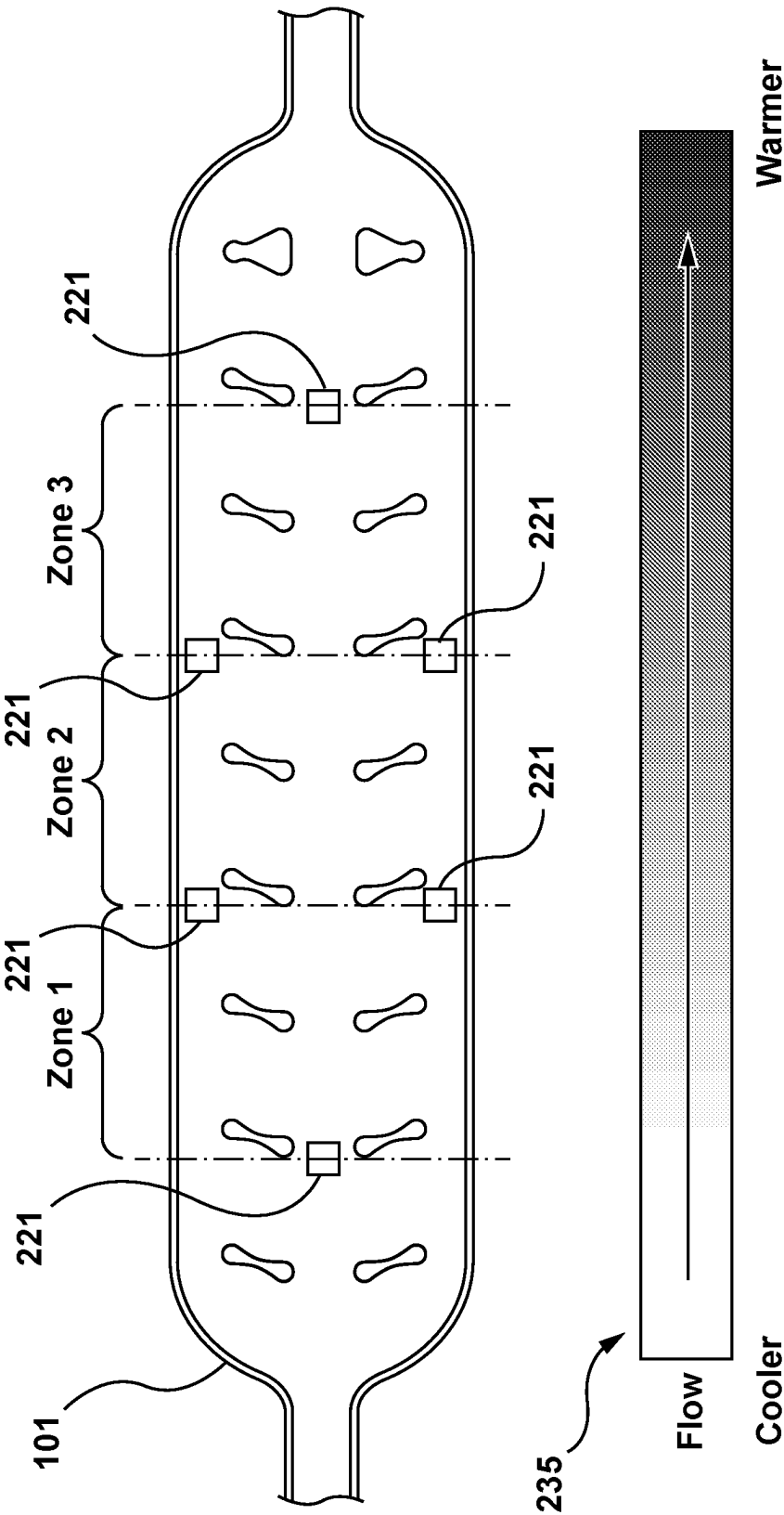


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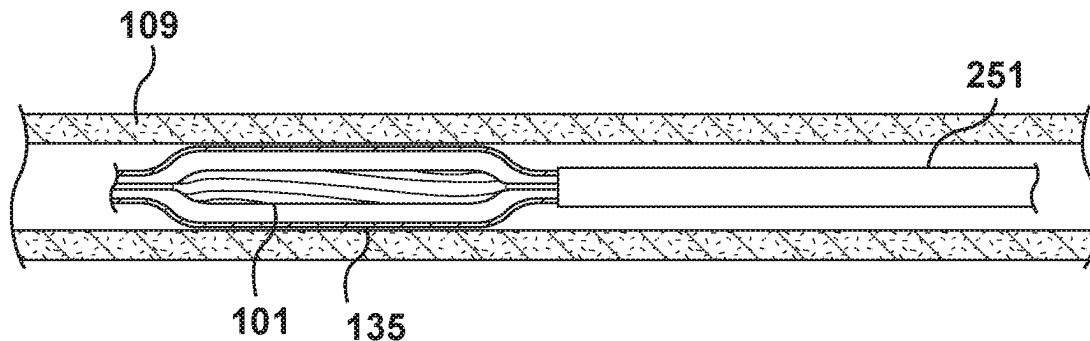


FIG. 66A

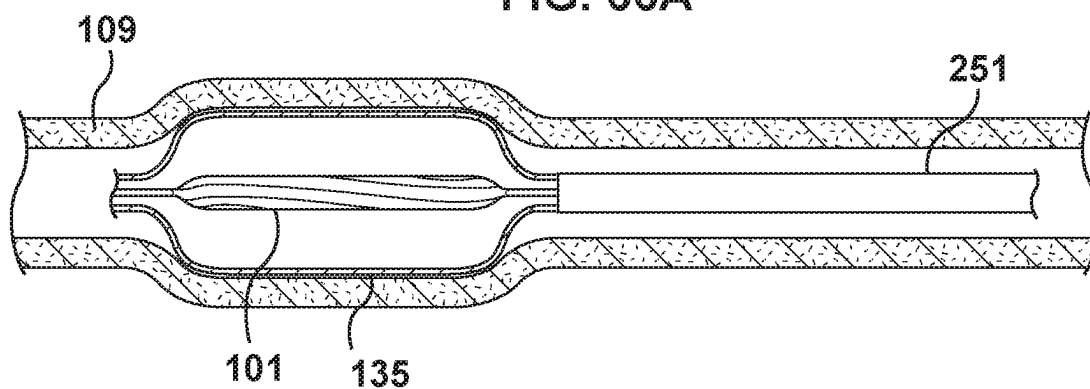


FIG. 66B

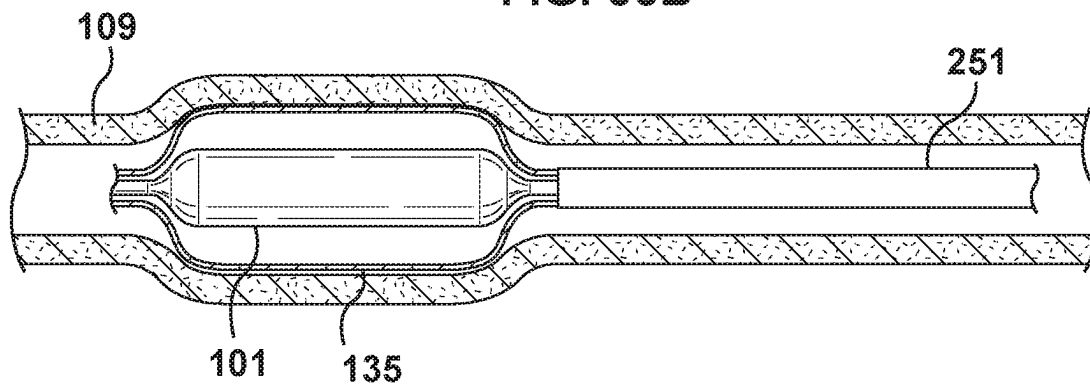


FIG. 66C

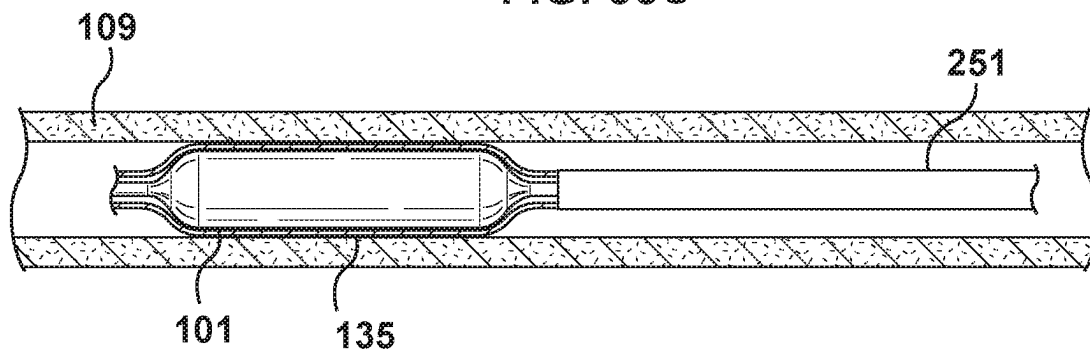


FIG. 66D

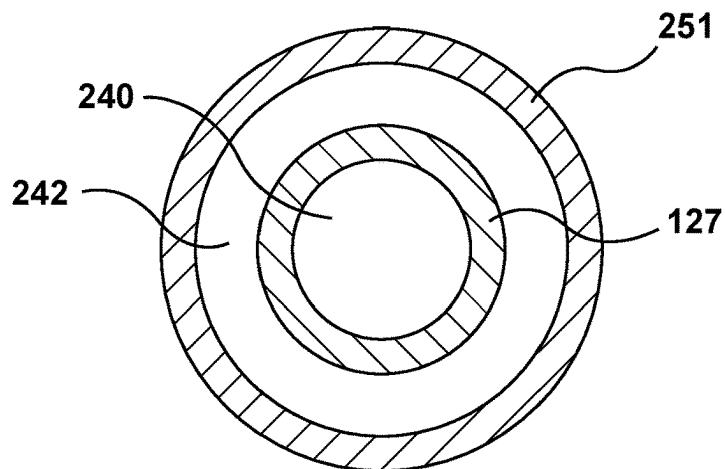


FIG. 67A

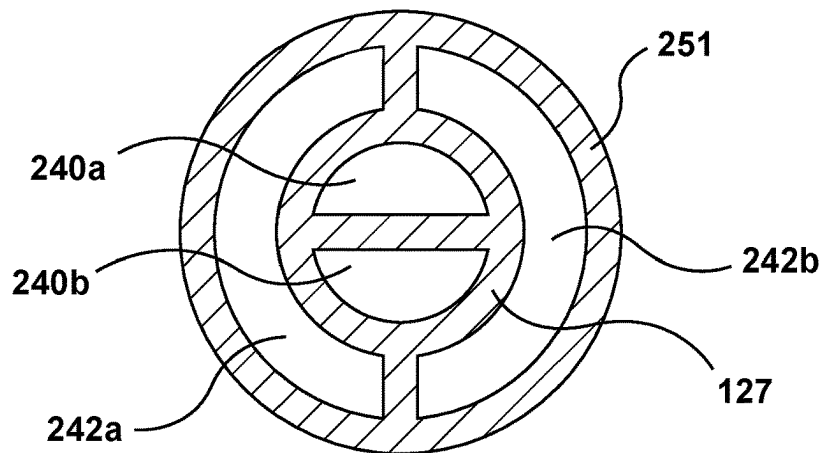


FIG. 67B

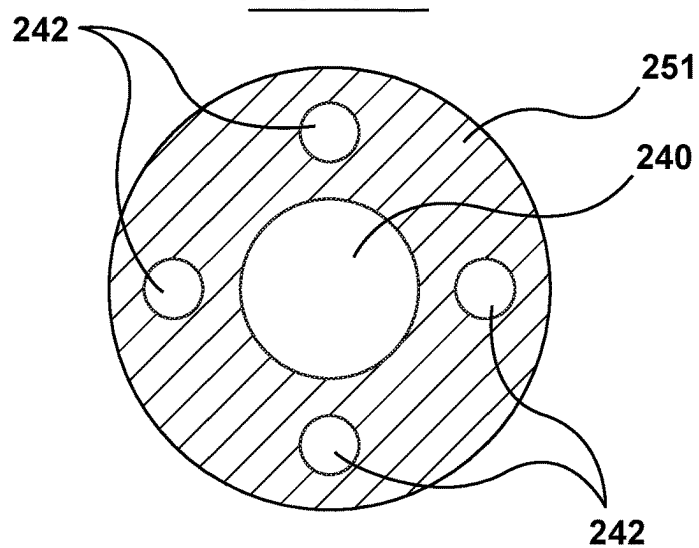


FIG. 67C

1

HEAT EXCHANGE AND TEMPERATURE SENSING DEVICE AND METHOD OF USE

TECHNICAL FIELD

The disclosure relates to the field of heating and cooling tissue, in particular the temperature management of tissue using a heat exchange and temperature sensing device.

SUMMARY OF THE DISCLOSURE

The problem of preventing injury to an esophagus caused by heat or cold being delivered to the heart or other nearby tissue may be solved by regulating the temperature of the esophagus using a heat exchange device having a heat exchanger which has a cross section substantially corresponding with the collapsed/relaxed/natural cross section of the inside of the esophagus. In some embodiments, the heat exchanger is a balloon, and inflation of the balloon substantially maintains the esophagus in its natural shape and location (i.e., the esophagus is not displaced towards the left atrium). Some alternative embodiments includes providing a heat exchanger that substantially conforms to or corresponds with the cross section of an esophagus by means other than inflation while substantially maintaining the natural shape and location of the esophagus.

In a first broad aspect, embodiments of the present invention are for a heat exchanging device for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium of a heart, the heat exchanging device comprising: an elongated shaft comprising a distal end and a proximal end; a heat exchanger attached to the distal end of the elongated shaft, the heat exchanger comprising an insertable configuration and a heat exchanging configuration; wherein a cross-section of the heat exchanger in the insertable configuration is smaller than a cross-section of the heat exchanger in the heat exchanging configuration, and wherein said cross-section of the heat exchanger in the heat exchanging configuration substantially conforms to and corresponds with a cross-section of an inside surface of the esophagus such that the esophagus is substantially maintained in its natural shape and location when the heat exchanger is in its heat exchanging configuration.

In some embodiments of the first broad aspect, the heat exchanger comprises a distal end, a proximal end, and a cavity therebetween. In some such embodiments, the elongated shaft defines at least a first lumen and a second lumen, and at least a portion of said cavity is in fluid communication with the first lumen and the second lumen of the elongated shaft.

In some embodiments of the first broad aspect, the shape of the heat exchanger is constrained by a weld pattern, wherein the weld pattern comprises at least one weld, wherein said at least one weld attaches at least part of an anterior surface of the heat exchanger and a posterior surface of the heat exchanger. In some such embodiments, the at least one weld comprises a plurality of welds which are oriented as a series of consecutive chevrons along the length of the heat exchanger. In some examples, at least one chevron of the series of consecutive chevrons is comprised of a pair of dumbbell-shaped welds.

In some examples, at least one chevron of the series of consecutive chevrons is comprised of a pair of welds which are generally curved in shape with hook-shaped ends. In some examples including a plurality of welds, the heat exchanging device further comprises an alternative pair of welds which have an alternative shape selected from the

2

group consisting of C-shaped welds, generally V-shaped welds, and welds which have a generally triangular-shaped portion and a circular portion.

In some embodiments of the first broad aspect, the elongated shaft is a multi-lumen tube which defines an inlet lumen and a first outlet lumen. In some such embodiments, the inlet lumen comprises a fluid inlet port to allow heat exchange fluid to flow into said heat exchanger. In some embodiments having the multi-lumen tube, the first outlet lumen comprises a fluid outlet port to allow heat exchange fluid to flow out of said heat exchanger. Some embodiments having the multi-lumen tube further comprise a second outlet lumen. Some embodiments further comprise the first outlet lumen having a larger diameter than the inlet lumen. Some embodiments further comprise a utility lumen.

Some embodiments of the first broad aspect further comprise a radiopaque material positioned on the heat exchanger. In some embodiments, the radiopaque material defines at least one circular shaped marker. In some embodiments, the radiopaque material defines a bordering frame on a side of the heat exchanger wherein the bordering frame borders at least a portion of the surface of the side. In some embodiments comprising a radiopaque material positioned on the heat exchanger, the radiopaque material is configured to indicate which side of the balloon is facing a heart of a patient. Some embodiments further comprise at least one temperature sensor on a side of the heat exchanger and the radiopaque material being configured to indicate which direction the at least one temperature sensor is facing.

In some embodiments of the first broad aspect, the first lumen is a fluid inlet tube and an inner layer of the fluid inlet tube defines an inlet fluid pathway, and the elongated shaft defines a series of air pockets surrounding the inlet fluid pathway to thereby thermally insulate the fluid inlet tube. In some such embodiments, an outer layer of the fluid inlet tube and the inner layer of the fluid inlet tube define the series of air pockets therebetween.

In some embodiments of the first broad aspect, the second lumen is a fluid outlet tube and an inner layer of the fluid outlet tube defines an outlet fluid pathway, and the elongated shaft defines a series of air pockets surrounding the outlet fluid pathway to thereby thermally insulate the fluid outlet tube. In some such embodiments, an outer layer of the fluid outlet tube and the inner layer of the fluid outlet tube define the series of air pockets therebetween.

In some embodiments of the first broad aspect, the heat exchanger comprises a first zone and a second zone which can be identified under imaging. In some such embodiments, the first zone has a first temperature sensor and the second zone has a second temperature sensor, whereby the heat exchanging device is operable to be moved such that a selected zone with a preferred temperature is adjacent to a heart of a patient.

In some embodiments of the first broad aspect, the heat exchanger is comprised of at least one balloon.

In some embodiments of the first broad aspect, the heat exchanger is comprised of at least one balloon, and the heat exchanging device further comprises an outer balloon surrounding the heat exchanger. In some such embodiments, the elongated shaft comprises an inner lumen which is in fluid communication with the at least one balloon of the heat exchanger and an outer lumen which is in fluid communication the outer balloon. In some embodiments having an outer balloon surrounding the heat exchanger, the elongated shaft contains a tubing wherein an inner surface of the tubing defines the inner lumen, and an outer surface of the tubing and an inner surface of the shaft define the outer lumen.

3

In a second broad aspect, embodiments of the present invention are for a method of regulating a temperature of an esophagus when heat or cold is delivered to a left atrium. The method comprises the steps of: (1) delivering a heat exchange device to a target site within an esophagus; (2) confirming a desired location of the heat exchange device; and (3) exchanging heat with the esophagus; wherein step (2) comprises a user (a) identifying a first zone and a second zone of the heat exchanger under imaging, (b) measuring a first zone temperature and a second zone temperature, (c) comparing the first zone temperature against the second zone temperature and selecting a preferred temperature, and (d) positioning the heat exchanger such that a selected zone having the preferred temperature is adjacent to a heart of a patient.

In a third broad aspect, embodiments of the present invention are for a heat exchanging device for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium of a heart, the heat exchanging device comprising: a heat exchanger which is elongate, the heat exchanger comprising a distal end, a proximal end, a cavity therebetween; the heat exchanging device further comprising a fluid inlet port and a fluid outlet port in fluid communication with the cavity, and a temperature sensor for measuring a temperature.

In some embodiments of the third broad aspect, the temperature sensor is attached to the heat exchanger. In some such embodiments, the temperature sensor is attached to the distal end of the heat exchanger.

In some embodiments of the third broad aspect, the heat exchanging device comprises a heat exchanging configuration wherein a side of the heat exchanger faces a heart of a patient and the temperature sensor is on the side of the heat exchanger which faces the heart. Some such embodiments further comprise an imaging marker for positioning the heat exchanging device in the heat exchanging configuration.

In some embodiments of the third broad aspect, the temperature sensor is spaced apart from the heat exchanger such that the temperature sensed by the temperature sensor is of an esophagus of a patient. In some embodiments, the temperature sensor is spaced apart from the heat exchanger such that the temperature sensed by the temperature sensor is of a stomach of a patient.

In some embodiments of the third broad aspect, the temperature sensor is mounted on a part of a structural frame that is external to the heat exchanger. Some embodiments further comprise an elongated shaft wherein the elongated shaft comprises at least two struts and the temperature sensor is mounted on a strut. In some embodiments, the temperature sensor is affixed to or woven into a textile which surrounds the heat exchanger. In some embodiments, the temperature sensor is affixed to a strand connected to an end of the heat exchanger. In some embodiments, an array of temperature sensors are operable to be used to measure a plurality of temperatures.

In some embodiments of the third broad aspect, the heat exchanger is a balloon. In some such embodiments, the temperature sensor is attached to the balloon, and in some further embodiments, the temperature sensor comprises conductive ink on the surface of the balloon.

In a fourth broad aspect, embodiments of the present invention are for method of installing a heat exchanging device in an esophagus wherein the heat exchanging device comprises an elongated shaft having a distal end, a heat exchanger with an inner balloon attached to the distal end of the elongated shaft, an outer balloon surrounding the inner balloon, and the elongated shaft having an inner lumen in

4

fluid communication with the inner balloon and an outer lumen in fluid communication with the outer balloon; the method comprising the steps of: (a) extending the heat exchanger with the inner balloon and the outer balloon out of a sheath to a location inside of an esophagus; (b) inflating the outer balloon with a first fluid which flows through the outer lumen to thereby expand the esophagus; (c) inflating the inner balloon with a second fluid which flows through the inner lumen; and (d) withdrawing the first fluid in the outer balloon through the outer lumen to thereby create a vacuum which collapses the outer balloon.

BRIEF DESCRIPTION OF THE DRAWINGS

In order that the invention may be readily understood, embodiments of the invention are illustrated by way of examples in the accompanying drawings, in which:

FIG. 1 is an illustration of a heat exchange device;

FIG. 2 is an illustration of a cross section of an esophagus;

FIG. 3 is an illustration of a cross section of a balloon heat exchanger;

FIG. 4 is an illustration of a balloon heat exchanger expanded in an esophagus;

FIG. 5 is an illustration of three balloons side-by-side;

FIG. 6 is an illustration of balloons with centered necks;

FIG. 7 is an illustration of balloons with offset necks;

FIG. 8 is an illustration of a serpentine welded balloon;

FIG. 9 is an illustration of a welded balloons with ties;

FIG. 10 is an illustration of a welded balloon with fins;

FIG. 11 is an illustration of a welded balloon with pockets;

FIG. 12 is an illustration of a coiled tube heat exchanger;

FIG. 13 is an illustration of a multiple-tube heat exchanger;

FIG. 14 is an illustration of a helical-tube heat exchanger;

FIG. 15 is an illustration of a serpentine-tube heat exchanger;

FIG. 16 is an illustration of an inlet port with a single hole;

FIG. 17 is an illustration of an inlet port with multiple holes;

FIG. 18 is an illustration of temperature sensors affixed to balloon surface;

FIG. 19 is an illustration of temperature sensors mounted on the embodiments depicted in FIGS. 13 and 14;

FIG. 20 is an illustration of temperature sensors mounted on struts made from a catheter shaft

FIG. 21 is an illustration of temperature sensors affixed to textile;

FIG. 22 is an illustration of temperature sensors affixed to strands;

FIG. 23 is an illustration of a heat exchanger with an insulating air balloon inside;

FIG. 24 is an illustration of a heat exchanger with an insulating air balloon outside;

FIG. 25 is an illustration of open irrigation of a fluid with suction;

FIG. 26 is a flowchart of a method;

FIG. 27 is an illustration of a balloon with a weld line;

FIG. 28 is an illustration of a balloon with tack welds;

FIG. 29 is another embodiment of a heat exchange device which includes an outer sheath;

FIG. 30 is an exploded view of a balloon;

FIG. 31 is an illustration of a coiled tube heat exchanger with an oblong cross-sectional profile;

FIG. 32 is an illustration of a multiple-tube heat exchanger with an oblong cross-sectional profile;

5

FIG. 33 is an illustration of a helical-tube heat exchanger with an oblong cross-sectional profile;

FIG. 34 is an illustration of a heat exchanger with an insulating portion;

FIG. 35 is an illustration of a heat exchanger with a shaping lumen and a heat exchanging lumen;

FIG. 36 is an illustration of temperature sensors mounted on the embodiment depicted in FIG. 32;

FIG. 37 is an illustration of temperature sensors mounted on the embodiment depicted in FIG. 33;

FIG. 38 is an illustration of a balloon with a pair of wavy weld lines;

FIG. 39 is an illustration of a further embodiment of a balloon with a pair of wavy weld lines;

FIG. 40 is an illustration of a balloon with a pair of wavy weld lines and multiple tack welds;

FIG. 41 is an illustration of a balloon with a pair of curved weld lines;

FIG. 42 is an illustration of a further embodiment of a balloon with tack welds;

FIG. 43 is an illustration of a balloon with a pair of broken weld lines;

FIG. 44 is an illustration of a further embodiment of a balloon with a pair of broken weld lines;

FIG. 45 is an illustration of yet another embodiment of a balloon with a pair of broken weld lines;

FIG. 46 is an illustration of a balloon with a pair of broken weld lines and a pair of tack welds;

FIG. 47 is an illustration of a balloon with chevron pattern welds;

FIG. 48 is an illustration of a further embodiment of a balloon with chevron pattern welds;

FIG. 49 is an illustration of yet another embodiment of a balloon with chevron pattern welds;

FIG. 50 is an illustration of a welded balloon with an outer balloon;

FIG. 51 is an illustration of an irrigation heat exchanger;

FIG. 52 is an illustration of the embodiment of FIG. 51 disposed in a body lumen;

FIG. 53 is a side view of the embodiment depicted in FIG. 52;

FIG. 54 is an exploded view of a an embodiment;

FIG. 55 is an illustration of a heat exchanger comprising a pocket;

FIG. 56 is a cross-sectional plan view of a heat exchanger attached to a shaft and an inlet tube;

FIG. 57 is a cross-sectional plan view of a heat exchanger attached to a shaft, an inlet tube, and an outlet tube;

FIG. 58 is an illustration of another embodiment of a balloon with chevron pattern welds and a pair of alternatively shaped welds;

FIG. 59 is an illustration of a further embodiment of a balloon with chevron pattern welds and a pair of C-shaped welds;

FIG. 60 is an illustration of a yet further embodiment of a balloon with chevron pattern welds and a pair of V-shaped welds;

FIG. 61 is a cross-section of a multi-lumen shaft;

FIG. 62 is an illustration of an embodiment of a balloon with radiopaque material;

FIG. 63 is an illustration of another embodiment of a balloon with radiopaque material;

FIG. 64 is an illustration of an embodiment of a shaft with air pockets;

FIG. 65 is an illustration of an embodiment of a balloon with zones;

6

FIGS. 66A to 66D are illustrations of an embodiment of a heat exchange device with inner and outer balloons; and

FIGS. 67A to 67C are illustrations of shaft embodiments which could be used in the heat exchange device of FIGS. 66A to 66D.

DETAILED DESCRIPTION

Inadvertent thermal injury to the esophagus is a dangerous complication of left atrial ablation due to the close proximity of the esophagus to the posterior aspect of the human heart. These thermal injuries can include esophageal mucosal changes, tissue necrosis, ulcer formation, and atrial-esophageal fistula formation.

Current preventative options include reducing the power or duration of ablation when targeting the posterior wall of the left atrium, and monitoring luminal esophageal temperature during ablation so that the ablation can be stopped if there is an unacceptable temperature change in the esophagus. These options may reduce the effectiveness of an ablation treatment.

Attempts have been made in the past to protect the esophagus using cooling balloons. One of the limitations of such balloons is that the balloons typically expand and/or displace the esophagus. Sometimes, a balloon expands and displaces an esophagus to a position closer to the posterior wall of the heart which is the location of heating by delivery of energy for ablation. In such cases, the cooling by the balloon may not be sufficient to protect the esophagus from thermal injury.

The present inventors have conceived of and reduced to practice embodiments of a heat exchange and temperature sensing device and a method of use of said device which is able to prevent injury to an esophagus caused by heat or cold being delivered to the left atrium of the heart. The device regulates the temperature of the esophagus by providing a heat exchanger which can be placed in the esophagus. The heating/cooling balloon has an inflatable cross section corresponding with the collapsed/relaxed/natural cross section of the inside of the esophagus. Inflation of the balloon maintains the esophagus in its natural shape and location such that the esophagus is not displaced towards the left atrium.

In its collapsed or insertable state, the balloon is low in profile and flexible so that it can be inserted into the nose or mouth and advanced to the esophagus. Once positioned in the esophagus, it is expandable to take on a shape with a profile and dimensions corresponding to the collapsed/relaxed/natural cross section of the internal lumen defined by a human esophagus. When fully expanded, the heat exchange balloon makes contact with the endoluminal surface of the esophagus without substantially displacing it from its natural location.

The outer surface of the balloon is in intimate contact with the mucosal layer of the esophagus. It supplies or removes thermal energy in order to keep the esophagus at a desired temperature throughout an ablation procedure. This includes cooling the esophagus during heat-based ablation procedures, (such as radio frequency/RF or high intensity focused ultrasound ablation/HIFU), or warming the esophagus during cold-based ablation procedures (such as cryoablation).

This method and device may be used during left atrial ablation procedures, which are procedures for treating atrial fibrillation in humans. These procedures may include RF/HIFU ablations and cryoablations. In these types of procedures, ablations are performed to create lesions around the ostia of the pulmonary veins, some of which are typically

very close to the esophagus. Before the veins are ablated, the balloon portion of the device is positioned in the esophageal lumen and posterior to the left atrium. Once activated, the device either removes thermal energy from the esophagus, or delivers thermal energy to the esophagus to keep it in a desired temperature range throughout the procedure.

The invention can also be used in other cardiac procedures where the temperatures in the heart reach undesired levels. It can also be used in other areas of the body where temperature management is required to protect sensitive structures, for example ablation of the prostate to treat cancer. Additionally, the invention can be used to control patient temperature, for example to induce and maintain hypothermia in critically ill patients, or to warm patients with body temperatures below normal, such as when they are under general anesthesia and undergoing surgery.

With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of certain embodiments of the present invention only. Before explaining at least one embodiment of the invention in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments or of being practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein is for the purpose of description and should not be regarded as limiting.

Heat Exchanging Fluid Device

An example of a heat exchange and temperature sensing device **100** for use in the method described herein is illustrated in FIG. 1. Heat exchanging fluid device **100** comprises a main shaft **103** which has balloon heat exchanger **101** at one end, with temperature sensor **104** being associated with balloon heat exchanger **101**. Handle **105** is at the other end of main shaft **103**. The end of heat exchange device **100** having handle **105** also includes fluid inflow **106**, fluid outflow **107**, and temperature sensor connector **108**. Another embodiment of heat exchange and temperature sensing device **100** is shown in FIG. 29. As will be described in more detail below, heat is exchanged by allowing fluid to be circulated through the balloon **101** via fluid inflow **106** and fluid outflow **107**.

In one embodiment, the fluid is made substantially of water. For example, the fluid may be distilled water or saline. Alternately, the fluid may be a substance that is not substantially water, such as an oil based or petroleum product. In addition, the fluid may contain additives, for example a disinfectant, or stabilizer. The temperature, flow rate, and pressure of the fluid is managed through an external controller which includes a pump. The heat exchanging fluid device of the present invention is described in greater detail below.

The heat exchanging fluid device comprises inlet port(s) and outlet port(s). The inlet port(s) is the location where the fluid enters the heat exchanger (e.g. a balloon). There may be one or multiple inlet ports which service different locations in the heat exchanger. In one embodiment, the inlet port **116** is a hole on tube **127** located inside the heat exchanger **101** (e.g. FIG. 16. Inlet port with single hole). The fluid advances through fluid inflow **106** and tube **127** until it reaches the hole **116** and enters the heat exchanger **101**. Fluid inflow **106** and tube **127** are in fluid communication to allow fluid to be supplied to the heat exchanger via fluid inflow **106**. In some embodiments the tube **127** is made of plastic, possibly reinforced with materials such as a metal

coil or braid within the tube wall. The hole **116** may be at the distal end of the heat exchanger **101**, or the proximal end of the heat exchanger **101**, or at any location in between. In the embodiments depicted in FIGS. 1 and 29, inlet port(s) are in fluid communication with fluid inflow **106**. In some embodiments, inlet port(s) are part of fluid inflow **106**.

In some embodiments, heat exchange fluid is circulated in a closed loop. After heat exchange fluid leaves the heat exchange device via fluid outflow, the heat exchange fluid is re-heated/re-cooled then introduced back into the heat exchange device via fluid inflow. Heat exchange fluid may thus be continuously recirculated.

In other embodiments, heat exchange fluid is circulated in an open loop. Heat exchange fluid leaving the heat exchange device is discarded or disposed of.

In an alternate embodiment, the tube has multiple holes spaced along the tube (e.g. FIG. 17. Tube **127** with multiple inlet ports **116**). The fluid advances through the tube until it reaches one of the multiple holes, and enters the heat exchanger in multiple locations simultaneously. The holes may be spaced linearly at regular intervals, or in a helical pattern around the tube, or in any other type of pattern along the tube. Typically, the holes are located to optimize one of the features of the heat exchange. For example, the holes may be located to maximize the thermal performance of the heat exchanger, or to control pressure inside the heat exchanger.

The outlet port is the location where the fluid exits the heat exchanger. There may be one or multiple outlet ports which service different locations in the heat exchanger. In one embodiment, the outlet port is a tube with a single hole located inside the heat exchanger. The fluid enters the heat exchanger at the inlet port(s), travels through the heat exchanger, and exits at the outlet port. In some embodiments, the tube is made of plastic, possibly reinforced with materials such as a metal coil or braid within the tube wall. The hole may be at the distal end of the heat exchanger, or the proximal end of the heat exchanger, or at any location in between. In the embodiment depicted in FIGS. 1 and 29, outlet port(s) are in fluid communication with fluid outflow **107**. In some embodiments, outlet port(s) are part of fluid outflow **107**.

In another embodiment, the tube **127** has multiple holes spaced along its length. The fluid in the heat exchanger exits through one of the multiple holes simultaneously. The holes may be spaced linearly at regular intervals, or in a helical pattern around the tube, or in any other type of pattern along the tube. Typically, the holes are located to optimize one of the features of the heat exchange. For example, the holes may be located to maximize the thermal performance of the heat exchanger, or to control pressure inside the heat exchanger.

The heat exchange device **100** may comprise a sheath or sleeve. A sheath **152** is depicted in FIG. 29. As will be discussed in greater detail below, a heat exchanger **101** may be collapsed/wrapped/deflated around a shaft **103** such that the heat exchanger **101** may be received into a sheath **152**. Accordingly, sheath **152** is dimensioned to receive heat exchanger **101** when it is in a collapsed/wrapped/deflated configuration. This feature may be provided to avoid damage to the body lumen when the heat exchanger is being advanced therethrough. Heat exchanger **101** may be provided with radiopaque (RO) markers or electroanatomic mapping (EAM) **153** for using imaging techniques to determine the location of the heat exchanger **101** relative to known anatomical markers. The embodiment depicted in FIG. 29 is provided with a body temperature sensor **155**.

Body temperature sensor **155** is spaced apart from heat exchanger **101** such that the temperature sensed by sensor **155** is of the body lumen (e.g. the esophagus) and not of the heat exchanger **101**. Items **154** are additional electrodes for either pacing or detecting electric signals. For pacing, pacing electrodes would be provided. For detecting electric signals, electrocardiogram electrodes would be provided. The various sensors and electrodes may be connected to one or more external devices through connector **108**.

FIGS. **62** and **63** illustrate embodiments of the heat exchanger **101** in which, radiopaque material has been placed on the balloon to aid in visualization for positioning and verifying the correct position of the device. Examples of radiopaque materials which may be used for visualizing the balloon under fluoroscopy include films or extrusions with additives, radiopaque ink or paint, metal foil, and other materials known to those skilled in the art. The radiopaque material **220** can be implemented in different configurations, for example, as a bordering frame (FIG. **63**), dots or circular markers (FIG. **62**), lines, patterns, and other known configurations. Radiopaque material which has been placed on a balloon may be used to verify the balloon is: (1) positioned behind the left atrium, (2) fully deployed such that the balloon is flat and not folded or twisted, and (3) positioned such that temperature sensors are facing the heart. For the balloon to be positioned such that one side of the balloon is facing the heart, the radiopaque material **220** should be placed on the balloon in an asymmetrical manner, for example, the radiopaque material being placed in the four corners of the balloon, with a different configuration from the others, as shown in FIG. **62**.

FIG. **56** illustrates an embodiment of heat exchanger **101** with inlet and outlet ports. The heat exchanger **101** comprises proximal neck portion **146** and distal neck portion **134**. In this embodiment, inlet ports **116** are positioned proximate the distal end of heat exchanger **101**. Fluid travels through tube **127** along the arrows shown in the figure. When the fluid reaches the outlet ports **116**, fluid exits tube **127** and enters the heat exchanger **101**. The fluid then follows a return path towards a proximal end of the heat exchanger **101** (i.e., towards proximal neck portion **146**) and leaves the heat exchanger **101** via outlet ports **150**. In this embodiment, outlet ports **150** are formed by providing a circumferential gap between the proximal neck portion **146** of the heat exchanger **101** and the tube **127**. The diameter of tube **127** is somewhat narrower than the inner diameter of shaft **103** to permit fluid to flow there between and back towards the fluid outflow **107**. Tube **127** is attached via welding or other means to distal neck portion **146** such that fluid is prevented from escaping the heat exchanger **101** out from the distal neck portion **146**. Proximal neck portion **146** is attached via welding or other means to shaft **103** such that fluid is prevented from escaping the heat exchanger **101** and shaft **103**.

FIG. **57** illustrates a further embodiment of a heat exchanger **101** with inlet and outlet ports. In this embodiment, an inner inlet tube **152** and an outer outlet tube **153** are provided. Fluid flows into the heat exchanger **101** via ports **116**. Both the inner inlet tube **152** and the outer outlet tube **153** comprise ports **116** to allow fluid to pass therethrough. Means are provided to prevent fluid from flowing into the space between inner inlet tube **152** and outer outlet tube **153**. In this embodiment, O-rings **151** are placed on either side of ports **116**. The O-rings **151** prevent fluid from flowing into the space between the inner inlet tube **152** and the outer outlet tube **153**. Outer outlet tube **153** comprise outlet ports

150. Fluid leaving the heat exchanger **101** flows into the outlet ports **150** and towards a fluid outflow.

FIG. **61** is a cross section of an embodiment of the shaft of a heat exchanger wherein the shaft is a multi-lumen tube **251**. Multi-lumen tube **251** includes an inlet lumen **252**, at least one outlet lumen **253**, and a utility lumen **254**. Utility lumen **254** may be used to provide for balloon pressure measurement, temperatures sensors, conductor wires, and other uses known to one skilled in the art. The multi-lumen design of the shaft allows for ease of assembly, provides thermal insulation between the lumens, the ability to achieve lower balloon pressures than some other designs.

(b) A heat exchanger (a cavity for circulation of fluid). In one embodiment, the cavity is a balloon—this embodiment will be described in greater detail below. Some embodiments of balloons are made of a non-compliant material such as Nylon 12 or PET. Alternative balloon embodiments are made of a compliant material such as Pebax® or urethane.

(c) Features augmenting contact. To ensure appropriate heat exchange is occurring at the esophagus, the heat exchanger must maintain adequate contact with the tissue. The heat exchanger having contract with the esophagus also ensures any heat sensors on the heat exchanger are contacting the inner surface of the esophagus. The following optional features may be included for augmenting the contact between the heat exchanger and the tissue:

(c.1) Controllable heat exchanger size. This feature includes the heat exchanger being expandable or contractable to fit the size of the esophagus and promote contact with the tissue. The change in size may be controlled with pressure, such as the internal pressure in a balloon, or external pressure exerted by the anatomy on the device. Alternately, the size may be controlled with a mechanical expansion/contraction mechanism, which may further comprise a feedback loop from the forces exerted on the device (detected via force sensors) to achieve the optimal contact force.

(c.2) Conformable heat exchanger shape. This feature includes the expansion of the heat exchanger being constrained in one or more axes, using the balloon designs outlined above and through the use of compliant and non-compliant materials, thin films with ties or welds, and shape memory materials. In alternative embodiments, the heat exchanger is moldable to the esophagus shape through the use of compliant materials that respond to forces exerted by the tissue.

(c.3) Anchoring feature. The heat exchange device may have an anchoring feature or features such as notches, necks, collars, or hooks that allow the device to engage internally with anatomical features to hold it in place. In alternative embodiments, the device has an anchoring feature such as tape, Velcro, and straps that allow it to engage externally with other devices such as an endotracheal tube or a nasal bridle to hold it in place.

(c.4) Suction feature. The heat exchange device may incorporate suction to hold the tissue against the surface of the device to ensure appropriate tissue contact. Tissue suction may also be used to ensure that tissue is pulled away from the area where heat is being applied. For example, when esophagus tissue is pulled towards the heat exchanger, it may be consequently pulled away from the left atrium of the heart where ablation is taking place.

Also, force may be applied to the esophagus or the device to maintain adequate tissue contact. This may be a force external to the patient, or applied from within the patient from the heat exchange device, or from another device (for example, by suction feature as described above).

11

An additional technique to ensure proper heat exchange is to assess the amount of tissue contact between the heat exchanger and the tissue at the target site. The heat exchange device may comprise force sensors to measure the amount of force between the tissue and the heat exchanger. This force may be used in a feedback loop in communication with the device to maintain optimal force between the heat exchanger and the tissue.

Yet another technique to ensure proper heat exchange is to use heat flux sensors to measure the heat flux at any given part of the tissue at the target site. A greater heat flux measurement represents greater heat transfer between the tissue and the heat exchanger.

Balloon Heat Exchanger

Some embodiments of the heat exchanging fluid device described herein comprise a balloon heat exchanger **101** (FIG. 1). The balloon **101** comprises a cavity for circulation of fluid. Embodiments of such heat exchange balloons are illustrated in FIGS. 2, 5-15, 18-24, 26-30, 34-35, and 38-50.

The inflated cross-sectional shape of such a balloon mimics the natural shape of the inside of a human esophagus. In its collapsed shape, a human esophagus **109** typically has a cross-section of around 1.5-3 cm wide and around 0-0.5 cm high (e.g. FIG. 2, cross section of esophagus). The balloon of the invention (e.g. FIG. 3, cross section of balloon heat exchanger **101**) maintains a cross-section of similar dimensions in order to make intimate contact with the mucosal layer of the esophagus without displacing it, i.e., the balloon is expandable but is restrained in one or more axes to reduce forces exerted on the abutting surfaces of the esophagus (e.g. FIG. 4, balloon heat exchanger **101** expanded in esophagus **109**).

The desired shape of the balloon heat exchanger **101** can be realized in a number of ways. In one embodiment (see FIGS. 5-7), at least two cylindrical balloons are abutted and held side-by-side. For example, if 3 balloons (see FIG. 5) with an inflated diameter of 5 mm are placed side-by-side, the overall dimensions of the cross-section of the heat exchanger (when expanded) is approximately 15 mm wide and 5 mm tall. Thus, both the number of cylindrical balloons and the inflated diameter of the balloons can be varied to vary the overall dimensions of the cross-section of the heat exchanger.

This approach may be used with any number of cylindrical balloons abutted side-by-side. In some embodiments, these balloons are cylindrical with balloon necks **110** in the middle of the balloon (FIG. 6, balloons with centered necks), or in some other embodiments, with offset balloon necks **110** located away from the center of the balloon (FIG. 7, balloons with offset necks). Balloon necks **110** may be in fluid communication with the main body of balloon **128**. Balloon necks **110** may be connected with input ports or output ports to allow fluid flow through the balloon.

In another embodiment, the desired shape of the balloon is achieved by welding thin films together. The films may be plastic such as urethane, or another material that is formable in thin film. In one embodiment, the films are welded in a serpentine shape. FIG. 8 illustrates a serpentine welded balloon having top and bottom films (when in the orientation of FIG. 8) welded together along weld lines **111**. The top and bottom films, when welded, result in a lumen **149** through which fluid may be circulated to perform the heat exchange.

In some embodiments the welding technique is used to add singular or multiple ties inside a balloon to prevent it from expanding in undesired axes. FIG. 9 shows two examples of welded balloons with ties with the balloon on

12

the left having a single tie **112** and the balloon on the right of the figure having two ties. Weld lines **111** weld the ties in place.

In other embodiments a balloon shape is constrained with welds. FIGS. 27-28, 30, and 38-50 feature balloon heat exchangers with a variety of weld patterns. Varying the weld patterns impact the lengthwise and widthwise inflatability and rigidity of the balloon as well as the flow of fluid through the balloon.

FIGS. 28 and 42 illustrate balloons with tack welds **126** (or spot welds). The weld pattern in this embodiment results in multiple fluid flow channels that extend along the length and the width of the balloon. These channels are "open" such that fluid flowing within one channel may flow to another channel. This allows fluid to flow into any particular area of the balloon, even if the balloon is bent, folded, or otherwise restricted from freely inflating in that area. These fluid channels allow the balloon to be more easily inflated and deployed in applications where the balloon is introduced into the esophagus deflated and wrapped around a central shaft (such as shown in FIG. 29) along the balloon's lengthwise axis. Also, these embodiments allow the mixing of flow amongst the various fluid channels, which promotes heat exchange across the entire surface of the balloon.

The balloon heat exchanger **101** of FIG. 27 includes a balloon with a weld line **111** creating two fluid flow channels. FIGS. 38, 39, and 40 includes a balloon **101** with two wavy welds creating three fluid flow channels—the wavy welds in these embodiments creates multiple hinge axis, and each axis resists hinge-like behavior giving the balloon added widthwise rigidity when the balloon is inflated, which may be desirable in certain applications. FIG. 41 includes a balloon **101** with two weld lines which are curved at their end. The curved ends correspond with an outer contour of the balloon. By providing these curves, the cross-sectional area at the ends of the balloon **101** are somewhat reduced, thereby reducing the stress on the material when inflated. The embodiments in FIGS. 27, 38, 39, 40, and 41 each create multiple fluid channels along the length of the balloon such that fluid which is introduced at one end (e.g., the distal end) may naturally flow through the channels towards the other end (e.g., the proximal end). Continuous fluid flow through the length of the balloon enables more efficient heat exchange as the target area is continuously provided with heated or cooled fluids.

FIGS. 43, 44, 45, and 46 illustrate balloons with broken line welds. This design enables the mixing of fluid flow between the lengthwise fluid channels, which may be desirable in applications where the fluid in a particular channel is being cooled or heated more than the fluid in the other lumens. These fluid channels are "open" such that fluid from one lengthwise channel may flow to a different lengthwise channel, the balloon may provide more even heat exchange to an area of the esophagus that is experiencing the most extreme temperatures. It also allows the balloon to be more easily inflated within the esophagus because there are multiple pathways for fluid to flow into a given area of the balloon.

FIGS. 47, 48, 49, 58, 59, and 60 illustrate balloons with a chevron pattern of internal welds. This pattern allows fluid flow mixing across the fluid flow channels. The diagonal alignment of the welds increases widthwise rigidity which allows the balloon to be more easily deployed and inflated after being introduced into the esophagus in applications where the balloon is introduced into the esophagus deflated and wrapped around a central shaft along the balloon's lengthwise axis. FIG. 58 includes short welds **211** which

13

define a plurality of chevron portions which are dumbbell shaped (or round ended). FIGS. 59 and 60 include short welds 211 which define a plurality of chevron portions which are curved with hook-shaped ends. All of FIGS. 58, 59 and 60 include a pair of short welds 211 (shown at the top of the drawings) which have alternative shapes. The alternative pair of the short welds 211 of FIG. 58 are generally triangular-shaped with each triangle having an extended circular corner portion (and could alternatively be described as being generally megaphone-shaped). The alternative pair of short welds 211 at the top of FIG. 59 are C-shaped and the alternative pair of short welds 211 at the top of FIG. 60 are generally V-shaped. In alternative embodiments of FIGS. 58, 59, and 60, the alternative pair of short welds 211 are at different locations along the balloon 101.

FIG. 30 is an exploded view of a balloon 101 before welding. In this embodiment, a pocket 144 is welded in between an anterior balloon surface 147 and a posterior balloon surface 148. This embodiment of balloon 101 further comprises a distal neck portion 145 and a proximal neck portion 146. Pocket 144 may comprise temperature sensors, heat flux sensors, force sensors, or other sensors (not shown). FIG. 54 is an exploded view of a further embodiment of balloon 101 before welding. In this embodiment, three pockets 144 are welded to the outer surface of the anterior balloon surface 147 (i.e., the surface closest to the target area), allowing various sensors to be spread across the width of the balloon. Weld lines 111 are provided creating three fluid flow channels. Pockets 144 are positioned along the fluid flow channels. Positioning pockets 144 on the outer surface of the balloon also allows the sensors to be closer to the target area. Other orientations and combinations of pockets may also be provided. In yet a further embodiment, FIG. 55 depicts a heat exchanger 101 comprising a pocket 144. Pocket 144 may be formed by attaching a piece of material on the outside of the heat exchanger 101, thereby creating a pocket 144 adjacent to lumen 149.

Some alternative embodiments of the heat exchanger 101 have fins or fingers that expand to the desired shape once inflated. The example of FIG. 10 includes a welded balloon with fins 113.

In other embodiments, multiple pockets are welded along the balloon and brought together with ties to hold the balloon in the desired shape. The FIG. 11 embodiment of balloon heat exchanger 101 comprises an inner film 131, an outer film 132, and a tie 112. The inner film 131 and outer film 132 are welded together to form a series of longitudinal pockets 114 (that is, along the length of the balloon). Fluid flows through the longitudinal pockets in order to perform heat exchange. Tie 112 is attached between two sides of an inner diameter of the balloon heat exchanger 101 to produce a desired cross-sectional shape. In FIG. 11, the cross-sectional shape of the balloon heat exchanger is circular. As previously mentioned, the balloon heat exchanger is more preferably oblong to better conform to the cross-sectional area of the collapsed esophagus and reduce the resulting displacement of the esophagus. The length and position of tie 112 may be adjusted to change the shape of the balloon heat exchanger when it is in its inflated or expanded configuration.

In addition to using welding to construct balloon heat exchangers, other means known to those skilled in the art may also be used. For example, other adhesive techniques or blow molding techniques may be employed.

In the embodiments where the outer edges of the balloon are welded, sometimes the outer edges may become sharp. In such cases, an outer balloon 135 without any sharp edges

14

may be provided and covers the inner welded balloon 101 (see FIG. 50). To avoid air or other fluids to be trapped between the inner balloon and the outer balloon 135, the outer balloon may be perforated or may be vacuum sealed against the inner balloon. The outer balloon may be constructed by flipping a welded balloon inside-out, blow molding, or other techniques known to those skilled in the art. A blunt tip 156 may be provided to prevent damage to the body lumen. In the embodiment shown in FIG. 50, a sheath 152 is also provided for receiving the heat exchanger 101.

Use of outer balloon 135 may also provide other advantages. Outer balloon 135 can provide a layer of protection for any electronics placed on the inner balloon (heat exchanger 101). The outer balloon 135 can be inflated to keep the esophagus open and allow the inner balloon to open without being obstructed by the esophagus. As to be explained below, when the outer lumen 242 is used to pull (i.e. create) a vacuum which collapses the outer balloon 135 around the inner balloon, a vacuum is also created in outer lumen 242 whereby the outer lumen functions to insulate the fluid flowing in inner lumen 240 and reduce heat exchange between the shaft and surrounding tissue.

The steps of FIGS. 66A to 66D are explained below with respect to the embodiment of shaft 251 illustrated in FIG. 67A. Shaft 251 of FIG. 67A contains tubing 127 whereby inner lumen 240 is defined inside of tubing 127 and outer lumen 242 is defined between tubing 127 and shaft 251. Inner lumen 240 is in fluid communication with inner balloon 101 and provides for the flow of fluid into and out of the inner balloon. Outer lumen 242 is in fluid communication with outer balloon 135 and provides for the flow of fluid into and out of the outer balloon. FIGS. 66A to 66D illustrate how the outer balloon 135 can keep the esophagus 109 open and allow the inner balloon to unfold without being obstructed by esophagus. FIG. 66A shows heat exchanger 101, outer balloon 135, and shaft 251 after being extended out of a sheath. The outer balloon 135 is not expanded and heat exchanger 101 is unfolded. Outer balloon 135 is inflated with fluid (possibly air) injected through outer lumen 242 to arrive at the configuration of FIG. 66B. FIG. 66B shows an expanded outer balloon 135 which is opening and expanding esophagus 109 to provide space around folded heat exchanger 101. Fluid (typically water but other fluids can be used) is injected through inner lumen 240 to expand heat exchanger 101 resulting in the situation of FIG. 66C wherein heat exchanger 101 has been expanded and unfolded. The fluid in outer balloon 135 is removed using a vacuum to thereby deflate outer balloon 135 so that it fits tightly around heat exchanger 101 (FIG. 66D). Creating a vacuum to collapse outer balloon 135 also creates a vacuum inside outer lumen 242 (FIG. 67A) whereby outer lumen 242 functions to insulate the flow of cooling fluid inside inner lumen 240 from surrounding tissue and thereby increase cooling (or heating for heating procedures) at the balloon esophagus interface. A heat exchanger 101 in the configuration of FIG. 66D could be rotated if needed so that a desired side faces the patient's heart.

FIG. 67B shows an alternative embodiment of shaft 251 which includes an inner inlet lumen 240a, an inner outlet lumen 240b, an outer inlet lumen 242a, and an outer outlet lumen 242b. FIG. 67C illustrates a multi-lumen extruded alternative embodiment of shaft 251 which comprises a single inner lumen 240 and a plurality of outer lumens 242 wherein all of the outer lumens 242 could be used for both inflow and outflow, or alternatively, some of the outer

15

lumens **242** could be dedicated inflow lumens and other outer lumens **242** dedicated outflow lumens.

In another embodiment, the outer edges of the welded balloon may comprise small cuts along the outer edge. By introducing small cuts along the outer edge, the rigid outer edge is rendered soft, and reduces the likelihood of damage to the esophagus while the balloon is being introduced through the esophagus. Other techniques may be used to blunt or soften the outer edge, including:

The outer edge may be widened such that the welded outer edge is softened.

The outer edge may be folded over and welded, glued, or bonded to create a rounded outer edge.

The outer edge may be melted to blunt the outer edge.

Other materials (sprays or dips) may be added to blunt the outer edge.

FIG. **35** illustrates a further embodiment of a balloon heat exchanger **101**. In this embodiment, the balloon **101** comprises a shaping lumen **132** and a heat exchanging lumen **133**. The shaping lumen **132** and heat exchanging lumen **133** are isolated from one another such that fluid in one lumen does not flow to or from the other. In operation, fluid flows through the heat exchanging lumen **133**. The temperature and flow rate of the fluid may be varied to change the rate at which heat is being exchanged between the heat exchanger **101** and the surrounding environment (i.e., the tissue in the esophagus when the heat exchanger **101** is inserted therein). In this embodiment, balloon **101** further comprises a shaping lumen **132**. Shaping lumen **132** may be supplied with a separate fluid (e.g., air or water) which inflates the shaping lumen **132** to its inflated form. Unlike the heat exchanging lumen **133**, fluid need not flow through the shaping lumen **132** in order for the shaping lumen **132** to perform its function. Once inflated, it is possible to maintain the shape of the shaping lumen **132** without providing any fluid flow. Thus, the shape of balloon **101** may be controlled independently from the fluid flow rate and pressure inside the heat exchanging lumen **133**. Those skilled in the art will appreciate that this allows greater flexibility in varying the parameters to arrive at an appropriate rate of heat exchange.

Tubular Heat Exchanger

In another embodiment of the heat exchanger, the cavity for circulation of fluid is an arrangement of thermally conductive tubes. The tubes are preferably arranged to fill a cross-sectional area with outside dimensions similar to the collapsed state of a human esophagus.

In some embodiments, the tubes are arranged in coils. FIG. **12** illustrates a tubular heat exchanger **102** having coils **115**. The profile of the tubular heat exchanger **102** of FIG. **12** is circular.

In some embodiments, the tubes are arranged in parallel and in a circular orientation, such as in the example of tubular heat exchanger **102** FIG. **13**. The tubular heat exchanger **102** of FIG. **12** includes a number of exposed tubes **129** while alternative embodiments may include separate lumens in a single tube (not shown).

In some embodiments, the tubes **129** are arranged in a helix (e.g. FIG. **14**. Helical-tube heat exchanger). In FIG. **14**, each tube **129** is spiral-shaped and is helically arranged adjacent to other spiral-shaped tubes. In the embodiments shown in FIGS. **13** and **14**, heat exchanger **102** further comprise a pair of end portions **130**. Each of the tubes **129** are fixed between the two end portions **130** to maintain the relative orientation between the tubes.

Each of the embodiments illustrated in FIGS. **12**, **13**, and **14** comprise a circular cross-sectional profile. More prefer-

16

ably, the cross-sectional profile of the heat exchanger **102** is oblong to better conform to the cross-sectional area of the inside of a collapsed human esophagus. Examples of such embodiments are illustrated in FIGS. **31**, **32**, and **33**.

Some alternative embodiments have a serpentine-shaped tube, such as shown in FIG. **15**.

Typically, the surface of the heat exchanger is thermally conductive to facilitate the transfer of heat at the desired treatment zone. In some examples, the surface is a film substantially thin enough to allow transfer of thermal energy, e.g. with a thickness between around 0.001" to around 0.003". In some alternate embodiments, the surface is made of a thermally conductive material, such as metal foil.

To further promote heat exchange, a thermally conductive gel or coating may be applied to the heat exchanger, or to the target tissue site. This may fill any gaps that might exist between the tissue and the heat exchanger.

Method of Using the Heat Exchanging Fluid Device

A method of regulating a temperature of an esophagus when heat or cold is delivered to a left atrium (FIG. **26**) includes the steps of:

- (1) measuring the esophagus and selecting a size of a heat exchange device which fits the esophagus;
- (2) delivering the heat exchange device to a target site;
- (3) confirming a desired location of the heat exchange device;
- (4) exchanging heat with the esophagus;
- (5) confirming that the target site is protected; and
- (6) retrieving the heat exchange device.

The steps of the method are described in more detail herein below.

Step 1: Measuring the Esophagus and Selecting a Size of a Heat Exchange Device which Fits the Esophagus

The esophagus is measured in order to select the appropriate device size for the patient. Ways of doing this include:

- (a) Using an internal measurement device. One example is a device that expands until optimal force, impedance, or another parameter indicative of size is measured by the device. Another technique is inserting a series of devices of different sizes into the esophagus until adequate force, impedance, or other parameter is measured by the device.
- (b) Using imaging, such as fluoroscopy, CT, MRI, EAM, etc. Measurements of the anatomy can be taken using methods known to those skilled in these areas of imaging.
- (c) Using a combination of internal measurement devices and imaging. For example, inserting devices of different sizes into the esophagus and viewing them with an imaging modality to determine proper fit. Another technique is inserting an internal ruler device into the esophagus and taking measurements with the imaging system.
- (d) Estimating the size of the esophagus based on external anatomical features.

Once the esophagus size is known, the heat exchange device of best fit can be chosen from a selection of devices that cover the range of most anatomical variations.

Step 2: Delivering the Heat Exchange Device to a Target Site

Delivering the heat exchange device to the target site in the esophagus includes inserting it through a small orifice such as the mouth or nostril, and then advancing the heat exchange device through tortuous path defined by the esophagus until the heat exchange device is positioned at the posterior aspect of the left atrium. A number of features enable the heat exchange device to enter a small orifice.

17

The heat exchanger may be collapsible, foldable, and wrappable such that it can be delivered through a substantially round hole with a diameter of about 0.2 cm to about 0.6 cm. In one embodiment, the heat exchanger is a balloon that can be deflated and wrapped or folded around a main shaft such that it can be delivered to the desired treatment area through a small orifice. Some embodiments of the heat exchange device **100** have an outer diameter equal to or less than 18F.

In an alternate embodiment, the heat exchanger is made of tubes that can be twisted, pulled, or otherwise re-arranged such that they maintain an outer diameter in the desired range and be delivered through a small orifice. Alternately, the tubes themselves may collapse when they are evacuated.

Alternately, the heat exchange device could have a folding or collapsing metal structure such as a stent-like configuration (see FIG. **20**).

The delivery orifice may be an access point on the patient, such as the nasal or oral passageway. Alternately, the delivery orifice may be a delivery tube. Once collapsed, the heat exchanger can be loaded inside the delivery tube, and the delivery tube delivered through an access point on the patient. Typical embodiments of the heat exchange balloon may be tapered at the ends to promote gradual dilation of the small orifice. Once in the desired treatment area, the heat exchanger (the balloon) can then be advanced to exit the tube. Alternately, instead of advancing the heat exchanger (the balloon) out of the delivery tube, the delivery tube could be retracted to expose the heat exchanger.

In addition to the above features, the delivery of the heat exchange device may be augmented by the addition of a lubricious coating on the outside surface of the heat exchange device or on the inner surface of the small orifice.

In order to advance the heat exchange device along a tortuous path, the flexibility of the device may be modifiable with a selection of features:

- (a) varying stiffness along the body of the device, and
- (b) bend points built into the device. For example, instead of a singular heat exchange balloon there may be a number of heat exchange balloons in series along the body of the device, with bend points between them. Alternately, there may be spring-like joints or bendy-straw style joints at desired bend points along the body of the device.

To overcome the difficulty of navigating a flexible device along a tortuous path, the heat exchange and temperature sensing device may have a selection of features:

- (a) steerable portions,
- (b) weighted portions, and/or
- (c) a stylet that may be removable. The stylet may be super-elastic, have a shape-set memory, may be steerable, or may change the shape of the heat exchange device as it is advanced and retracted within.

To avoid mechanical injury to tissue, the heat exchange device may have features to promote atraumatic delivery. These features may include floppy portions, tapered ends, soft portions, steerable portions, and a soft covering sheath.

If the heat exchange device is collapsed/folded/wrapped, it must be expanded once it reaches the target location of the esophagus. The heat exchange device may be expanded in a number of ways:

- (a) Expanded with pressure, such as with a balloon or tubes inflated with heat exchange fluid. In some embodiments, the device may operate at more than one pressure. For example, fluid provided at a first higher pressure may be used to expand or inflate the balloon or tube. Once the balloon or tubes have been expanded,

18

the heat exchange device may operate at a lower pressure so that the balloon or tubes are less rigid. A balloon or tubes which are less rigid are more likely to make good contact with the esophagus while minimizing displacement of the esophagus.

- (b) Expanded with shape memory. The heat exchange device may employ shape memory metals or polymers that may be expanded into the desired shape through thermal or electrical activation.
- (c) Expanded with a mechanical mechanism, such as with a stent-like configuration.
- (d) With any of these expansion methods, the heat exchange device may expand to perforate a delivery sheath that was holding the folded/collapsed/wrapped portions within.

Step 3: Confirming a Desired Location of the Heat Exchange Device

Once the heat exchange device has been delivered to the target site and expanded (if required) the user confirms that the device is in the correct location. This may be achieved by a number of means:

- (a) Device visualization relative to known anatomical markers. This can be achieved by having markers on the device, such as a ruler on the device body, orientation markers on the device body or handle, electrodes visible on an EAM system, or radiopaque markers on the device body (see item **153** in FIG. **29**), handle or stylet visible on fluoroscopy. Visualization of markers can be used to confirm the position and orientation of heat exchange device **100**. Markers are located on the heat exchange device such that they do not interfere with the desired use of the device, for example, located on the posterior aspect of the heat exchanger.

FIG. **65** illustrates apparatus for a method of confirming the location of the heat exchange device. In a balloon **101** with cooled water flowing in it, there is a temperature gradient from the inlet to the outlet. This gradient is due to the heating added to the fluid from the tissue touching the balloon. In the example of FIG. **65**, the cooling fluid flows from left (the inlet end) to right (the outlet end) whereby the fluid flow **235** is cooler on the left and warmer on the right. FIG. **65** further illustrates radiopaque markers **221** which under imaging allow a user to identify the regions of Zone **1**, Zone **2**, and Zone **3**, wherein Zone **1** is the coolest zone and Zone **3** is the warmest zone. Typically each zone has a temperature sensor associated with it. Such temperature sensors could be attached to balloon **101** within the zones; or temperatures could be determined by other means. The temperature sensors for the different zones are typically independent of each other. Depending on the case, the user may wish to adjust the cooling at a particular site (e.g. the zone closest to ablation does not have the desired temperature) by moving the balloon. The different cooling zones can be identified with radiopaque markers **221** and viewed under imaging. The user may then utilize the cooling zones by adjusting the position of the balloon (if needed) to achieve the desired cooling. The above example relates to heat ablation procedures. In the case of a cryoablation procedure, the gradient would be reversed, but the same method could be applied. Alternative embodiments have two, four, or more zones.

- (b) Measurement of a physiological parameter. Some embodiments of the heat exchange device are capable of measuring a physiological parameter indicative of location in the body through the use of sensors or electrodes. Examples of the parameter which may be

measured include ECG, tissue impedance, temperature, blood perfusion rate, oxygen saturation, and others.

Step 4: Exchanging Heat with the Esophagus

Option 1: Using a Heat Exchanging Fluid Device

As discussed above, heat may be exchanged within the esophagus using a heat exchange fluid device, such as those embodiments described above in the section titled "Heat Exchange Fluid Device". In one embodiment, the fluid used in the device is comprised substantially of water. For example, the fluid may be distilled water or saline. Alternatively, the fluid may be a substance that is not substantially water, such as an oil based or petroleum product. In addition, the fluid may contain additives, for example a disinfectant, or stabilizer. The temperature, flow rate, and pressure of the fluid is managed through an external controller which includes a pump.

In operation, fluid flows through an inlet port into the heat exchanger of the heat exchanging fluid device and circulates through the body of the heat exchanger. An outlet port is also provided to allow fluid to flow out of the heat exchanger. Fluid may continuously flow through the heat exchanger so that there is continuous heat exchange with the esophagus.

Option 2 for Exchanging Heat: Open Irrigation

In an alternate embodiment (e.g. FIG. 25 open irrigation of fluid with suction), the heat exchange fluid 125 is delivered directly to the desired treatment zone in an open-irrigated system. In one embodiment, the heat exchange device 100 is connected to an external controller that provides the heat exchange fluid 125. The fluid is delivered through a fluid spray tube 122 and sprayed circumferentially toward the endoluminal surface of the esophagus. Fluid is removed using fluid suction tube 123. In typical embodiments, the tube has multiple holes along its length and around its circumference in order to deliver an even spray of fluid to the desired treatment zone. In one embodiment, the fluid is allowed to travel through the esophagus to the stomach. Alternately, in some embodiments, the esophagus is blocked by an esophageal blocking balloon 124, and the fluid is collected cranial to the blocking balloon 124 and suctioned from the esophagus.

A further alternative embodiment is depicted in FIGS. 51-53. Similar to the embodiment depicted in FIG. 25, heat exchange fluid 125 is delivered directly to the desired treatment zone. The irrigation heat exchanger 136 delivers heat exchange fluid through an irrigation surface 137 with a series of irrigation ports 138 from which heat exchange fluid is sprayed. The heat exchanger 136 may further comprise a proximal blocking balloon 142 and a distal blocking balloon 143 which prevents fluid from escaping into the stomach or the larynx respectively. The heat exchanger 136 may further comprise a distal suction component 140 and a proximal suction 141 component which captures fluid after it has been sprayed from the irrigation ports 138. The heat exchanger 136 may be connected to or integral with a tube 139. Tube 139 may comprise an inlet tube and outlet tube (not depicted) for supplying fluid to and removing fluid from the desired treatment zone. The other end of tube 139 may be connected with an external controller that provides heat exchange fluid 125.

Option 3 for Exchanging Heat: Using a Thermoelectric Heat Exchange Device

In some embodiments, the heat exchanger is a Peltier device which may heat or cool the esophagus with thermoelectric heat exchange. The heat exchange device is connected to an external controller that powers the Peltier device.

Option 4 for Exchanging Heat: Using an Evaporative Cooling Device

An alternate method of cooling the esophagus is to deliver a coolant directly to the endoluminal surface of the esophagus. In one embodiment, the heat exchange device is connected to an external controller that provides the coolant. The coolant is sprayed in a mist mixed with a gas such as air or oxygen to the surface of the esophagus. The coolant rapidly evaporates due to the gas flow. The esophageal surface is cooled as a result of the evaporation.

Option 5 for Exchanging Heat: Using a Vortex Tube Heat Exchange

Some embodiments of the heat exchanger make use of a vortex tube, a mechanical device that separates a compressed gas into a hot stream and a cold stream. Either stream could be used for heat exchange, so this type of heat exchanger could be used to either warm or cool the esophagus.

Option 6 for Exchanging Heat: Endothermic/Exothermic Chemical Reaction.

Step 5: confirming that the target site is protected

Once the heat exchanger is positioned at the target site and adequate heat exchange is occurring between the esophagus and the heat exchanger, the user confirms that the tissue is protected. There are a number of options to make this confirmation:

- (a) Imaging modalities such as MRI or ultrasound may be used to monitor tissue changes in the esophagus. An absence of lesion growth or tissue changes supports the lack of tissue damage.
- (b) Monitoring a physiological parameter indicative of tissue viability/health. Examples of physiological parameters may include temperature, tissue impedance, blood perfusion rate, oxygen saturation, or nerve function (for example vagus or phrenic nerve). Some embodiments of the heat exchange device comprise a means to measure these parameters. The heat exchange device may be connected to an external controller that interprets/displays/analyses the signals produced from the heat exchange device. The measured physiological parameters may be used in a control loop to alert the user of unsafe levels. The control loop may be connected to the ablation therapy device to stop ablation before a critical level is reached. The control loop may include a mathematical model of changes in the physiological parameter that can predict when irreversible damage may occur, and stop the ablation energy before the dangerous levels are reached.

There are a number of options for measuring temperature. The temperature measured may be one of a number of temperatures, including the temperature of the desired treatment area, or the patient's core body temperature. Temperature may be measured by any of a number of sensors, including thermocouples, thermistors, fiber optics, or by another method such as ultrasound, MRI, infrared, or microwave radiometry.

In one embodiment, the means of measuring temperature is affixed to the heat exchange surface. For example, individual thermocouple pairs, or a flexible circuit containing thermocouples and/or thermistors, or a fiber optic cable may be affixed to the surface of the heat exchanger with adhesives. Alternately, the temperature sensors may be spray or dip coated onto the surface of the heat exchanger with a flexible material such as urethane. Alternately, the temperature sensors may be laminated onto the surface of the heat exchange surface with a thin film, or they may be laminated between two thin film layers, which may then be used to

21

create the heat exchanger. Alternately, the temperature sensors may be positioned inside pockets welded to the surface of the heat exchanger. When positioned on the surface of the heat exchanger, the temperature sensors measure the temperature of the desired treatment area once the heat exchanger makes contact with the desired treatment area (e.g. FIG. 18. Temperature sensors **104** affixed to balloon surface of balloon heat exchanger **101**).

In another embodiment, the temperature sensors are drawn onto the surface of the balloon with conductive ink. For example, the temperature sensors of some embodiments are thermocouples made by crossing a line of conductive silver ink with a line of conductive nickel ink.

In another embodiment, the temperature sensors are affixed to the shaft with adhesives, thermal welding, or another means. For example, a temperature sensor may be added to the distal end of the shaft, which is positioned in the patient's stomach to monitor core body temperature.

In another embodiment, the temperature sensors **104** are mounted on a structural frame **117** that is separate from the heat exchanger. For example, the structural frame **117** may be made of expandable and collapsible struts that can be deployed around the heat exchanger to measure the temperature of the desired treatment area (e.g. FIG. 19 temperature sensors mounted on structural frame separate from heat exchanger). The struts may be in one of a number of configurations, such as linear (top of FIG. 19), helical (bottom of FIG. 19), intersecting, or asymmetrical. The struts may be expanded and collapsed with the use of a mechanical mechanism such as a pull wire. The struts may be made of a number of materials, for example, a flexible metal such as Nitinol, or a plastic such as Pebax, or a shape memory alloy or shape memory polymer. The shape memory polymer may be activated to take on the desired shape by thermal or electrical inputs.

In another embodiment, the struts may be part of the shaft. The embodiment of FIG. 20 includes temperature sensors **104** mounted on struts **118** made from main shaft **103**.

As previously mentioned, the balloon heat exchanger is more preferably oblong to better conform to the cross-sectional area of the collapsed esophagus and reduce the resulting displacement of the esophagus. Accordingly, the embodiments illustrated in FIGS. 19 and 20 may be provided with a more oblong cross-sectional shape. For example, FIGS. 36 and 37 illustrate further embodiments comprising tubes **129** and temperature sensors **104** which feature a more oblong cross-sectional profile.

In another embodiment, the temperature sensors **104** are affixed to or woven into a textile **119** (i.e. a fabric material) that surrounds the heat exchanger (e.g. FIG. 21 temperature sensors affixed to textile). When the heat exchanger is expanded into its desired shape, the fabric may expand around it, allowing the temperature sensors to make contact with and measure the temperature of the desired treatment area.

In another embodiment, the temperature sensors **104** are affixed to strands **120** connected at one end of the heat exchanger so that they hang freely about the other end of the heat exchanger. The example of FIG. 22 includes temperature sensors **104** affixed to strands **120** which are attached to main shaft **103**. Strands **120** are flexible and atraumatic such that as they are advanced through the esophagus, the esophagus is not damaged.

In order to obtain meaningful temperature data, and array of temperature sensors may be used to measure a plurality of temperatures. The sensors may be positioned in such a way that an algorithm may be used to interpolate the tempera-

22

tures between the sensors in order to produce a temperature map of the esophageal surface. Alternately, a temperature map may be produced using IR or microwave temperature measuring modalities.

One concern some users may have with respect to the sensors is what is known as the antenna effect. There is some published literature indicating that metal electrodes in the esophagus may promote thermal injury as a result of electrical or thermal interactions with the ablation catheter. To eliminate this risk, the electrodes on some embodiments of the heat exchange device are insulated, or made of a non-conductive material. Alternately, the electrodes may be positioned such that the electrical or thermal interactions will not affect them, for example, the electrodes may be located on the posterior wall of the heat exchanger so that the heat exchanger insulates the electrodes from the interactions. In addition, filters may be built into the external device where the signals are interpreted and displayed to eliminate these interactions.

Step 6: Retrieving the Heat Exchange Device

After treatment, the heat exchanger is typically collapsed for removal from the patient. In one embodiment, the heat exchanger is evacuated by pulling a vacuum at the outlet port or the inlet port. Once evacuated, the heat exchanger can be pulled back through the delivery orifice and removed from the patient. In alternative embodiments, the heat exchanger is collapsed using a sleeve around the heat exchanger. This sleeve may comprise a fabric mesh structure, a metal structure, such as a structure similar to a stent, or a polymer cage. In some embodiments the sleeve is a sheath. In one embodiment, the sleeve is collapsed using a mechanical mechanism. In another embodiment, the sleeve is collapsed using shape memory material properties.

Once the heat exchanger is collapsed, the heat exchange device may be pulled into the delivery orifice, or the delivery orifice may be advanced over the heat exchange device. The heat exchange device may be inverted (inside-out) as it is pulled into the delivery orifice. In some embodiments of the method, the delivery orifice is the patient's nose or mouth. In another embodiment, the delivery orifice is a sheath separate from the device. The sheath may have a telescoping feature. The sheath may be integrated with the heat exchange device. For example, it may comprise expanding and contracting struts that are part of the body of the heat exchange device, or it may be a translating portion of the heat exchange device body.

Once inside the delivery orifice, the heat exchange device is removed from the patient.

Patient's Body Core Temperature

The user may be concerned about affecting the patient's core body temperature as a result of exchanging heat in the esophagus. There are a number of optional features and surgical techniques to mitigate this risk.

(a) Focus heat exchange at areas of highest risk. This may be achieved by monitoring a physiological parameter at different locations on the esophagus and using a control loop in the external controller to determine the high-risk areas and focus heat exchange in those areas.

(b) Counteract heat exchange at esophagus with opposite and optionally equal heat exchange at another body location. This may be achieved by measuring the amount of heat exchanged by the heat exchange device in the esophagus and using a separate device (such as a warming or cooling blanket) to exchange an equal and opposite amount of heat at a location distinct from the esophagus. A control loop may be used to automatically balance the heat exchanged. Alternatively,

23

the heat exchange device may be used to supply opposite and optionally equal heat exchange while the ablation therapy is not being applied.

- (c) Only exchange heat at the esophagus while an ablation is being performed. This may be achieved by a communication link between the ablation therapy device and the heat exchange device. The heat exchange device is activated only when the ablation therapy is applied.
- (d) Some embodiments of the device may comprise the following insulative features at the non-therapy areas to minimize overall heat exchange and focus heat exchange only in the target area:
 - (i) a coating of insulative material,
 - (ii) an insulating lubricant or gel,
 - (iii) an air filled lumen or space, or
 - (iv) an air filled balloon inside (or outside) of the heat exchanger. FIG. 23 illustrates a balloon heat exchanger 101 with an insulating balloon 121 on the inside. FIGS. 24 and 34 illustrates a balloon heat exchanger 101 with an insulating balloon 121 on the outside.

In an ablation procedure, the “non-therapy area” is the side of the esophagus farthest away from the heart. By insulating the side of the esophagus farthest away from the heart, the heat exchange directed away from the “non-therapy area” and focused on the target area, which is the side of the esophagus closest to the heart. FIG. 34 illustrates another embodiment of a heat exchanger 101 which comprises an insulating portion 131. The insulating portion 131 may comprise one or more of an insulating lubricant or gel, a coating of insulating material, or air.

FIG. 64 illustrates the example of air pockets around the shaft tubing of the balloon catheter minimizing heat loss at a non-therapy area. The outer layer 232 of tubing 230 insulates the shaft by creating air pockets 233 between the outer layer 232 and the inner layer 231 (which surrounds fluid pathway 235). The air pockets 233 reduce the cooling of the mouth and the part of the esophagus which is not in the ablation area, thereby increasing the thermal efficiency of the design.

- (e) Monitoring core body temperature. The heat exchange device may have a temperature sensor at a location away from the heat exchange area to monitor core body temperature. For example, in some embodiments the temperature sensor is at the distal end of the device and is positioned in the patient’s stomach. A control loop may be used to feedback the patient’s core temperature to the user and alert the user of dangerous temperatures. Alternately, the control loop could be used to control the amount of heat being exchanged in the patient.
- (f) Determining a safe heat exchange operating range based on patient characteristics, the bio-heat equation, and other pertinent information. Some embodiments of the method include monitoring the amount of heat exchanged by the heat exchange device and confirming that it does not exceed the calculated safe amount.

These techniques can be performed during step 4 (FIG. 26) of the above described method.

Ablation Therapy

It is also important that an ablation therapy is not adversely affected by the heat exchange at the esophagus. To eliminate this risk, the user may monitor lesion growth or a physiological parameter at the therapy site using methodologies described above. A feedback loop may also be used to maximize the therapeutic energy delivered while the esophagus is not in danger. This may be achieved by monitoring a physiological parameter indicative of tissue

24

health/viability as described above, and using that data in a control loop to stop or decrease ablative therapy when the esophageal tissue is in danger, and increase/optimize ablative therapy when the esophageal tissue is not affected. The data may also be used to focus the heat exchange at high risk areas in the esophagus to minimize the impact on the therapeutic energy delivery. The data may also be used to decrease or stop the heat exchange during ablations when the esophagus is not at risk. These techniques can be performed during step 4 (FIG. 26) of the above described method.

Other Steps

Other additional steps in the method may include pacing the heart and performing a cardiac EP exam using the heat exchange device. To facilitate these steps, some embodiments of the heat exchange device comprise pacing and ECG electrodes on the body of heat exchange device. This technique can be performed during step 3 (FIG. 26) of the above described method.

Injury to an esophagus caused by heat or cold being delivered to the left atrium is prevented by regulating the temperature of the esophagus using embodiments of a heat exchange device having a heating/cooling balloon (or sac) which has an inflated cross section corresponding with the collapsed/relaxed/natural cross section of the inside of the esophagus whereby inflation of the balloon maintains the esophagus in its natural shape and location and avoids not displacing the esophagus towards the left atrium. Some alternative embodiments includes altering a configuration of the balloon to conform to or correspond with the cross section of an esophagus by means other than inflation.

The embodiments of the invention described above are intended to be exemplary only. The scope of the invention is therefore intended to be limited solely by the scope of the appended claims.

It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable subcombination.

Although the invention has been described in conjunction with specific embodiments thereof, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the broad scope of the appended claims. All publications, patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention.

EXAMPLES

Example 1. A method of regulating a temperature of an esophagus when heat or cold is delivered to a left atrium comprises altering a heat exchange device from an insertable configuration to a heat exchanging configuration which substantially conforms and corresponds with a cross-section of an inside of the esophagus such that the esophagus is

25

substantially maintained in its natural shape and location whereby the esophagus is substantially not displaced towards the left atrium.

Example 2. A method of regulating a temperature of an esophagus when heat or cold is delivered to a left atrium comprises (a) inflating a heat exchange device from an collapsed configuration to an inflated configuration which substantially conforms and corresponds with a cross-section of an inside of the esophagus such that the esophagus is substantially maintained in its natural shape and location whereby the esophagus is substantially not displaced towards the left atrium and (b) regulating the temperature of the esophagus using the heat exchange device.

Example 3. A method of regulating a temperature of an esophagus when heat or cold is delivered to a left atrium includes the steps of:

- (1) measuring the esophagus and selecting a size a heat exchange device which fits the esophagus;
- (2) delivering the heat exchange device to a target site;
- (3) confirming a desired location of the heat exchange device;
- (4) exchanging heat with the esophagus;
- (5) confirming that the target site is protected; and
- (6) retrieving the heat exchange device.

Example 4. The method of example 3, wherein step (1) comprises using imaging such as fluoroscopy, CT, MRI, or EAM.

Example 5. The method of example 3, wherein the heat exchange device comprises a balloon and a main shaft, and the method includes, before step (2), the step of deflating or collapsing the balloon and wrapping or folding the balloon around the main shaft.

Example 6. The method of example 3, wherein the heat exchange device comprises a balloon, and the method includes, before step (2), the step of priming the heat exchange device to replace air with fluid.

Example 7. The method of example 3, wherein step (2) comprises advancing the heat exchange device through a nostril.

Example 8. The method of example 3, wherein the heat exchange device further comprises imaging markers and step (2) includes using an imaging system to position the heat exchange device.

Example 9. The method of example 3, wherein step (2) comprises advancing an outer sheath with the heat exchange device and pulling back on the outer sheath when the heat exchange device is positioned to expose the heat exchange device.

Example 10. The method of example 3, wherein step (3) comprises confirming an orientation of the heat exchange device relative to a known anatomical marker by imaging of imaging markers on the heat exchange device.

Example 11. The method of example 10, wherein the known anatomical marker is the left atrium.

Example 12. The method of example 3, wherein step (4) includes begin circulating a heat exchange fluid through the heat exchange device before heat or cold is delivered to the left atrium.

Example 13. The method of example 12, wherein step (4) includes stop circulating the heat exchange fluid through the heat exchange device after heat or cold is delivered to the left atrium.

Example 14. The method of example 3, wherein step (5) comprises imaging of a tissue of the esophagus to determine if the tissue has been changed.

26

Example 15. The method of example 3, wherein step (5) comprises monitoring a physiological parameter which indicates a health factor of a tissue of the esophagus.

Example 16. The method of example 13, wherein prior to step (6), the method includes vacuuming the heat exchange fluid from the heat exchange device.

Example 17. The method of example 9, wherein prior to step (6), the method includes advancing the outer sheath to cover the heat exchange device, thereby reducing a diameter of the heat exchange device.

Example 18. The method of example 3, wherein step (6) includes removing the heat exchange device from a patient.

Example 19. A method of monitoring a temperature of a tissue of an esophagus includes (a) inflating a device from an collapsed configuration to an inflated configuration which conforms and corresponds with a cross-section of an inside of the esophagus such that the esophagus is maintained in its natural shape and location whereby the esophagus is not displaced towards a left atrium and (b) monitoring the temperature of the tissue using sensors on an outside of the device.

Example 20. The method of example 19, wherein step (b) comprises using sensors on one side of the device.

We claim:

1. A heat exchanging device for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium of a heart, comprising:

an elongated shaft comprising a distal end and a proximal end, the elongated shaft defining at least a first lumen and a second lumen;

a heat exchanger attached to the distal end of the elongated shaft, the heat exchanger comprising an insertable configuration, a heat exchanging configuration, a distal end, a proximal end, and a cavity therebetween, at least a portion of said cavity being in fluid communication with the first lumen and the second lumen of the elongated shaft;

wherein a cross-section of the heat exchanger in the insertable configuration is smaller than a cross-section of the heat exchanger in the heat exchanging configuration, wherein said cross-section of the heat exchanger in the heat exchanging configuration substantially conforms to and corresponds with a cross-section of an inside surface of the esophagus such that the esophagus is substantially maintained in its natural shape and location when the heat exchanger is in its heat exchanging configuration, and wherein said shape of the heat exchanger is constrained by a weld pattern, wherein the weld pattern comprises at least one weld, and wherein said at least one weld attaches at least part of an anterior surface of the heat exchanger and a posterior surface of the heat exchanger.

2. The heat exchanging device of claim 1, wherein the at least one weld comprises a plurality of welds which are oriented as a series of consecutive chevrons along a length of the heat exchanger.

3. The heat exchanging device of claim 2, wherein at least one chevron of the series of consecutive chevrons is comprised of a pair of welds which are generally curved in shape with hook-shaped ends.

4. The heat exchanging device of claim 1, wherein the elongated shaft is a multi-lumen tube which defines an inlet lumen and a first outlet lumen, said inlet lumen comprises a fluid inlet port to allow heat exchange fluid to flow into said heat exchanger and said first outlet lumen comprises a fluid outlet port to allow heat exchange fluid to flow out of said heat exchanger.

27

5. The heat exchanging device of claim 4, further comprising a second outlet lumen.

6. The heat exchanging device of claim 4, wherein the first outlet lumen has a larger diameter than the inlet lumen.

7. The heat exchanging device of claim 1 further comprising a radiopaque material positioned on the heat exchanger.

8. The heat exchanging device of claim 7, wherein the heat exchanger is comprised of at least one balloon.

9. The heat exchanging device of claim 8, wherein the radiopaque material defines a bordering frame on a side of the heat exchanger, wherein the bordering frame borders at least a portion of a surface of the side of the heat exchanger.

10. The heat exchanging device of claim 8, wherein the radiopaque material is configured to indicate which side of the balloon is facing a heart of a patient.

11. The heat exchanging device of claim 7, further comprising at least one temperature sensor on a side of the heat exchanger and the radiopaque material being configured to indicate which direction the at least one temperature sensor is facing.

12. The heat exchanging device of claim 1, wherein the first lumen is a fluid inlet tube and an inner layer of the fluid inlet tube defines an inlet fluid pathway, and the elongated shaft defines a series of air pockets surrounding the inlet fluid pathway to thereby thermally insulate the fluid inlet tube.

13. The heat exchanging device of claim 12, wherein an outer layer of the fluid inlet tube and the inner layer of the fluid inlet tube define the series of air pockets therebetween.

14. The heat exchanging device of claim 1, wherein the second lumen is a fluid outlet tube and an inner layer of the fluid outlet tube defines an outlet fluid pathway, and the elongated shaft defines a series of air pockets surrounding the outlet fluid pathway to thereby thermally insulate the fluid outlet tube.

15. The heat exchanging device of claim 14, wherein an outer layer of the fluid outlet tube and the inner layer of the fluid outlet tube define the series of air pockets therebetween.

16. The heat exchanging device of claim 1, wherein the heat exchanging device further comprises an outer balloon surrounding the heat exchanger.

17. The heat exchanging device of claim 16, wherein the elongated shaft comprises an inner lumen which is in fluid communication with the outer balloon of the heat exchanger and an outer lumen which is in fluid communication the outer balloon.

18. A heat exchanging device for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium of a heart, comprising:

an elongated shaft comprising a distal end and a proximal end, the elongated shaft defining at least a first lumen and a second lumen;

a heat exchanger attached to the distal end of the elongated shaft, the heat exchanger comprising an insertable configuration, a heat exchanging configuration, a distal end, a proximal end, and a cavity therebetween,

28

at least a portion of said cavity being in fluid communication with the first lumen and the second lumen of the elongated shaft;

wherein a cross-section of the heat exchanger in the insertable configuration is smaller than a cross-section of the heat exchanger in the heat exchanging configuration, wherein said cross-section of the heat exchanger in the heat exchanging configuration substantially conforms to and corresponds with a cross-section of an inside surface of the esophagus such that the esophagus is substantially maintained in its natural shape and location when the heat exchanger is in its heat exchanging configuration, and wherein said shape of the heat exchanger is constrained by a weld pattern, wherein the weld pattern comprises at least one weld, and wherein said at least one weld attaches at least part of an anterior surface of the heat exchanger and a posterior surface of the heat exchanger;

wherein the heat exchanger comprises a first zone and a second zone which can be identified under imaging, the first zone having a first temperature sensor and the second zone having a second temperature sensor, whereby the heat exchanging device is operable to be moved such that a selected zone with a preferred temperature is adjacent to a heart of a patient.

19. A heat exchanging device for regulating a temperature of an esophagus when heat or cold is delivered to a left atrium of a heart, comprising:

an elongated shaft comprising a distal end and a proximal end, the elongated shaft defining at least a first lumen and a second lumen;

a heat exchanger attached to the distal end of the elongated shaft, the heat exchanger comprising an insertable configuration, an expanded heat exchanging configuration, a distal end, a proximal end, and a cavity therebetween, at least a portion of said cavity being in fluid communication with the first lumen and the second lumen of the elongated shaft;

wherein a cross-section of the heat exchanger in the insertable configuration is smaller than a cross-section of the heat exchanger in the heat exchanging configuration, wherein said cross-section of the heat exchanger in the heat exchanging configuration substantially conforms to and corresponds with a cross-section of an inside surface of the esophagus such that the esophagus is substantially maintained in its natural shape and location when the heat exchanger is in its heat exchanging configuration, wherein the heat exchanger moves from the insertable configuration to the heat exchanging configuration by inflating the cavity with heat exchange fluid, and wherein said shape of the heat exchanger is constrained by a weld pattern.

20. The heat exchanging device of claim 19, wherein the weld pattern comprises a plurality of welds that attach at least part of an anterior surface of the heat exchanger and a posterior surface of the heat exchanger to create multiple flow channels that extend along a length and width of the heat exchanger.

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